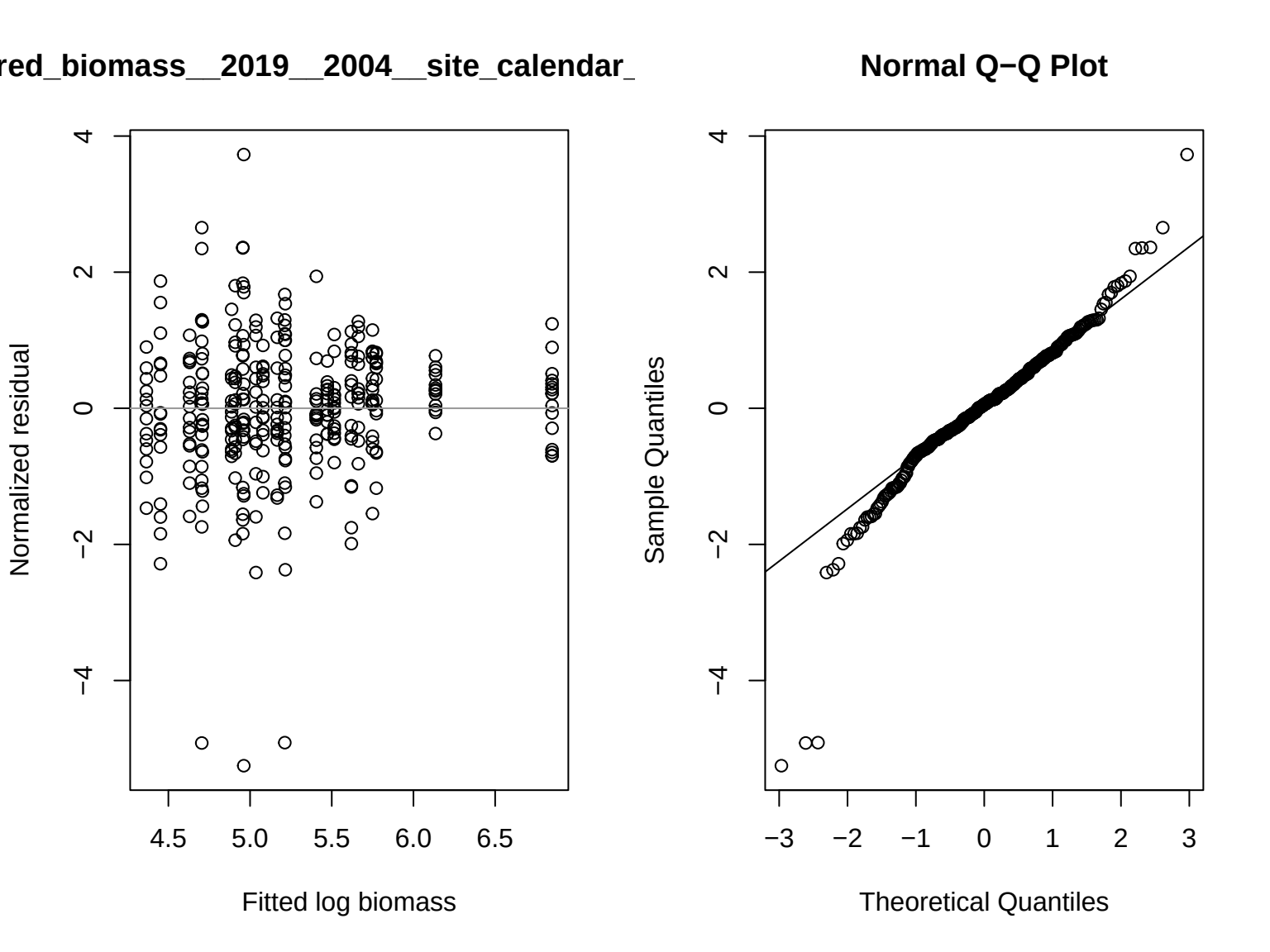
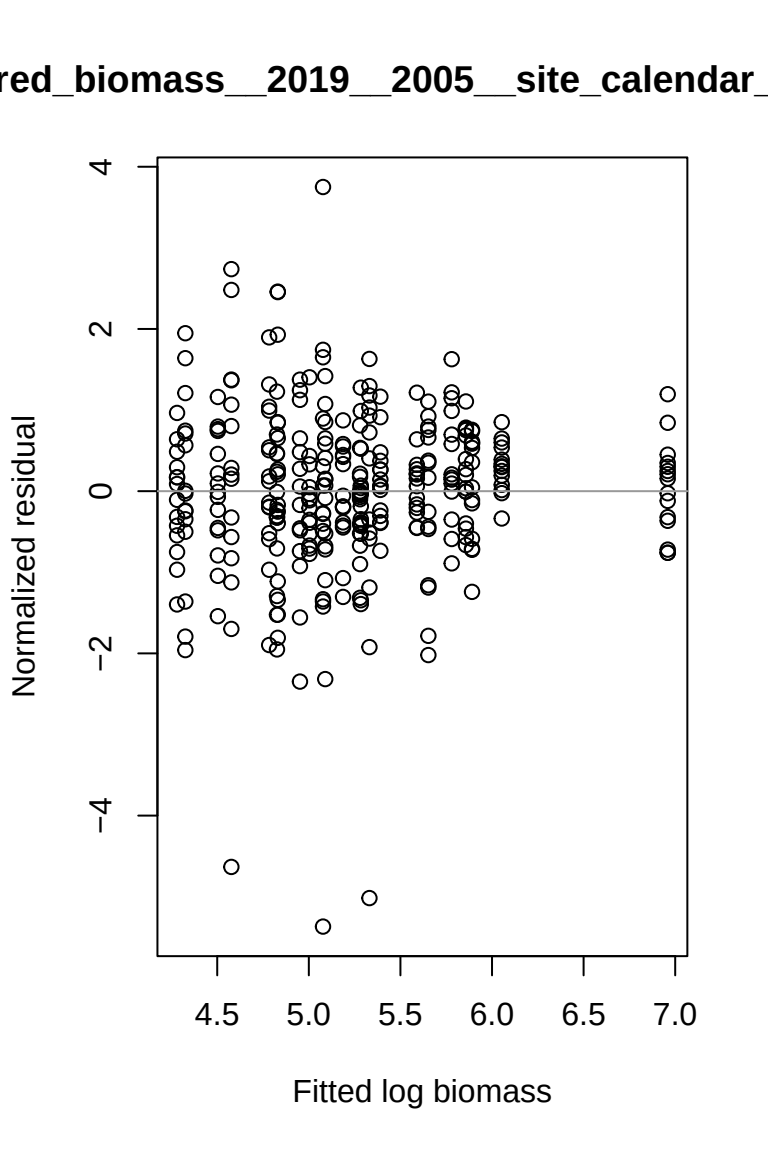
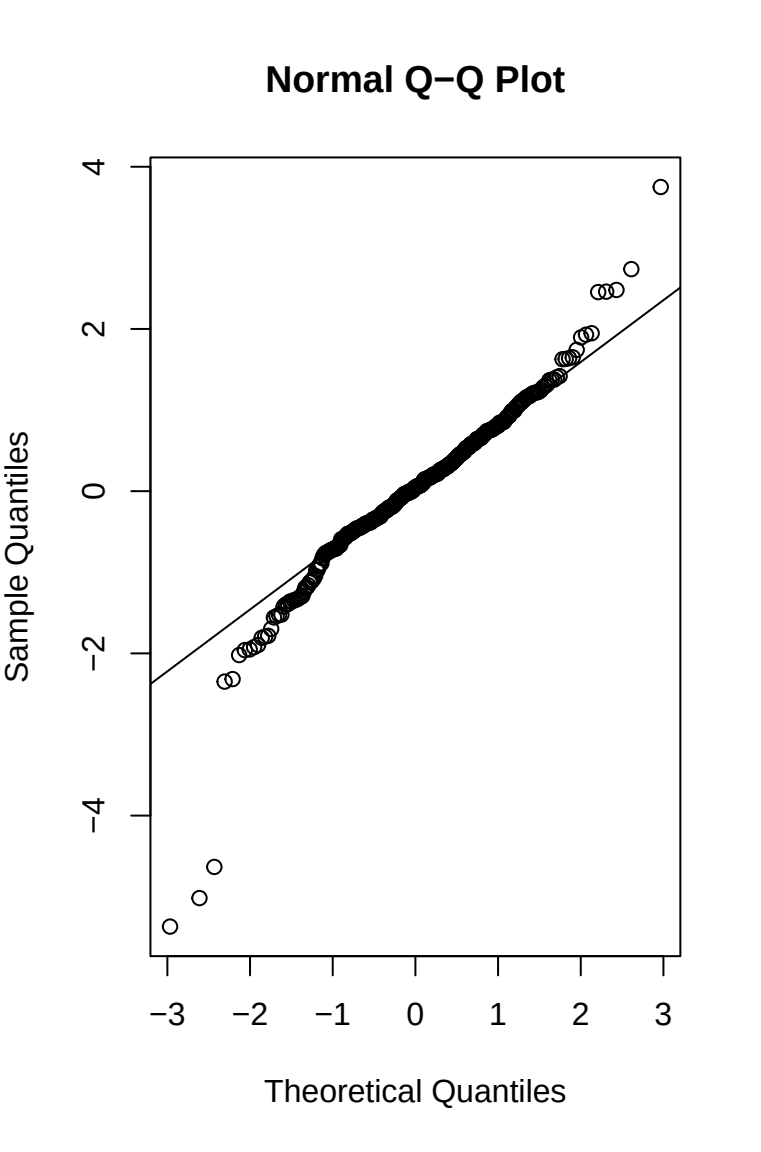
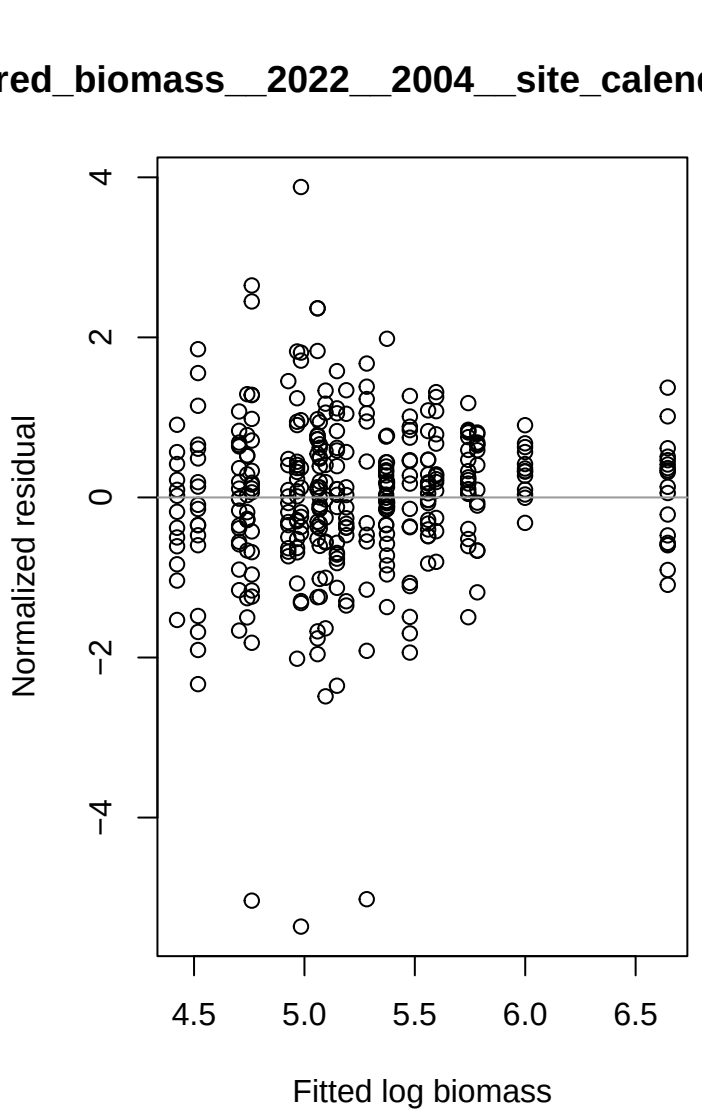
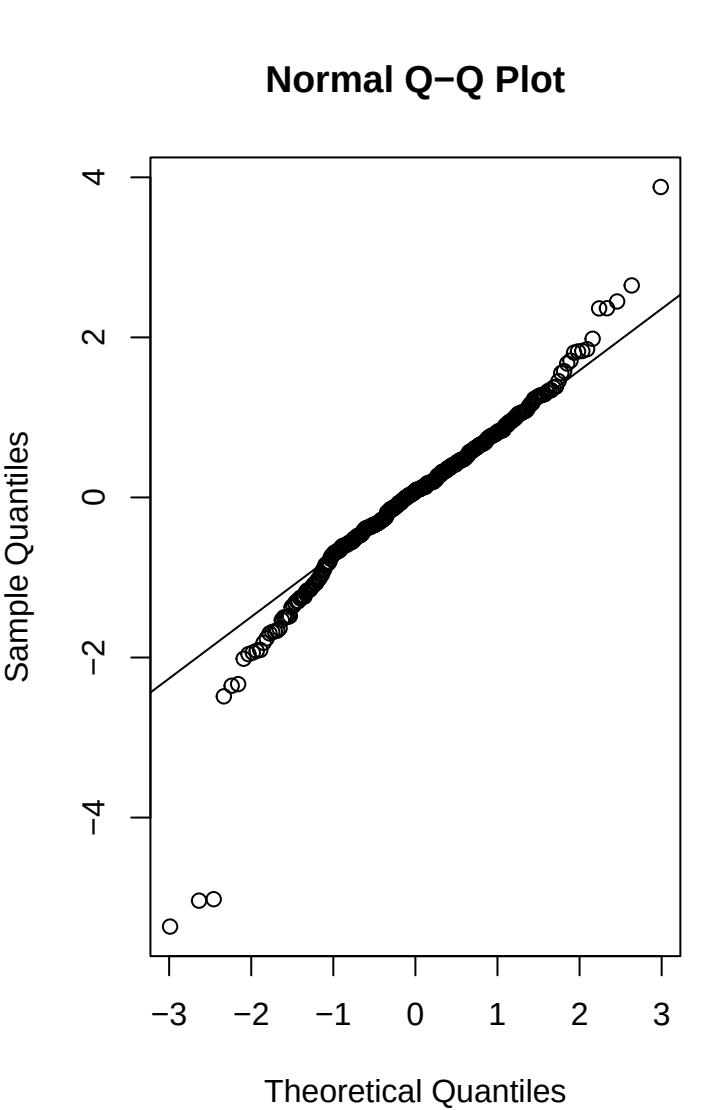
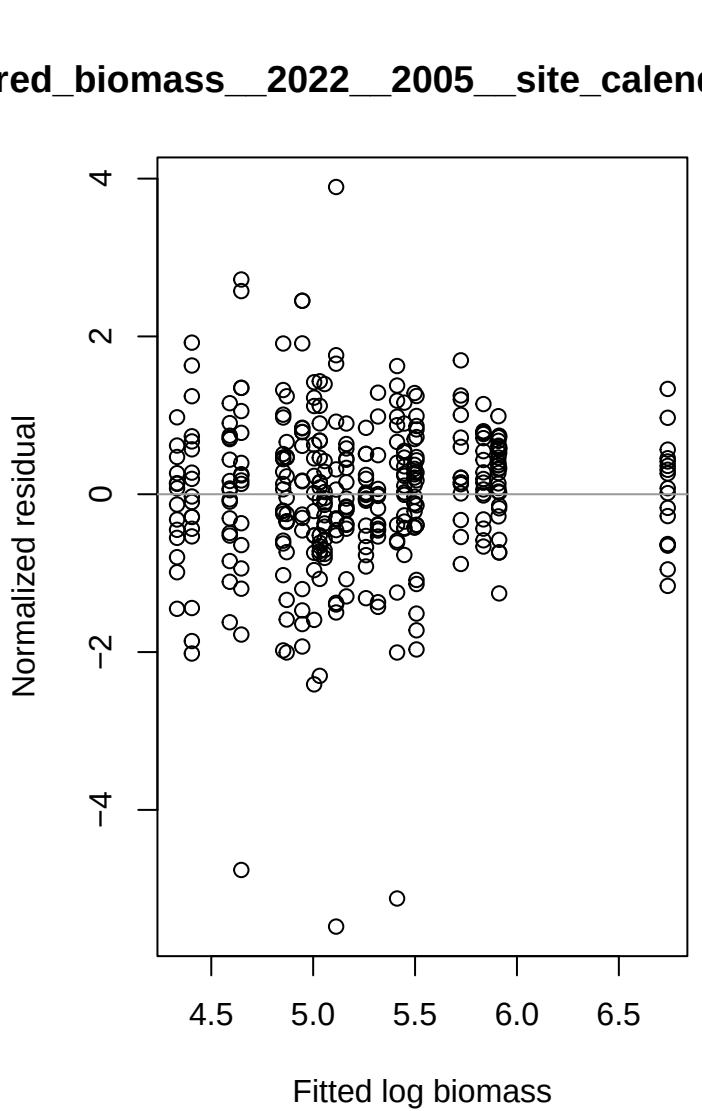

red_biomass__2019__2004__site_calendar_
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

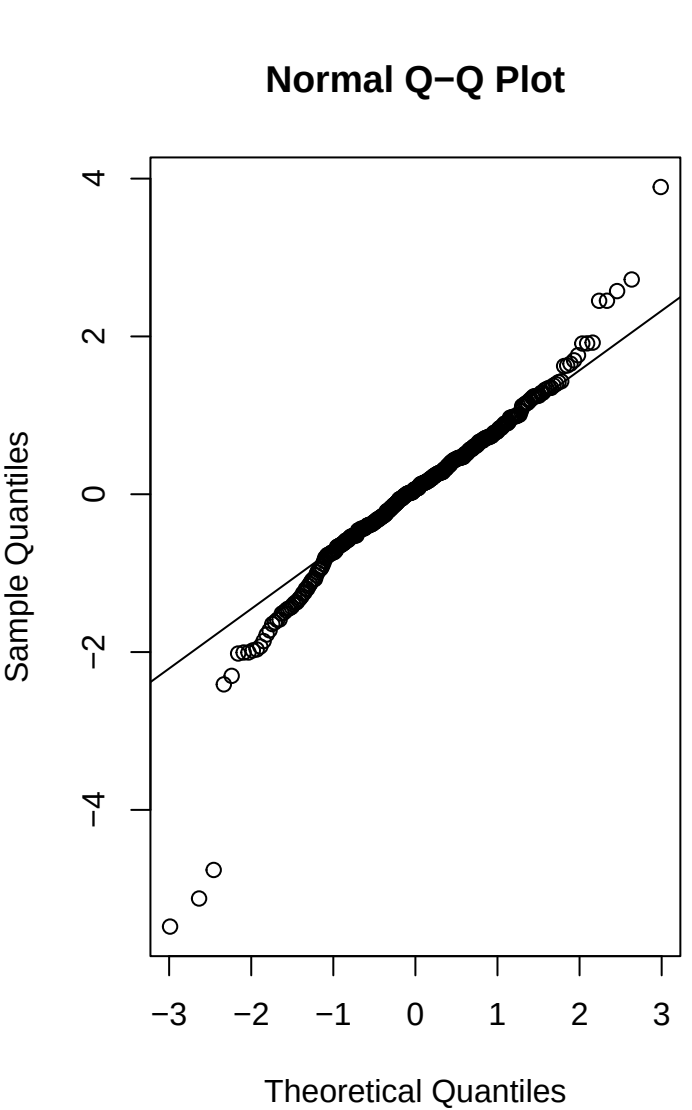

red_biomass__2019__2005__site_calendar_
Normalized residual
Fitted log biomass
4
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5
7.0

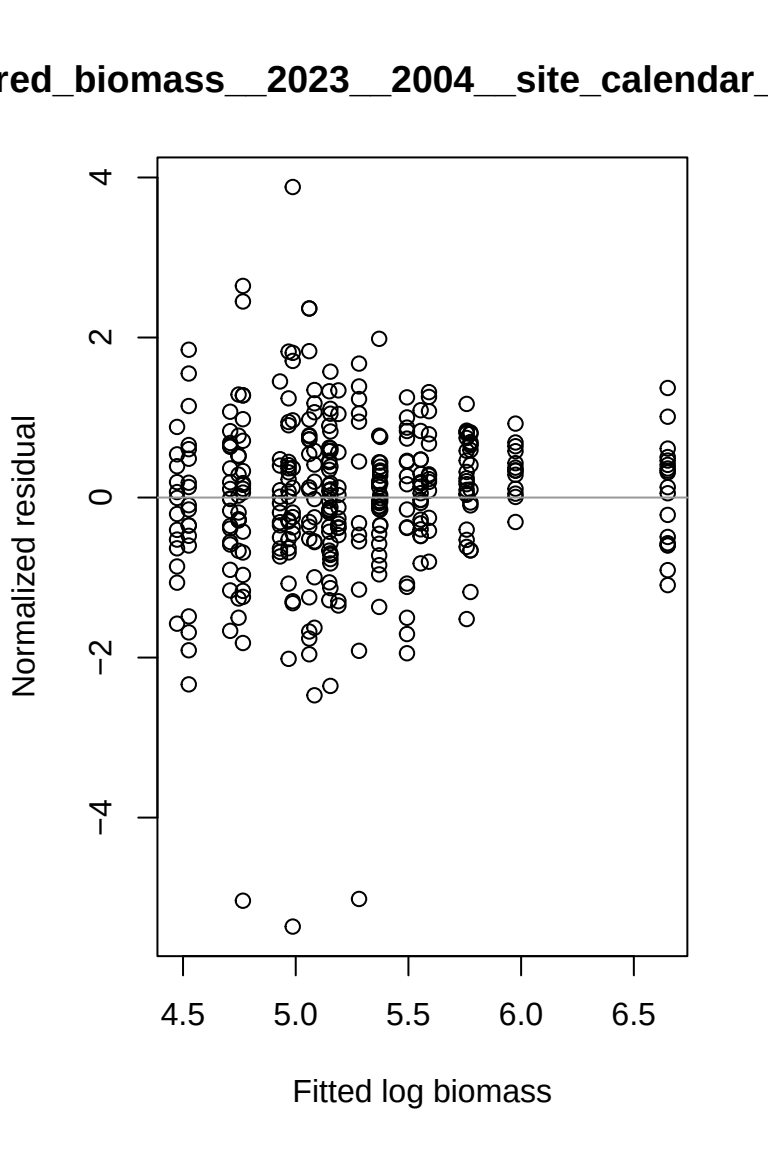

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3

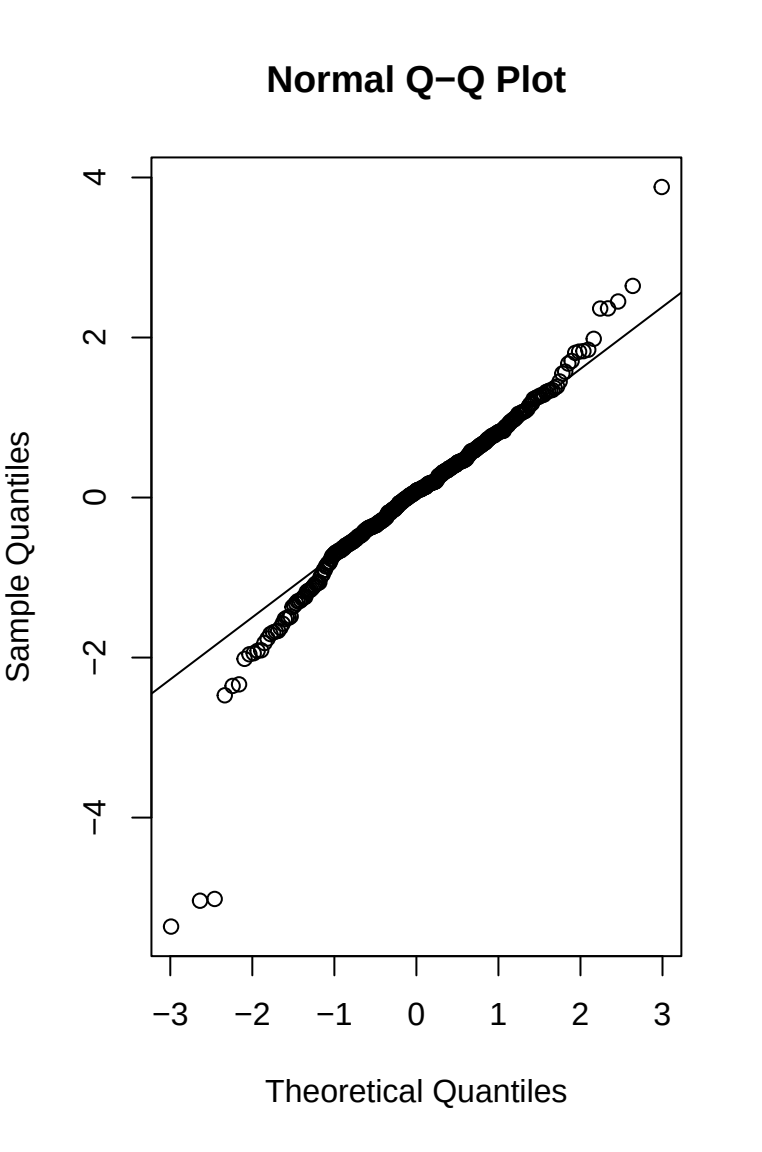

red_biomass__2022__2004__site_calendar_
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
4
2
0
−2
−4

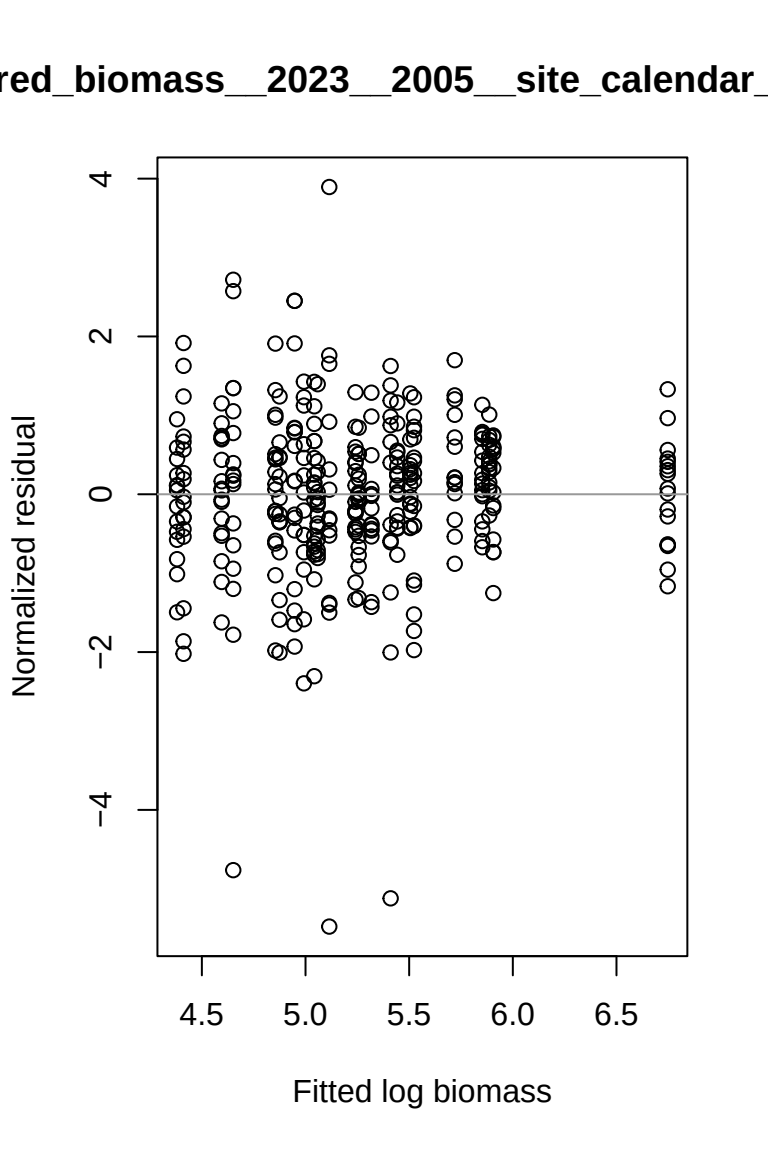

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
4
2
0
−2
−4

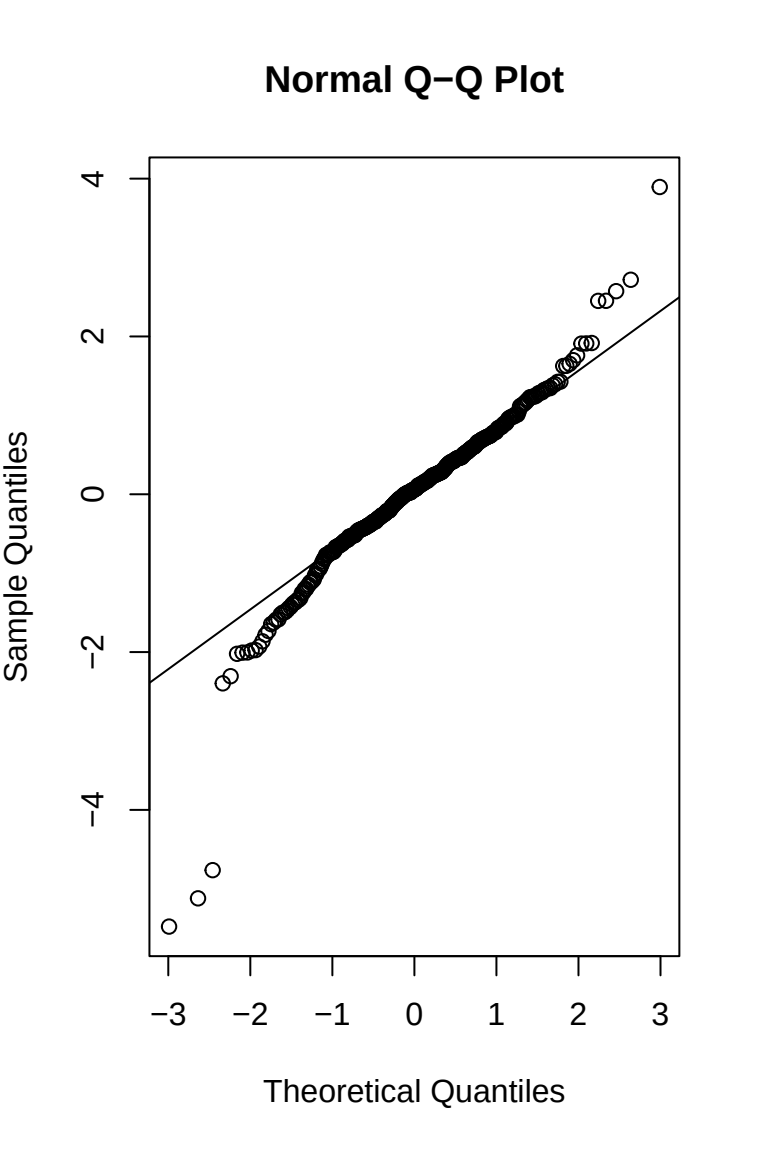

red_biomass__2022__2005__site_calendar_
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4

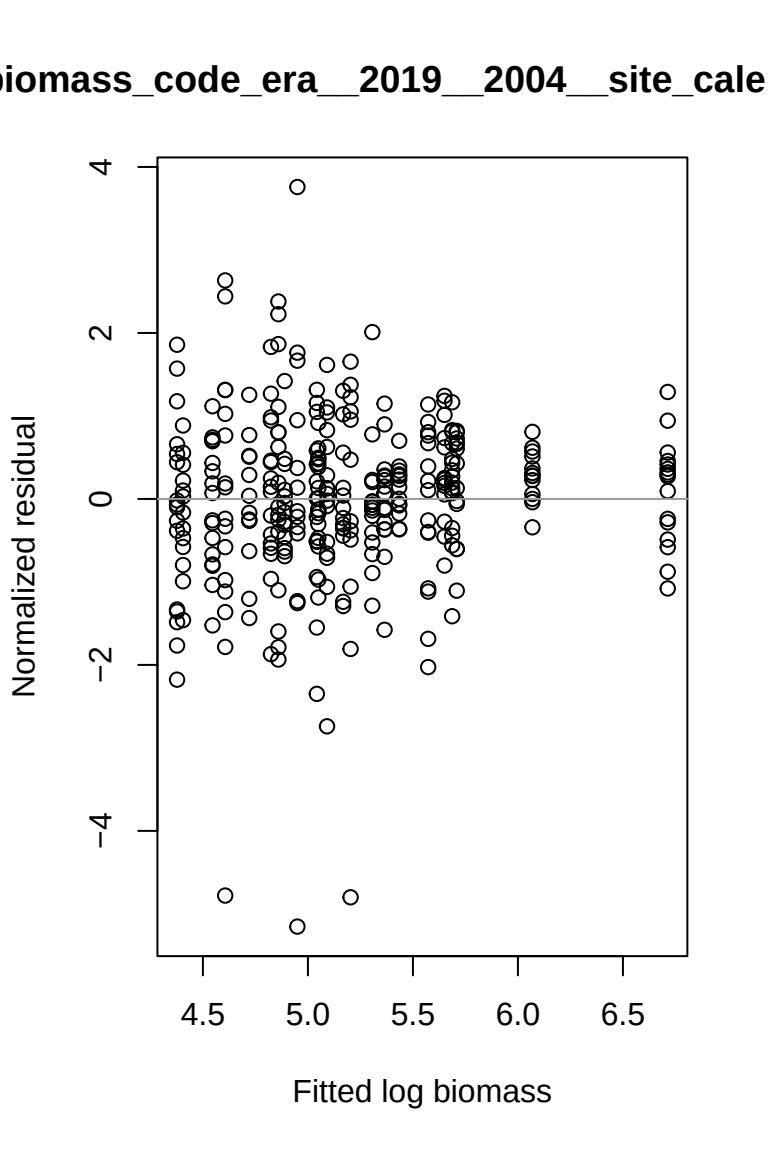

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2
4

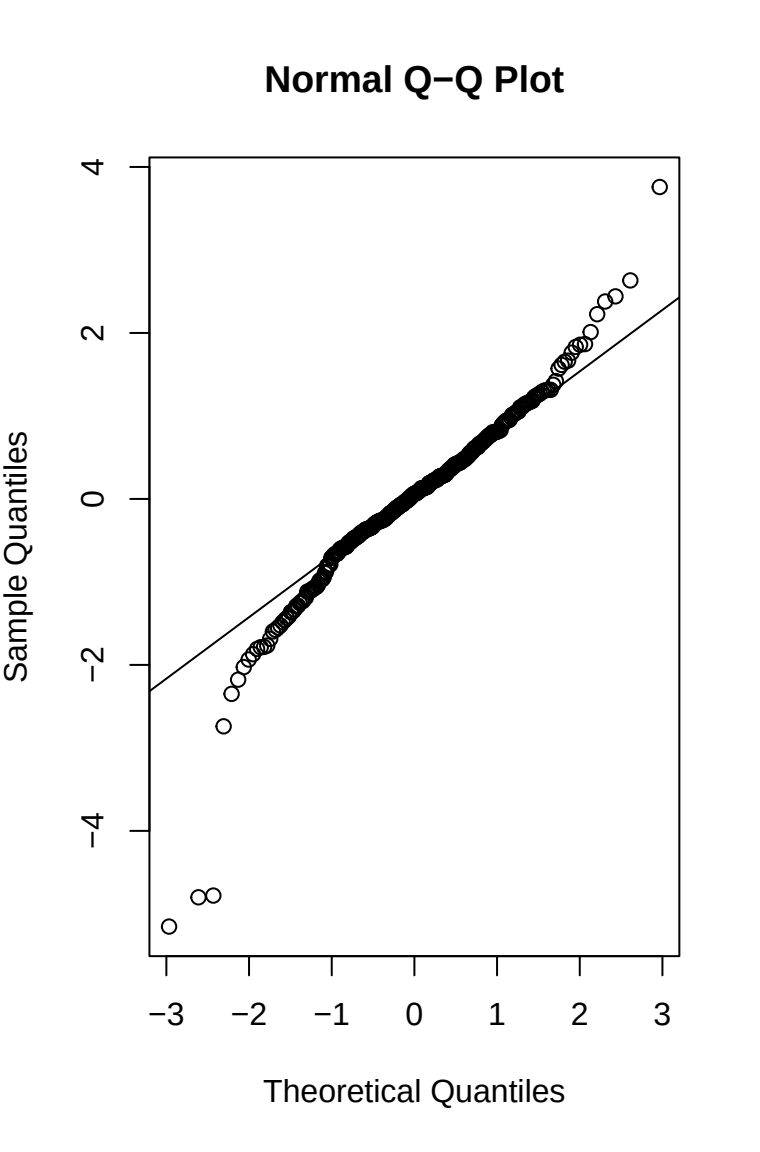

red_biomass__2023__2004__site_calendar_
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4

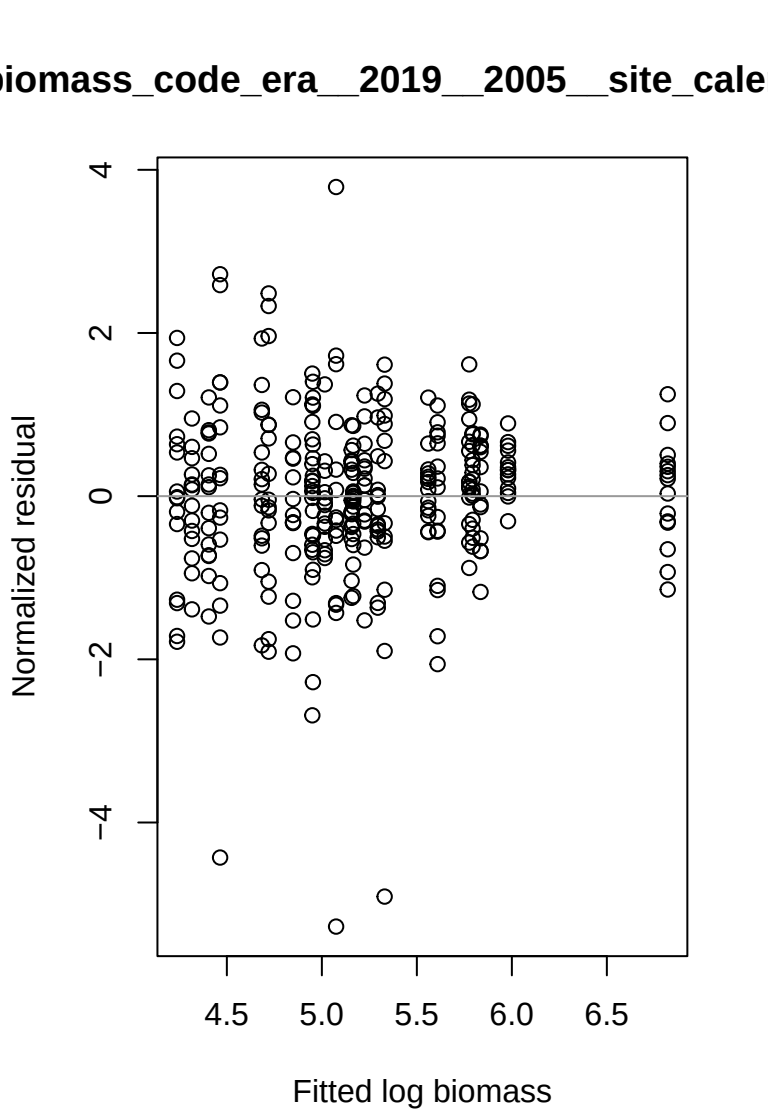

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2
4

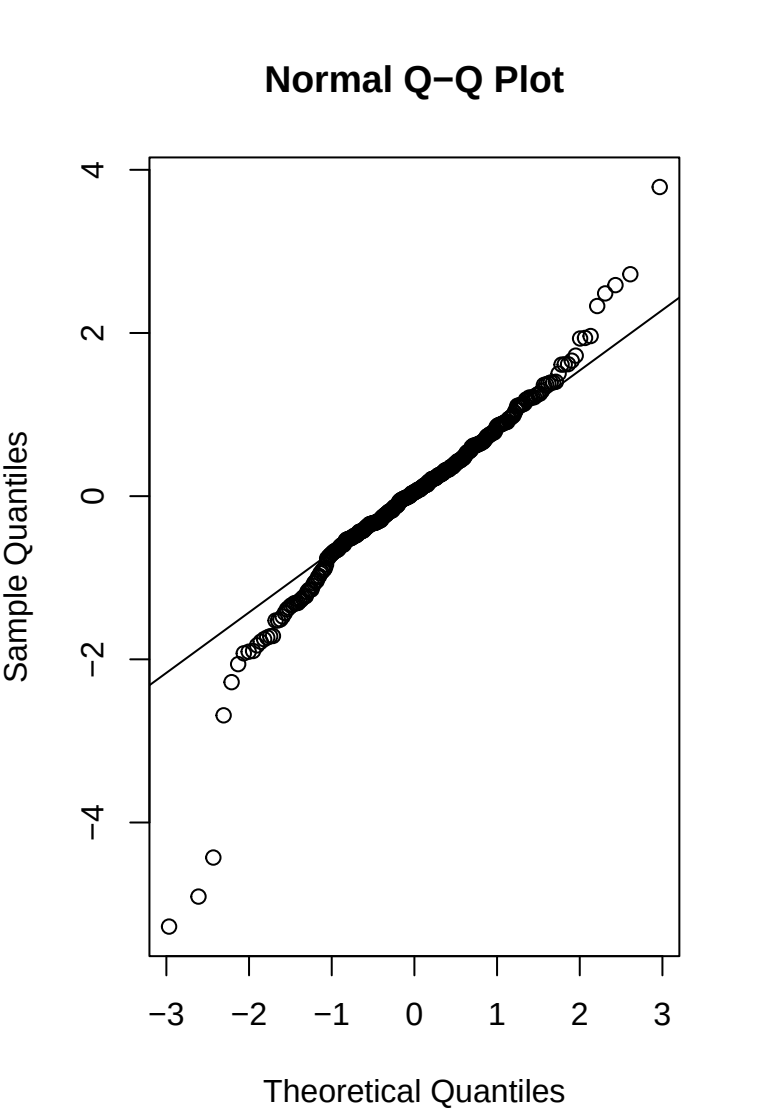

red_biomass__2023__2005__site_calendar_
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4

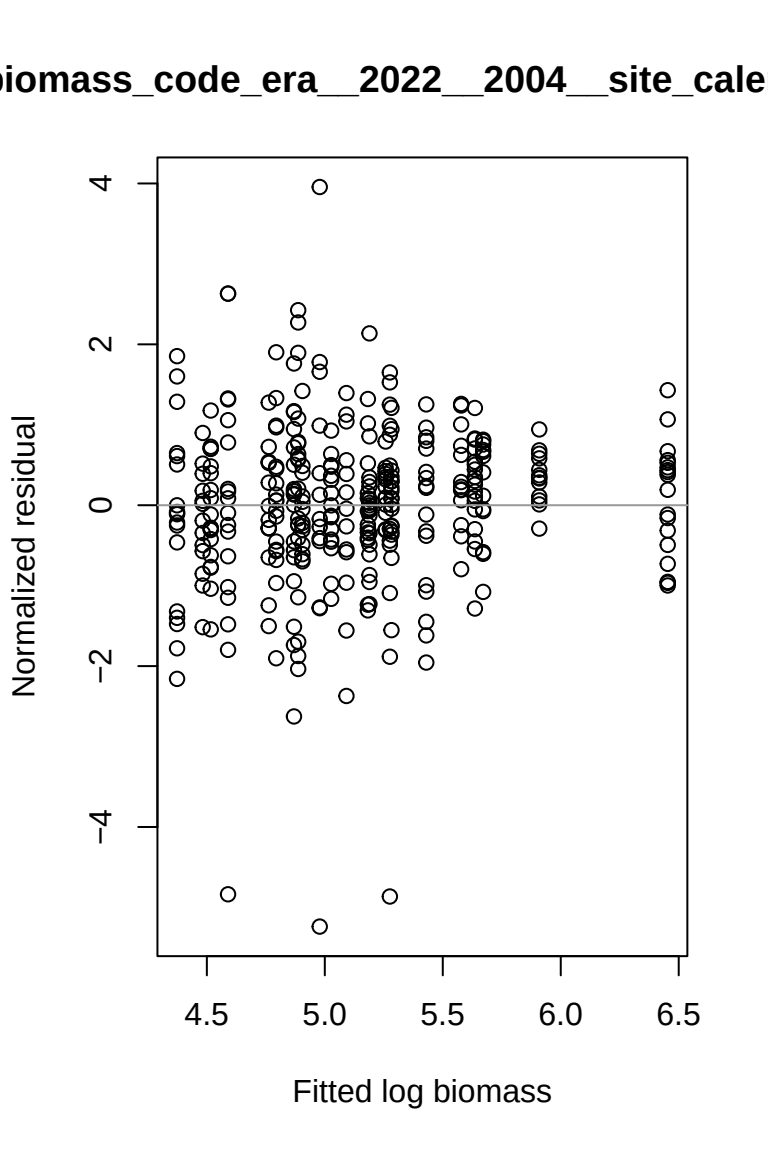

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2
4

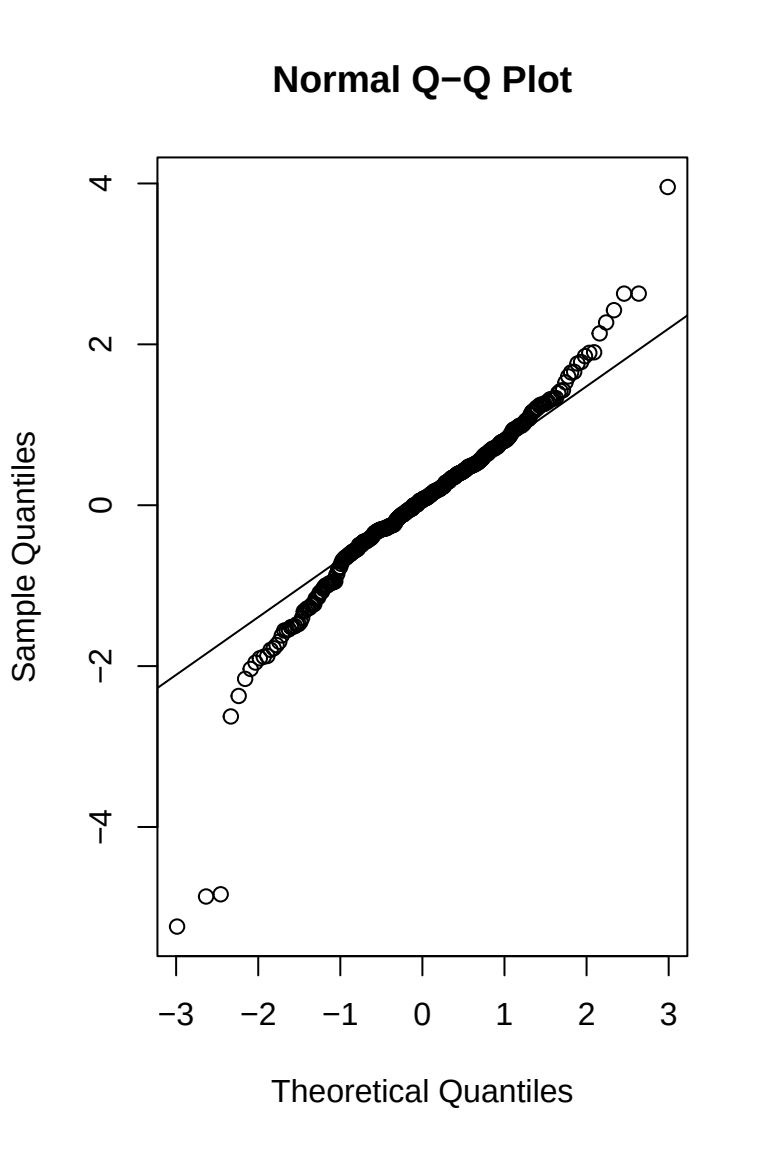

biomass_code_era__2019__2004__site_cale
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4

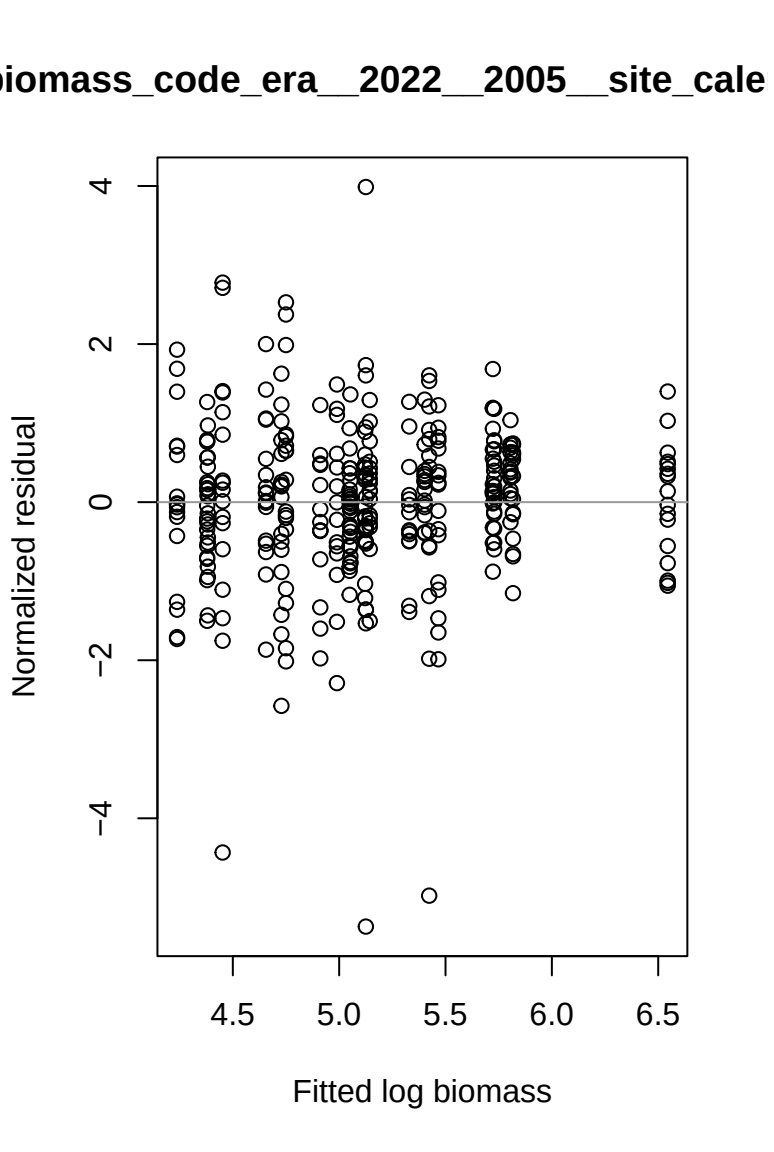

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2
4

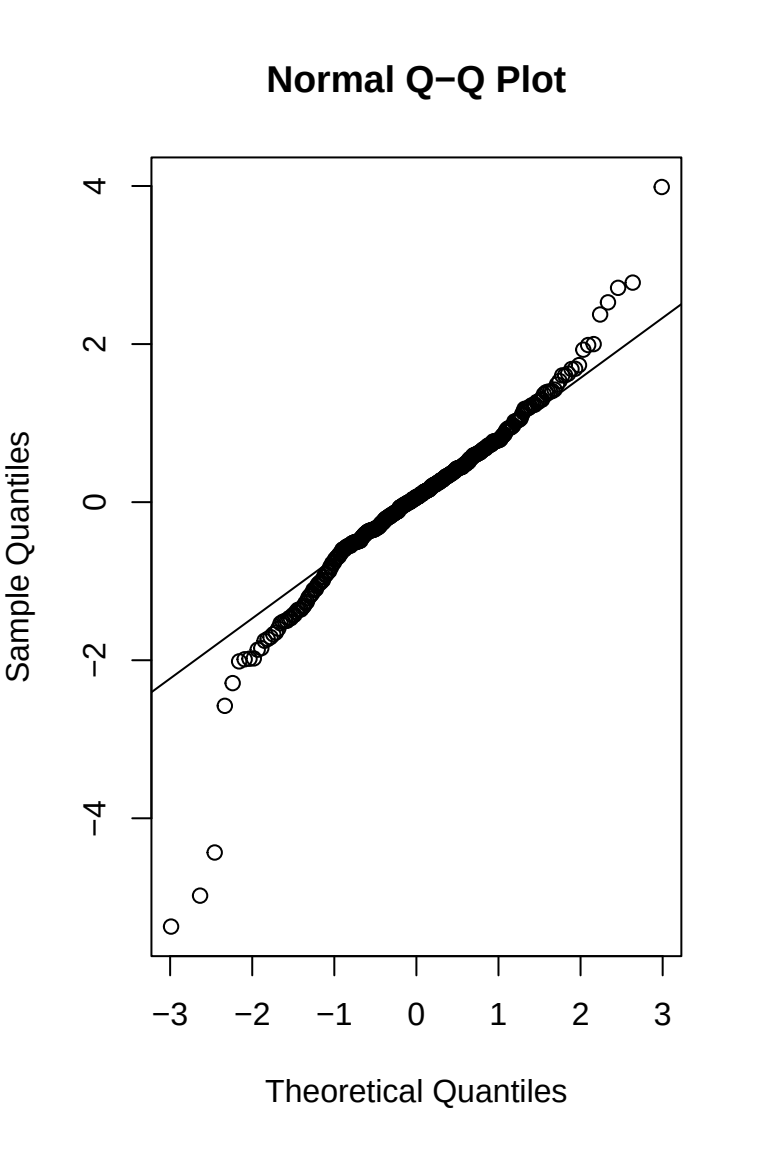

biomass_code_era__2019__2005__site_cale
Normalized residual
Fitted log biomass
4
2
0
−2
−4
4.5
5.0
5.5
6.0
6.5

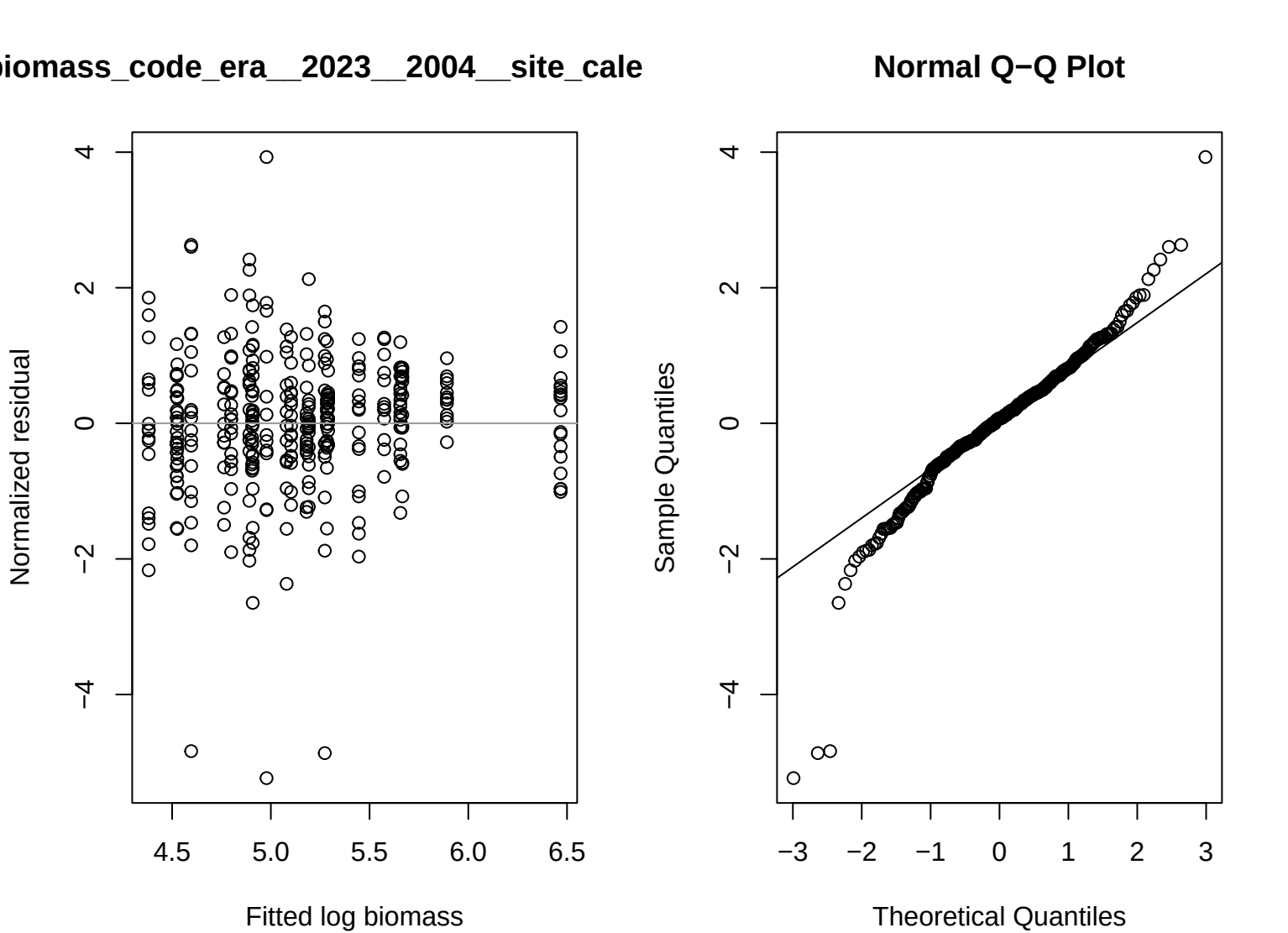

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3

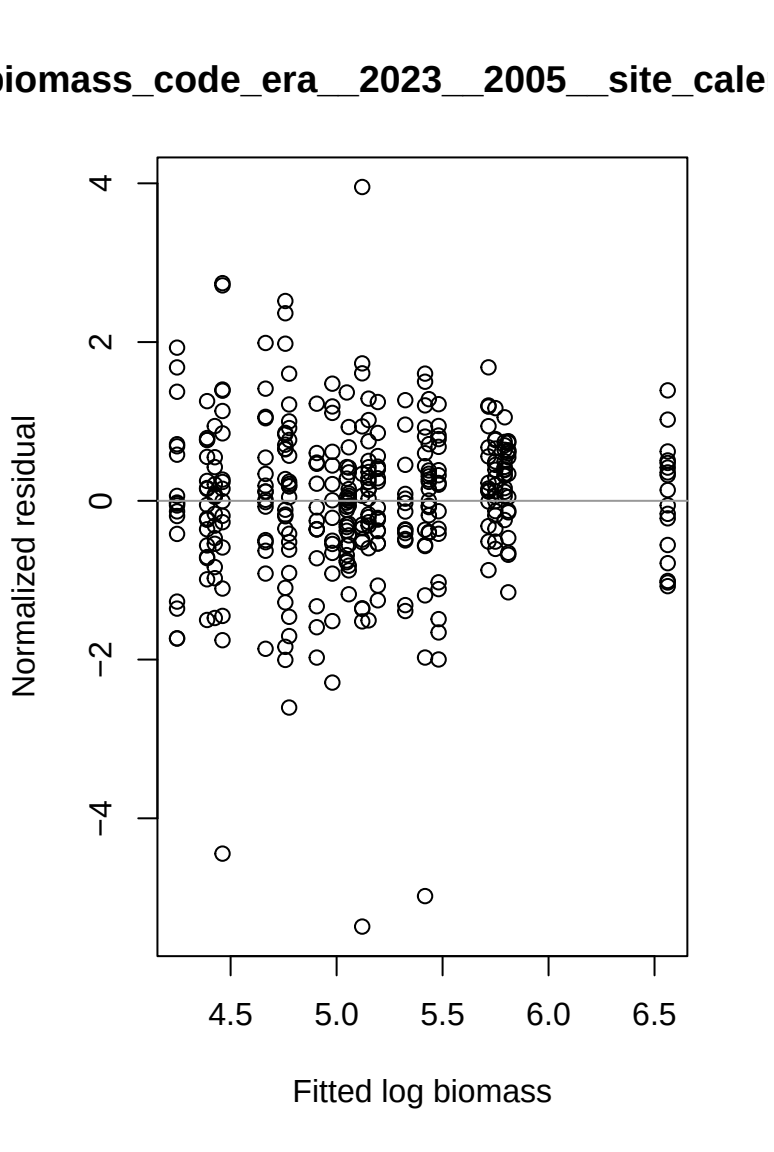

biomass_code_era__2022__2004__site_cale
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4

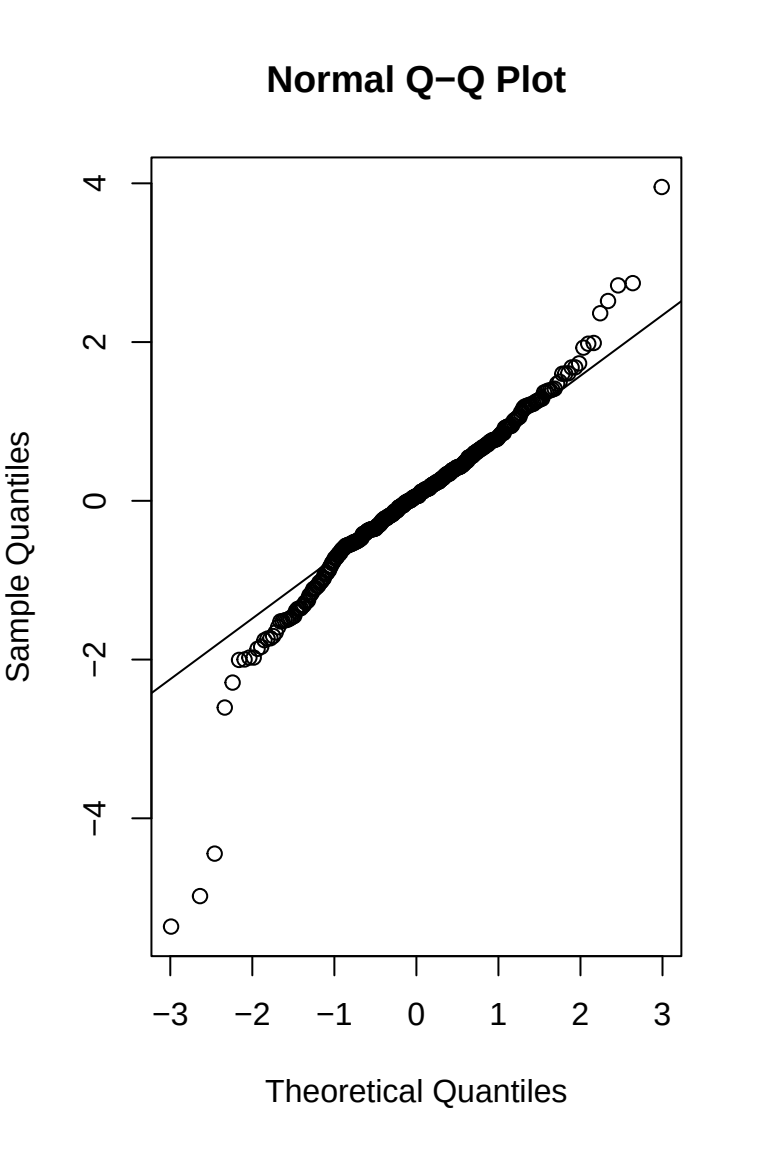

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2
4

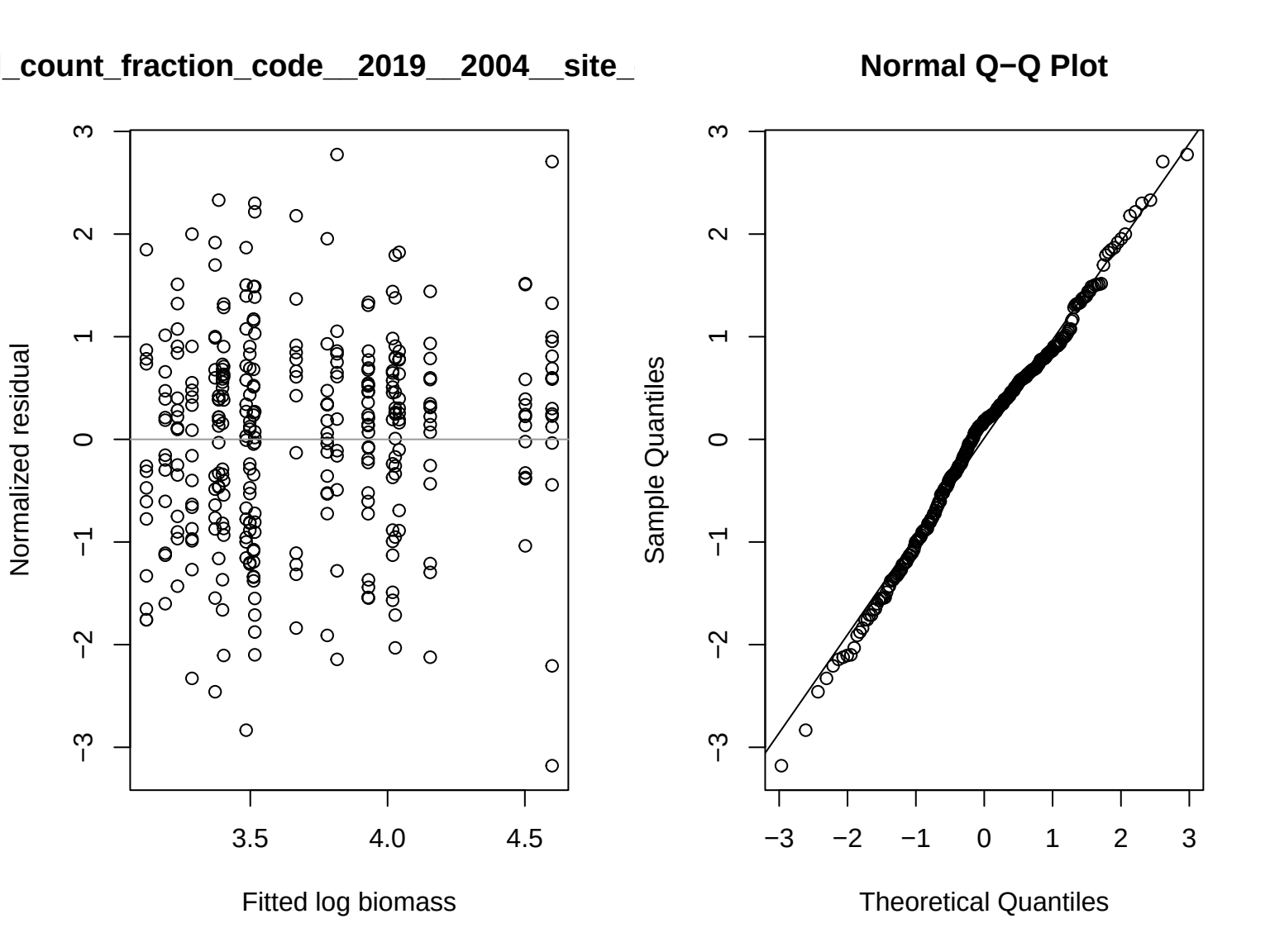

biomass_code_era__2022__2005__site_cale
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
4
2
0
−2
−4

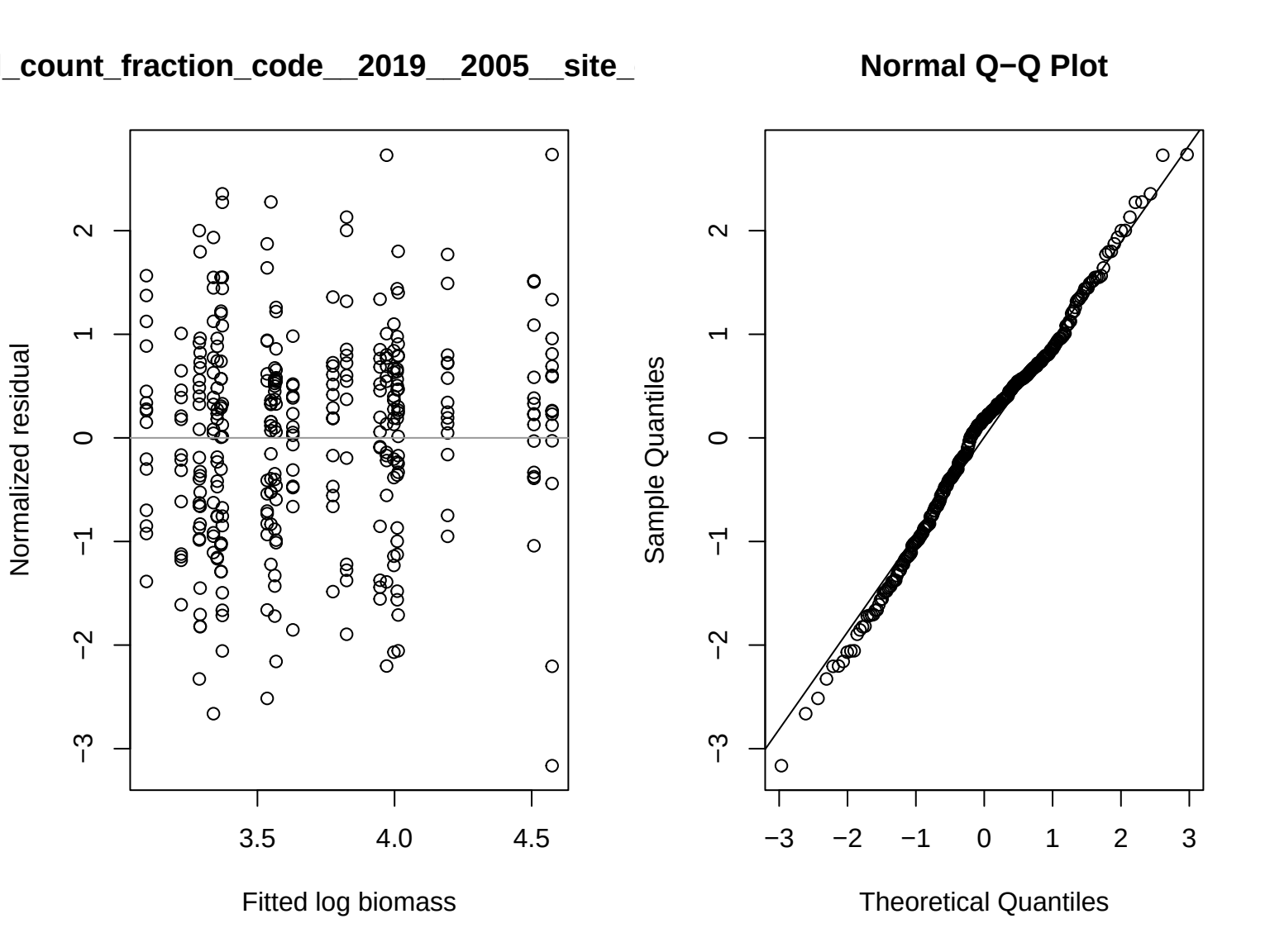

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
4
2
0
−2
−4

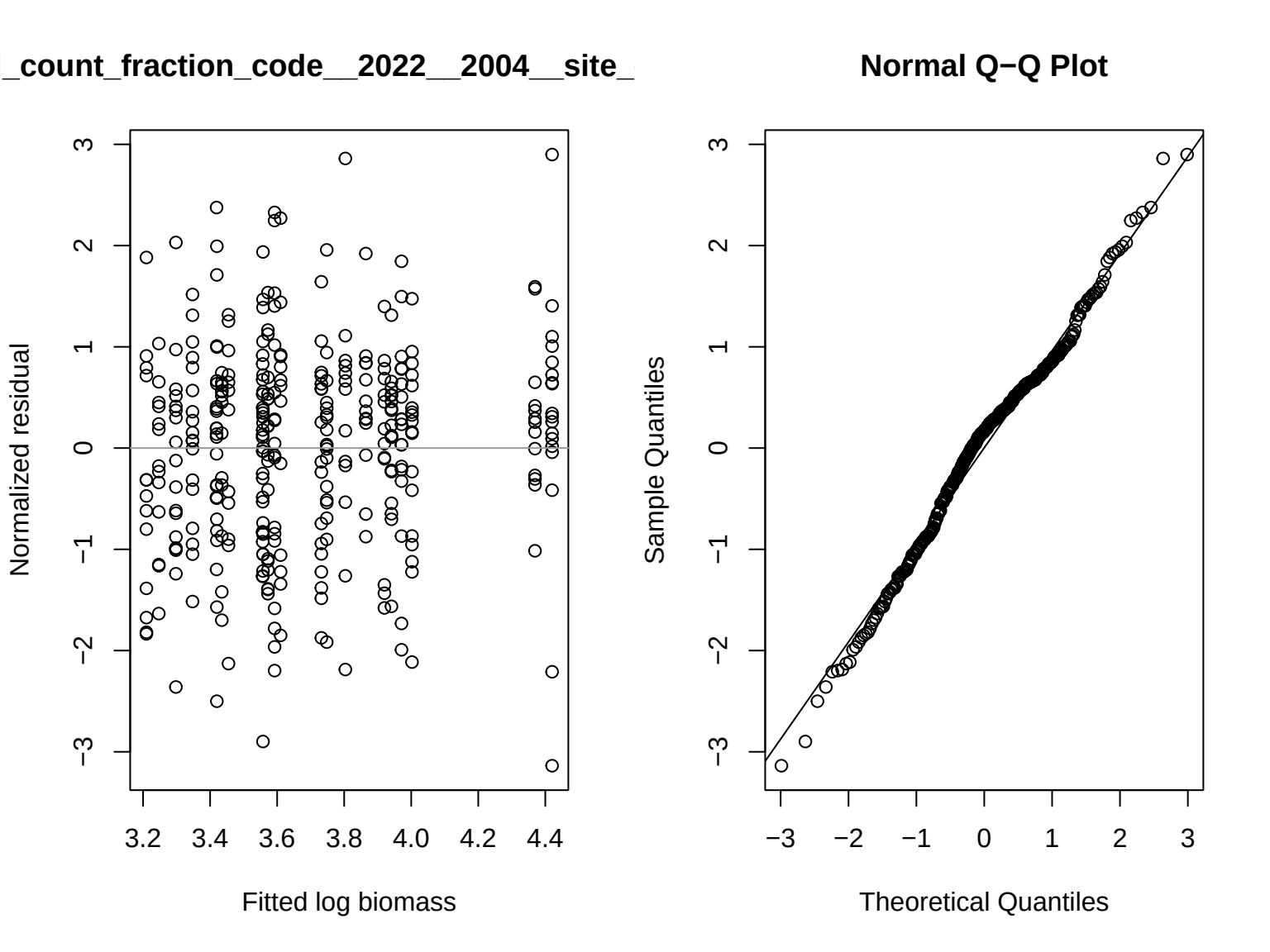

biomass_code_era__2023__2004__site_cale
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

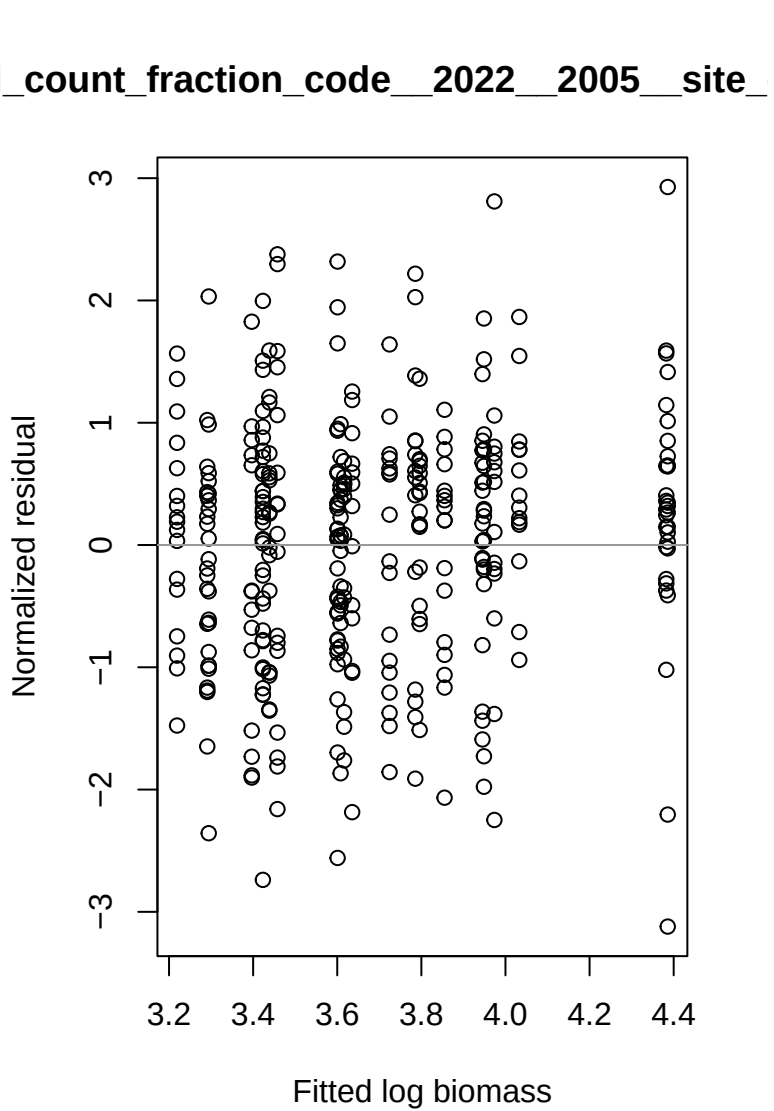

biomass_code_era__2023__2005__site_cale
Normalized residual
Fitted log biomass
4.5
5.0
5.5
6.0
6.5
−4
−2
0
2
4

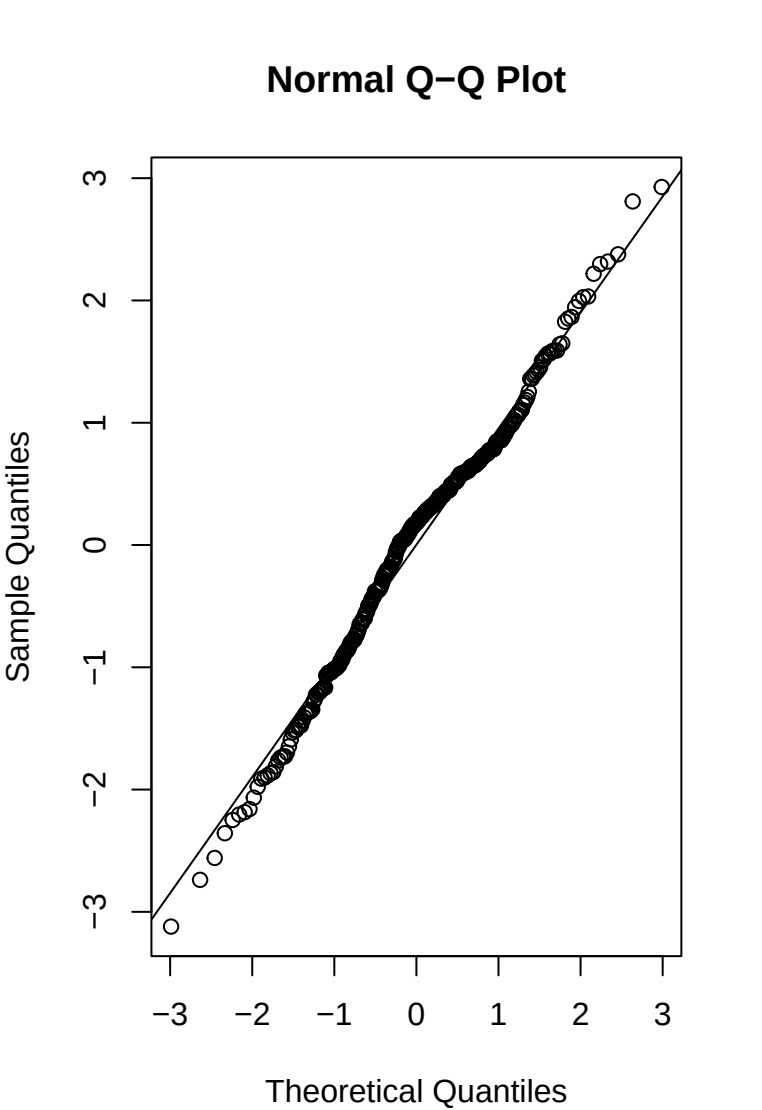

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2
4

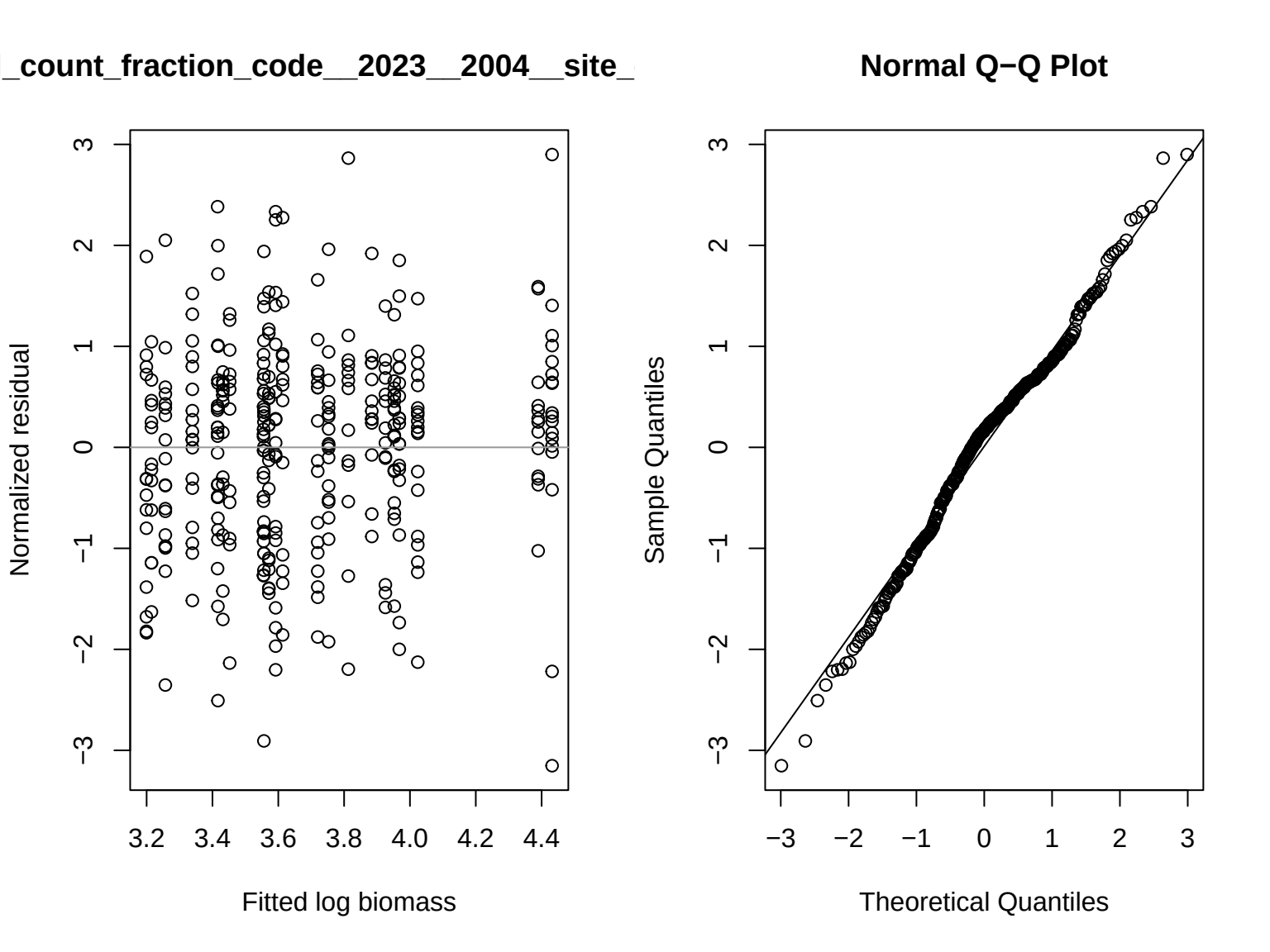

_count_fraction_code__2019__2004__site_
Normalized residual
Fitted log biomass
3.5
4.0
4.5
−3
−2
−1
0
1
2
3
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

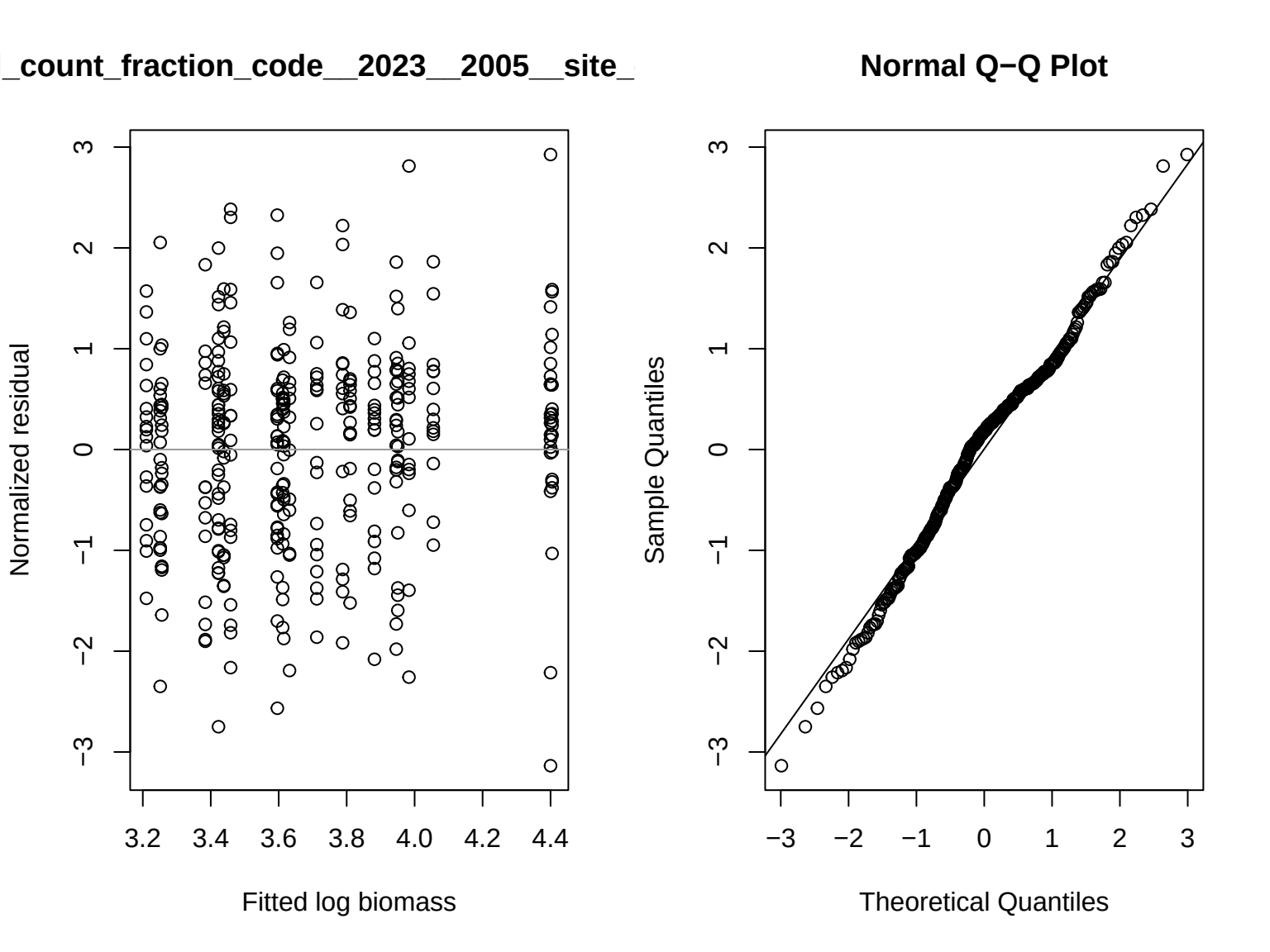

_count_fraction_code__2019__2005__site_
Normalized residual
Fitted log biomass
3.5
4.0
4.5
−3
−2
−1
0
1
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

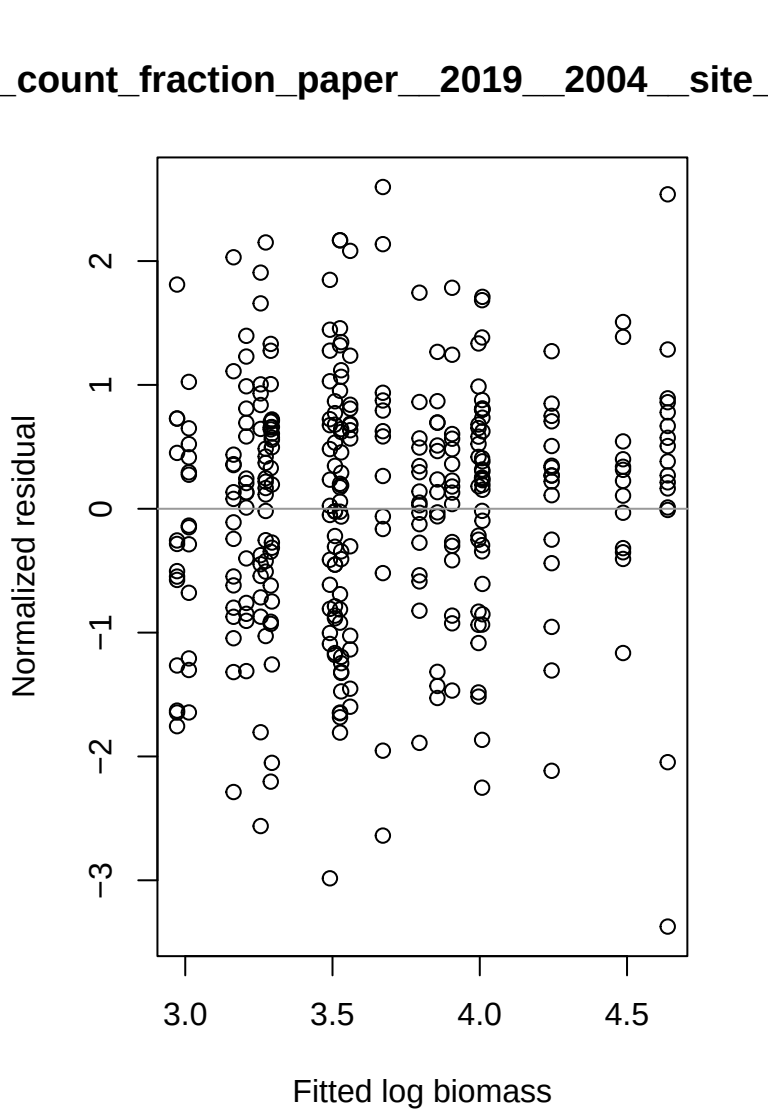

_count_fraction_code__2022__2004__site_
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
−3
−2
−1
0
1
2
3
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

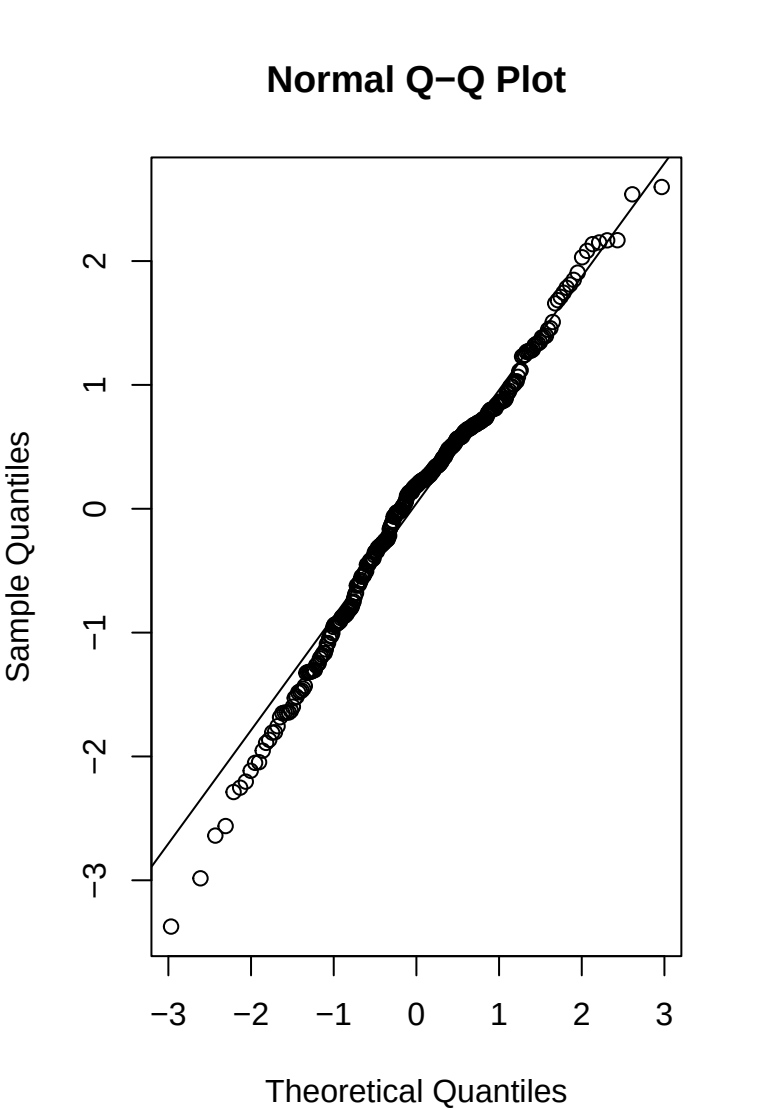

_count_fraction_code__2022__2005__site_
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
−3
−2
−1
0
1
2
3

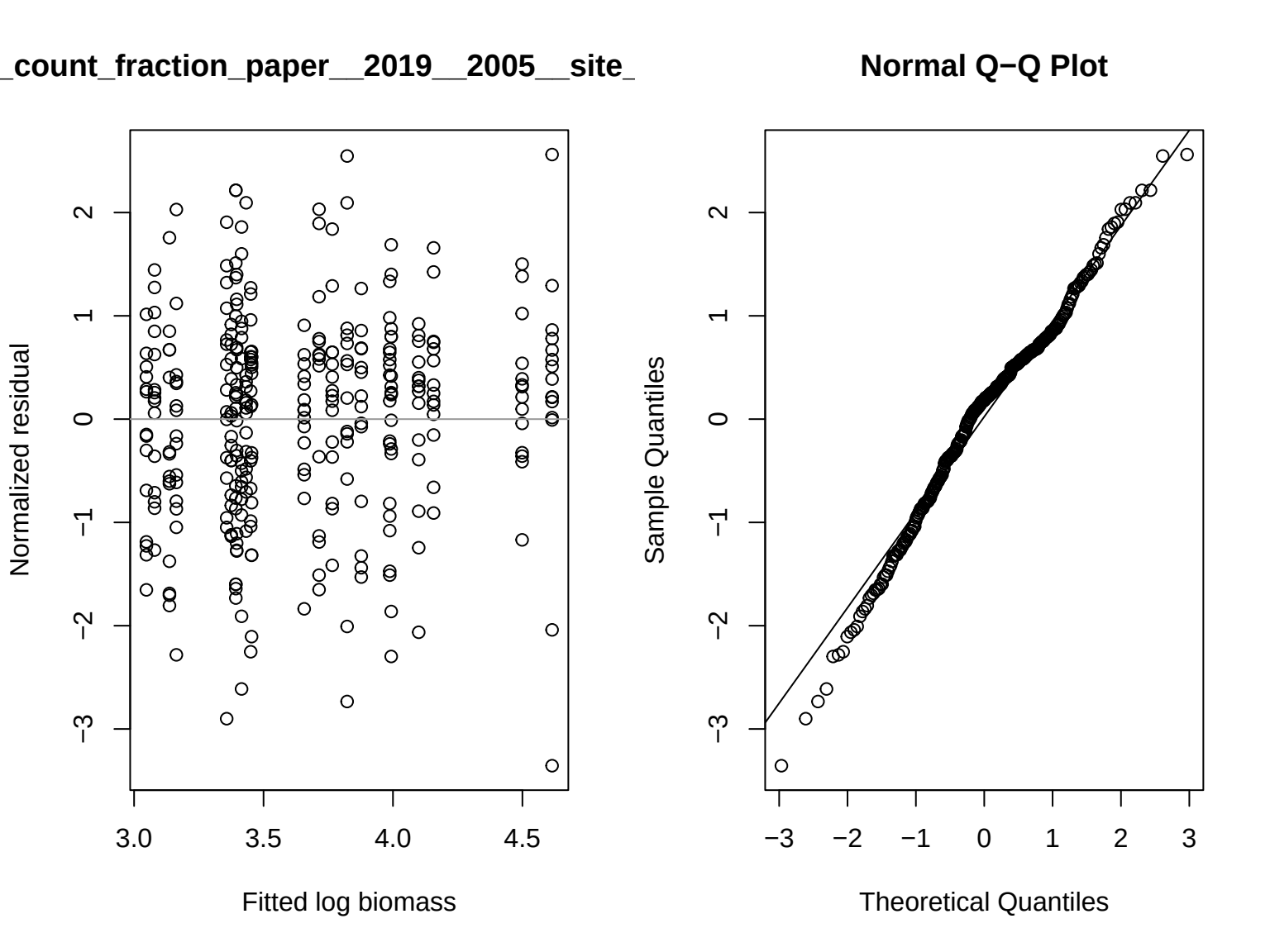

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

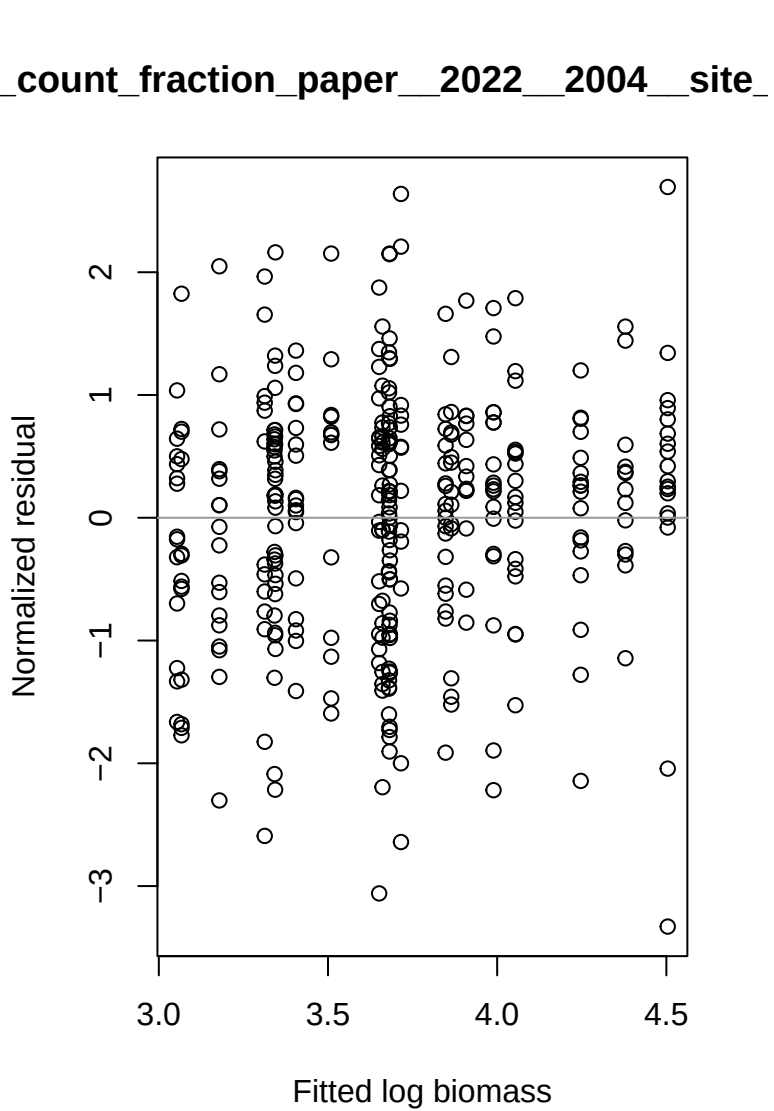

_count_fraction_code__2023__2004__site_
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
3
2
1
0
−1
−2
−3
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

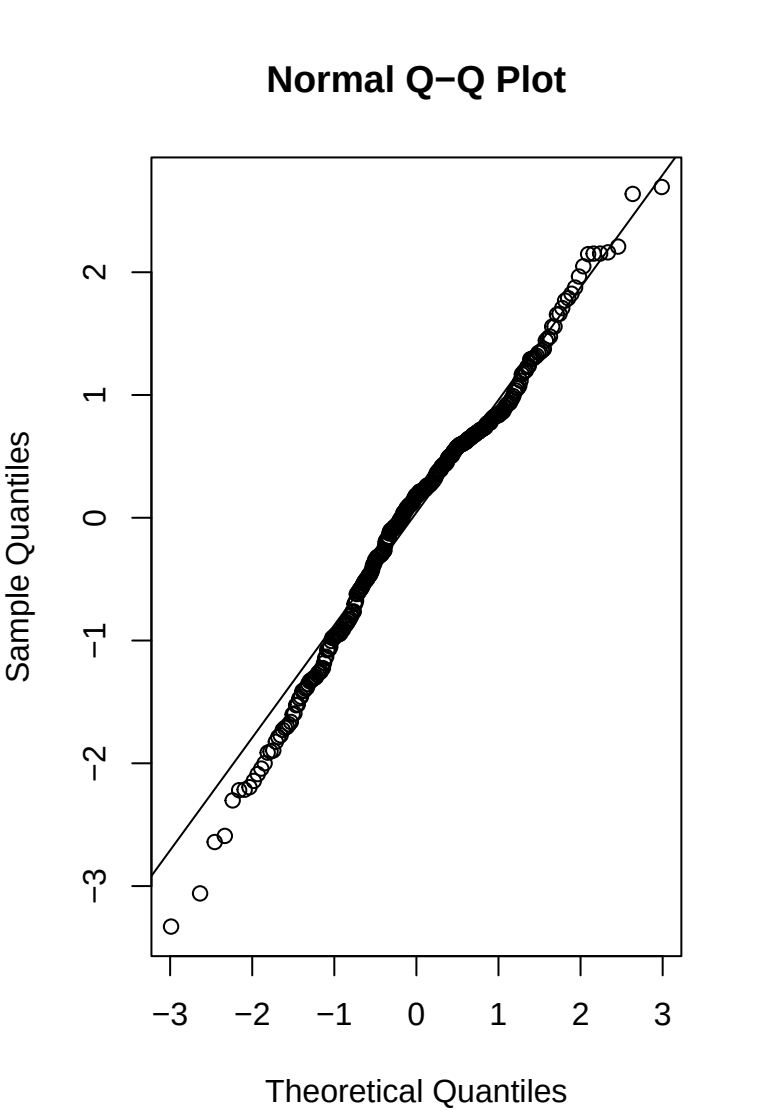

_count_fraction_code__2023__2005__site_
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
−3
−2
−1
0
1
2
3
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

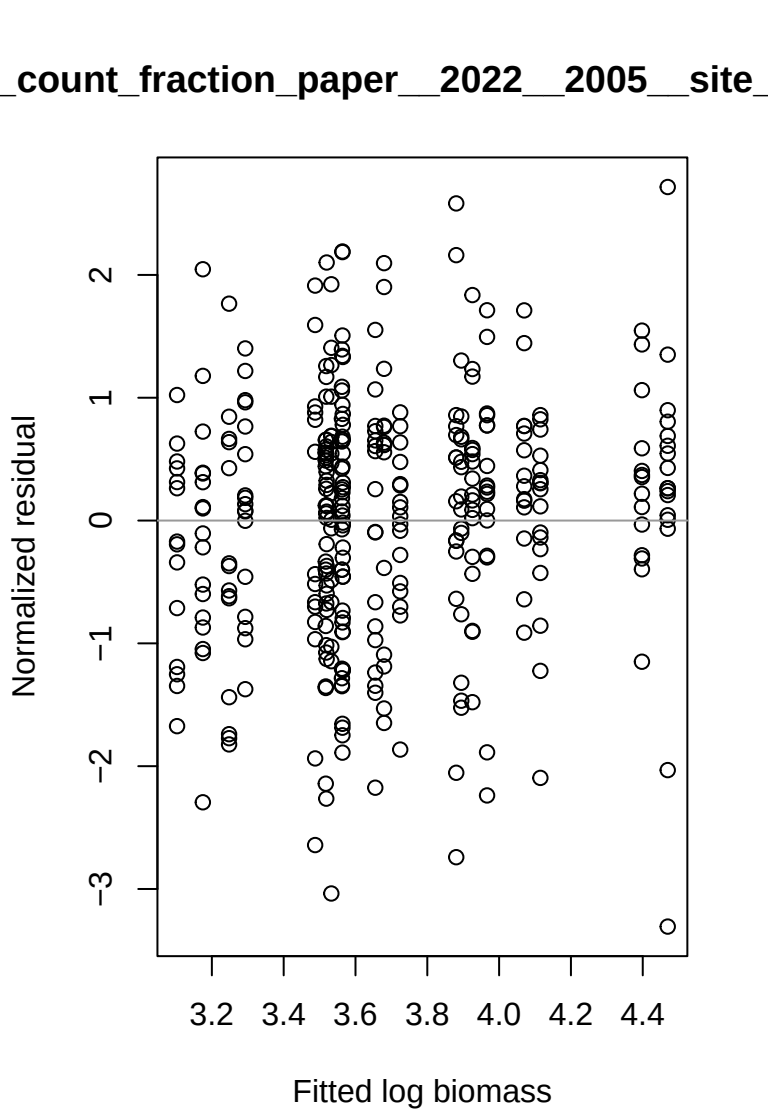

_count_fraction_paper__2019__2004__site_
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−3
−2
−1
0
1
2

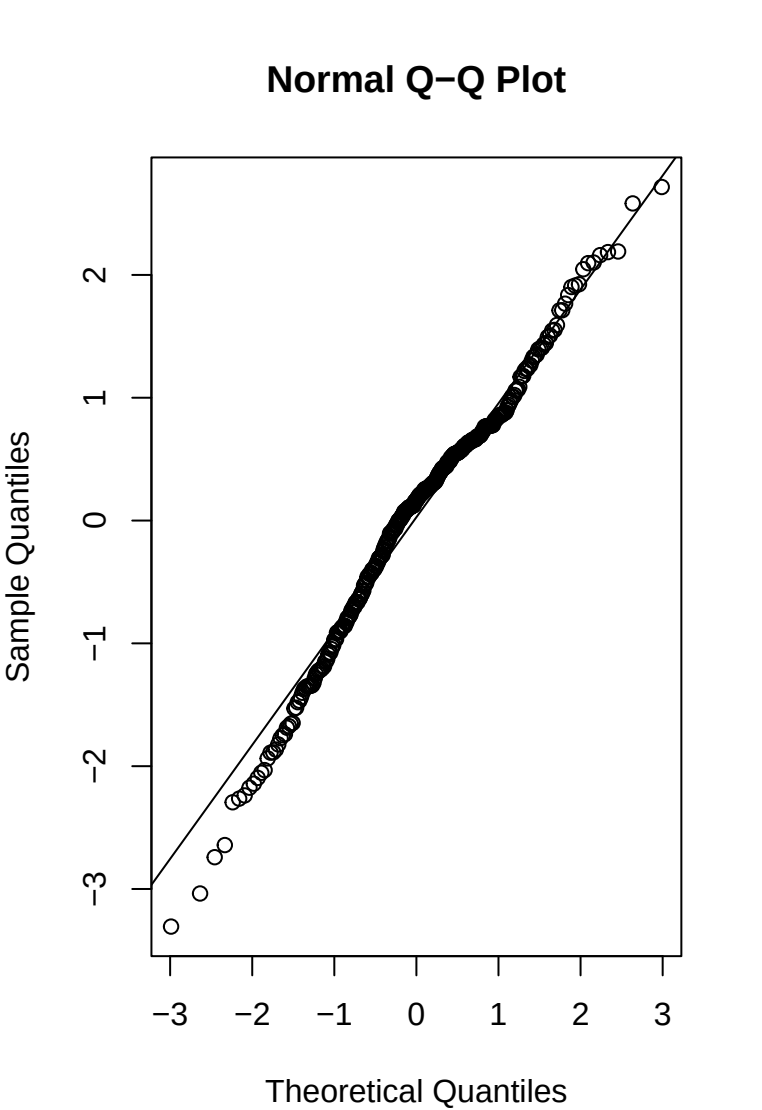

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

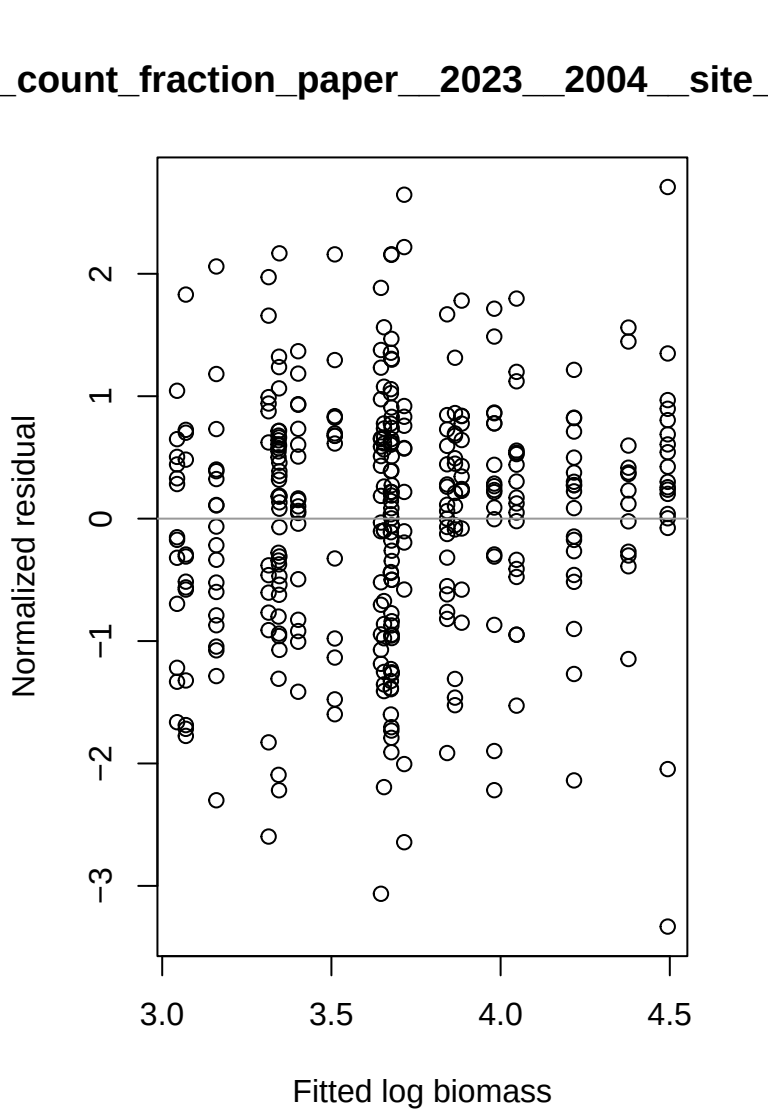

_count_fraction_paper__2019__2005__site_
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−3
−2
−1
0
1
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

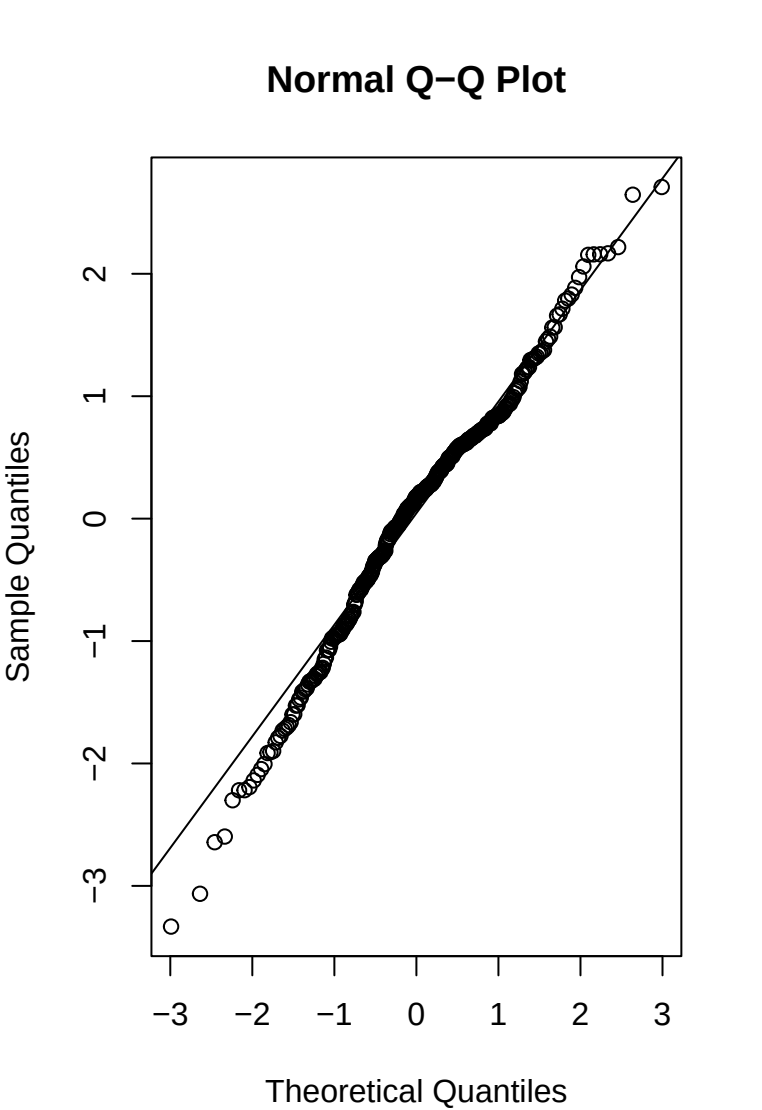

_count_fraction_paper__2022__2004__site_
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−3
−2
−1
0
1
2

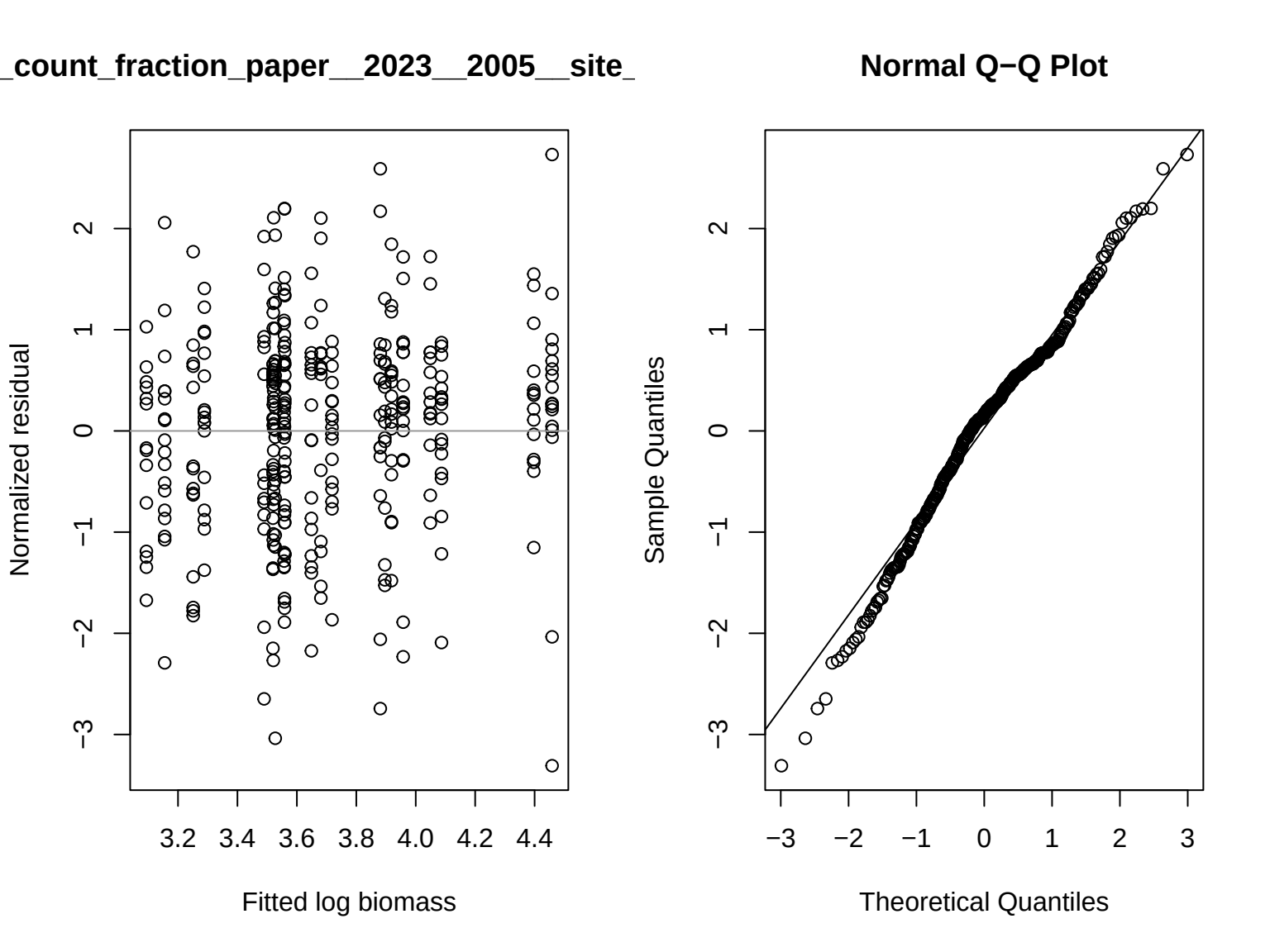

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

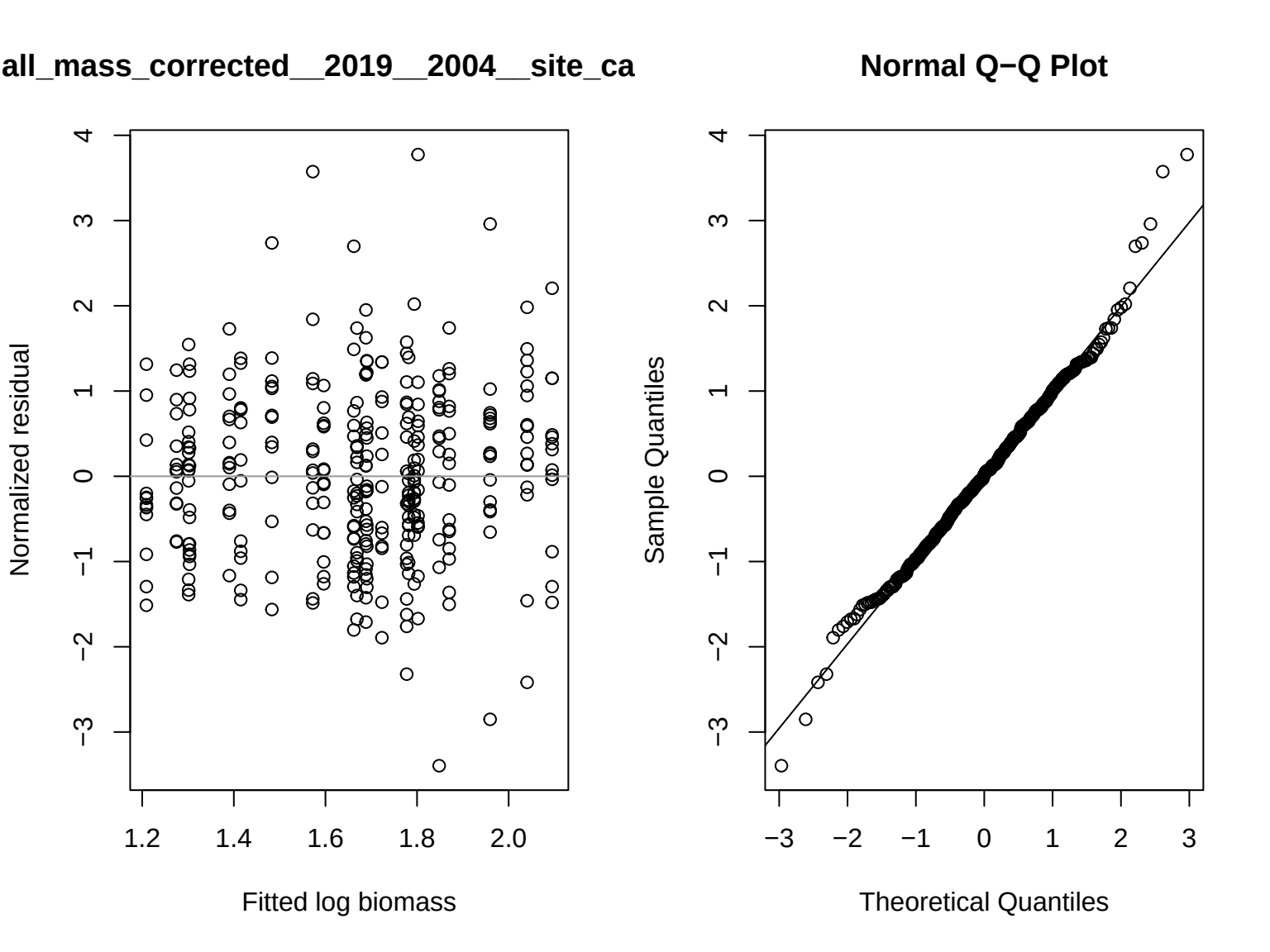

_count_fraction_paper__2022__2005__site_
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
−3
−2
−1
0
1
2

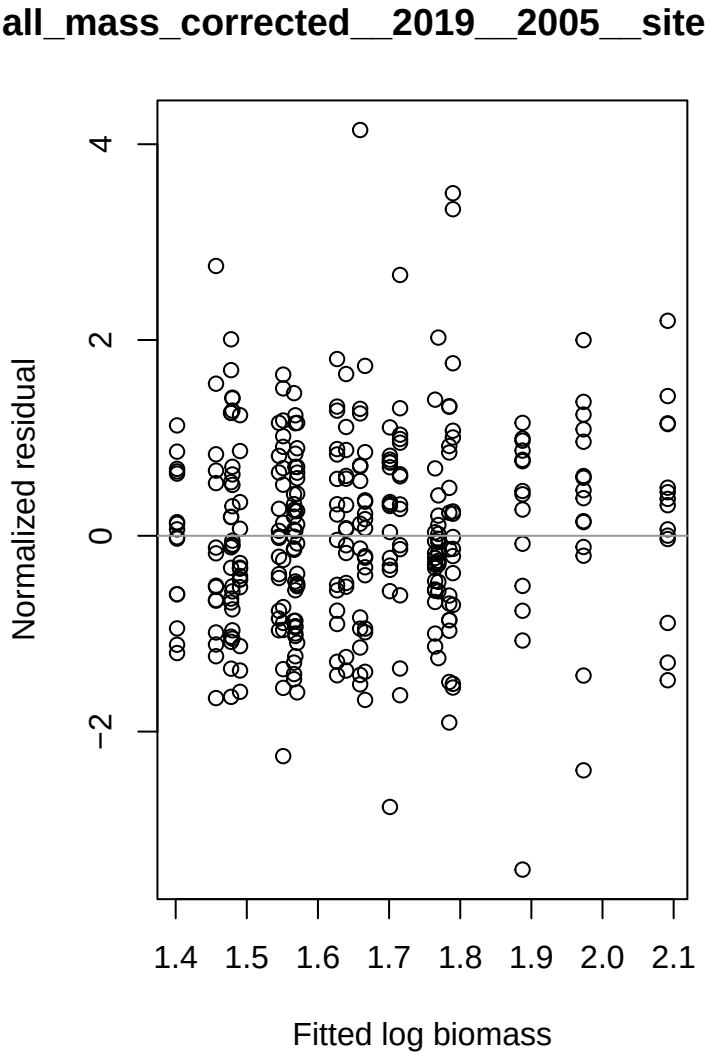

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

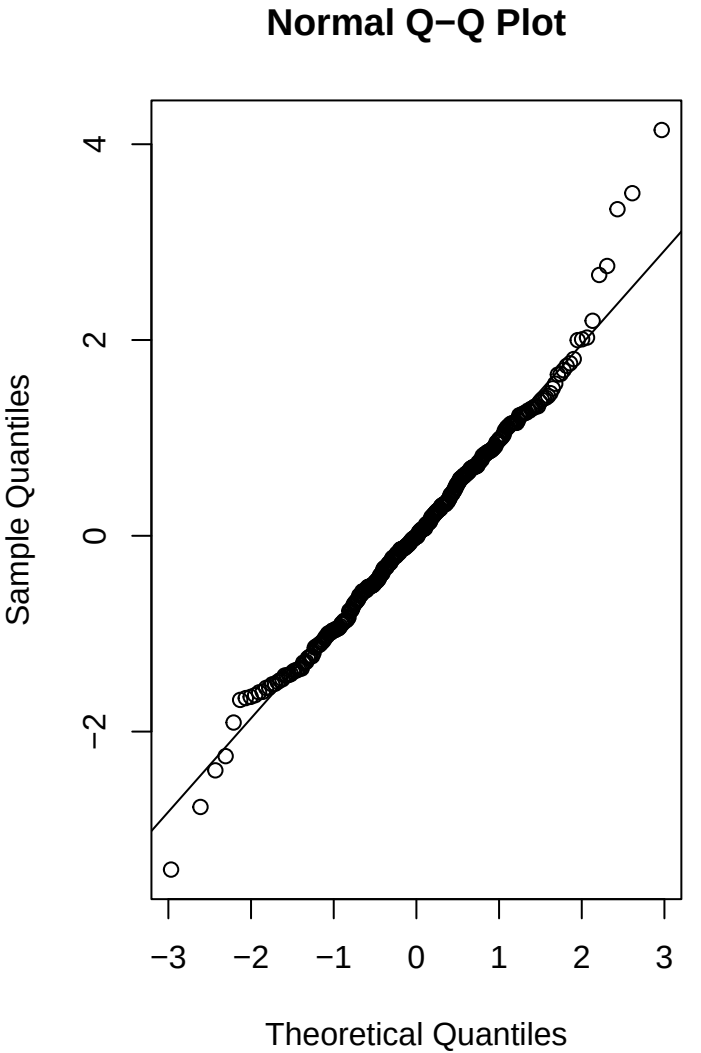

_count_fraction_paper__2023__2004__site_
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−3
−2
−1
0
1
2

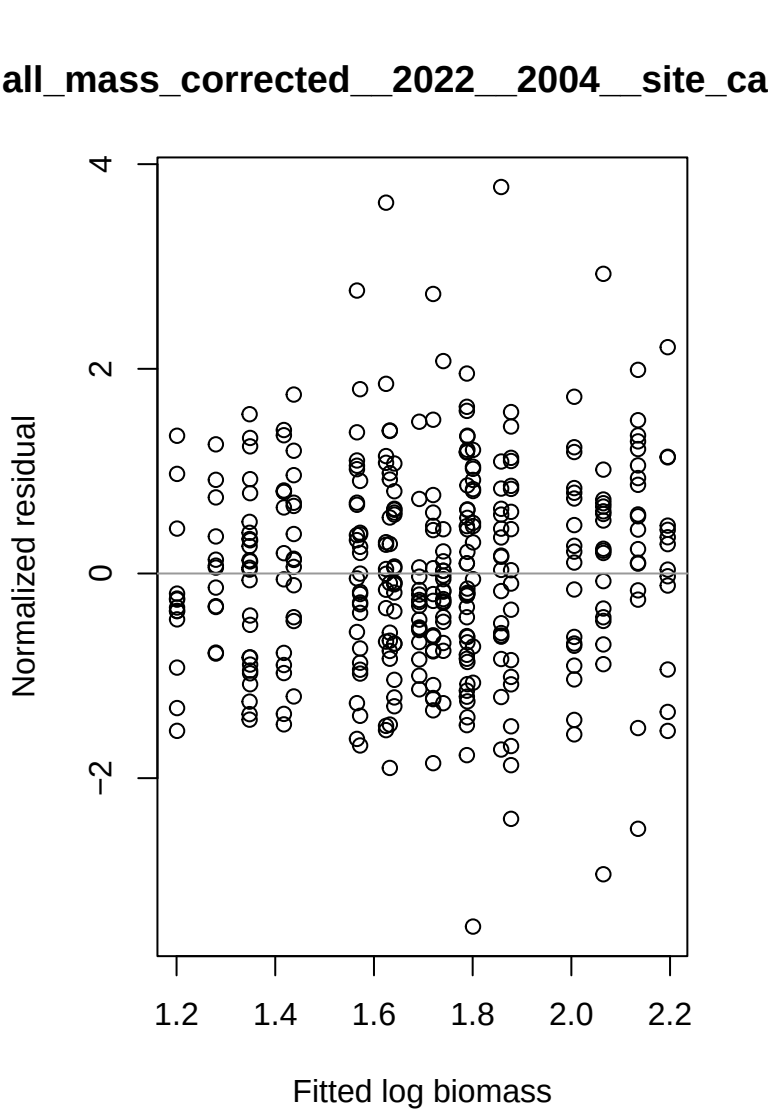

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

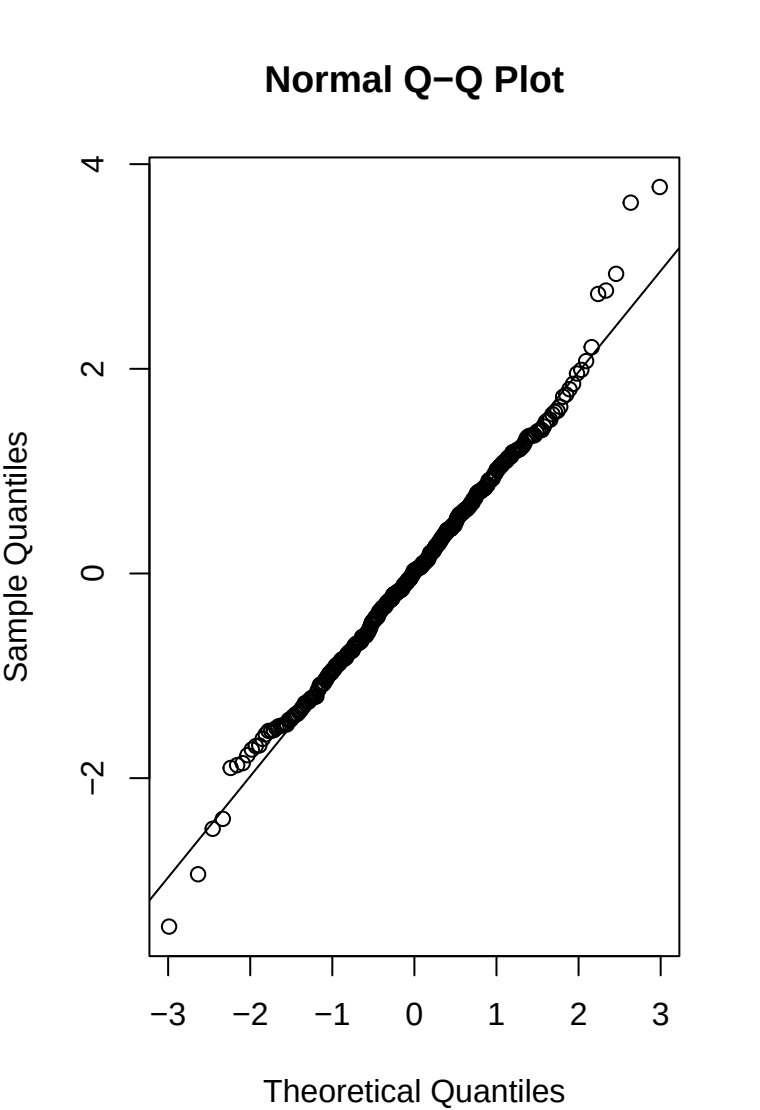

_count_fraction_paper__2023__2005__site_
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
−3
−2
−1
0
1
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

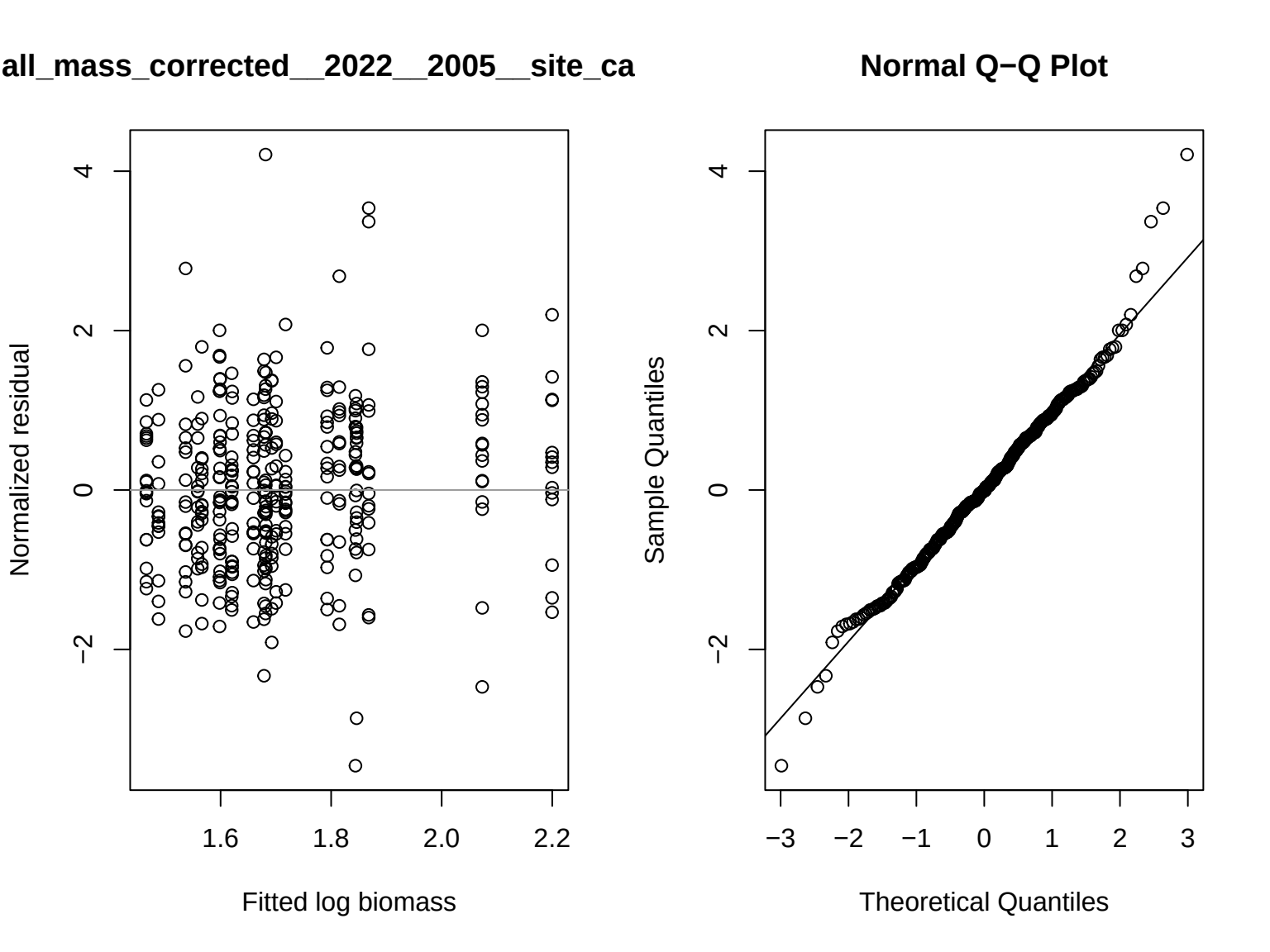

all_mass_corrected__2019__2004__site_ca
Normalized residual
Fitted log biomass
1.2
1.4
1.6
1.8
2.0
−3
−2
−1
0
1
2
3
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

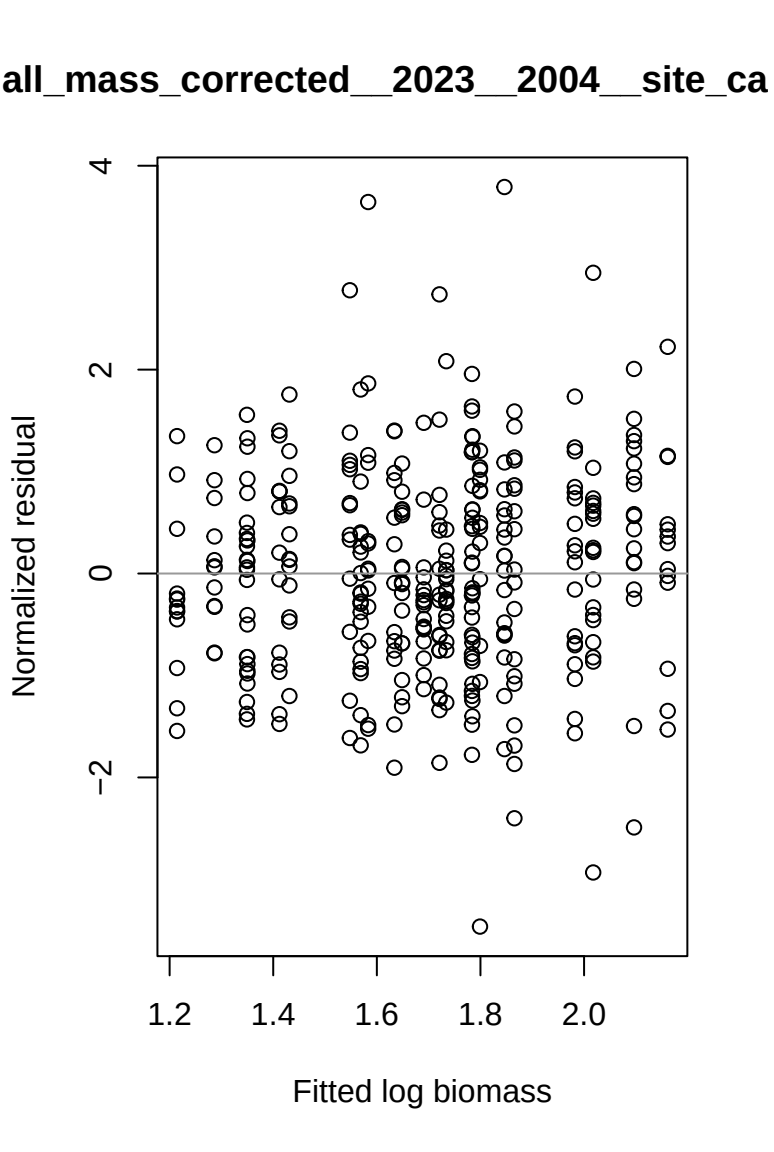

all_mass_corrected__2019__2005__site_ca
Normalized residual
4
2
0
−2
1.4
1.5
1.6
1.7
1.8
1.9
2.0
2.1
Fitted log biomass

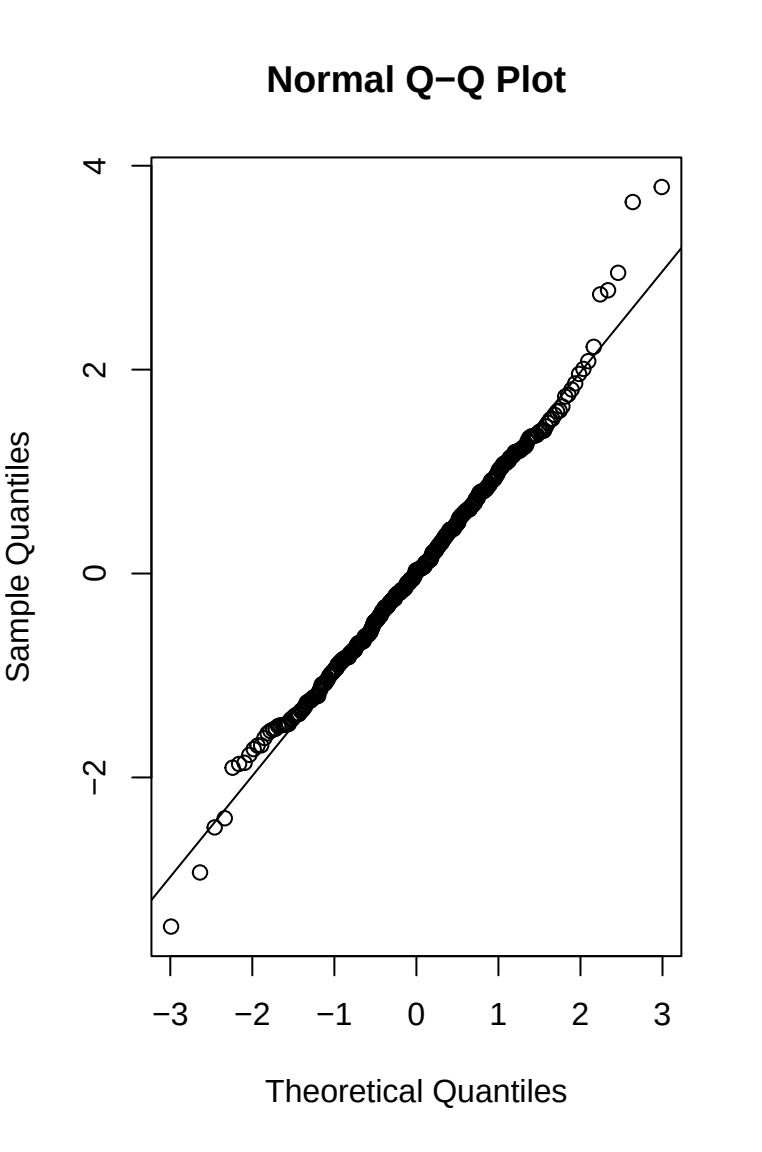

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

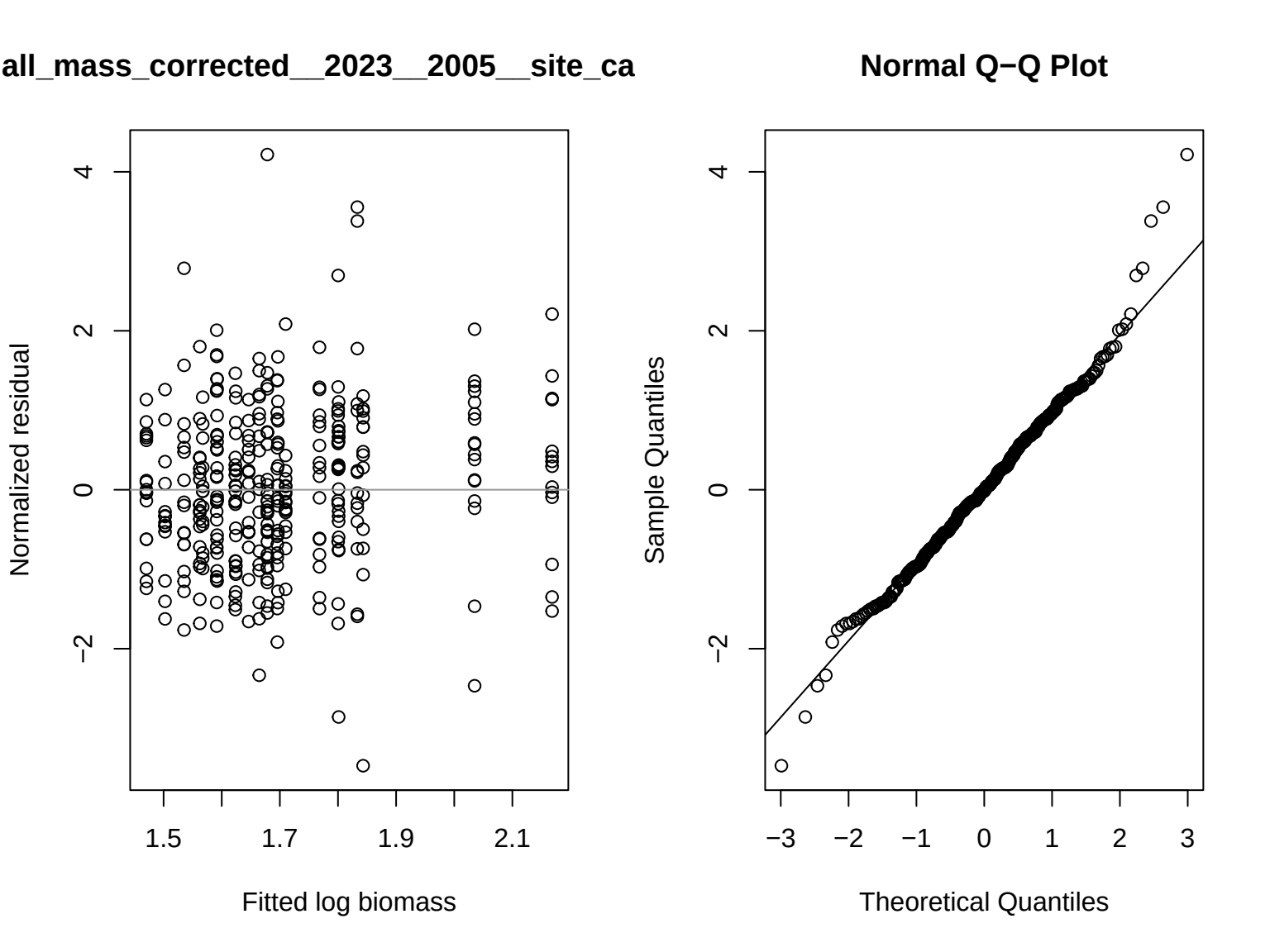

all_mass_corrected__2022__2004__site_ca
Normalized residual
4
2
0
−2
1.2
1.4
1.6
1.8
2.0
2.2
Fitted log biomass

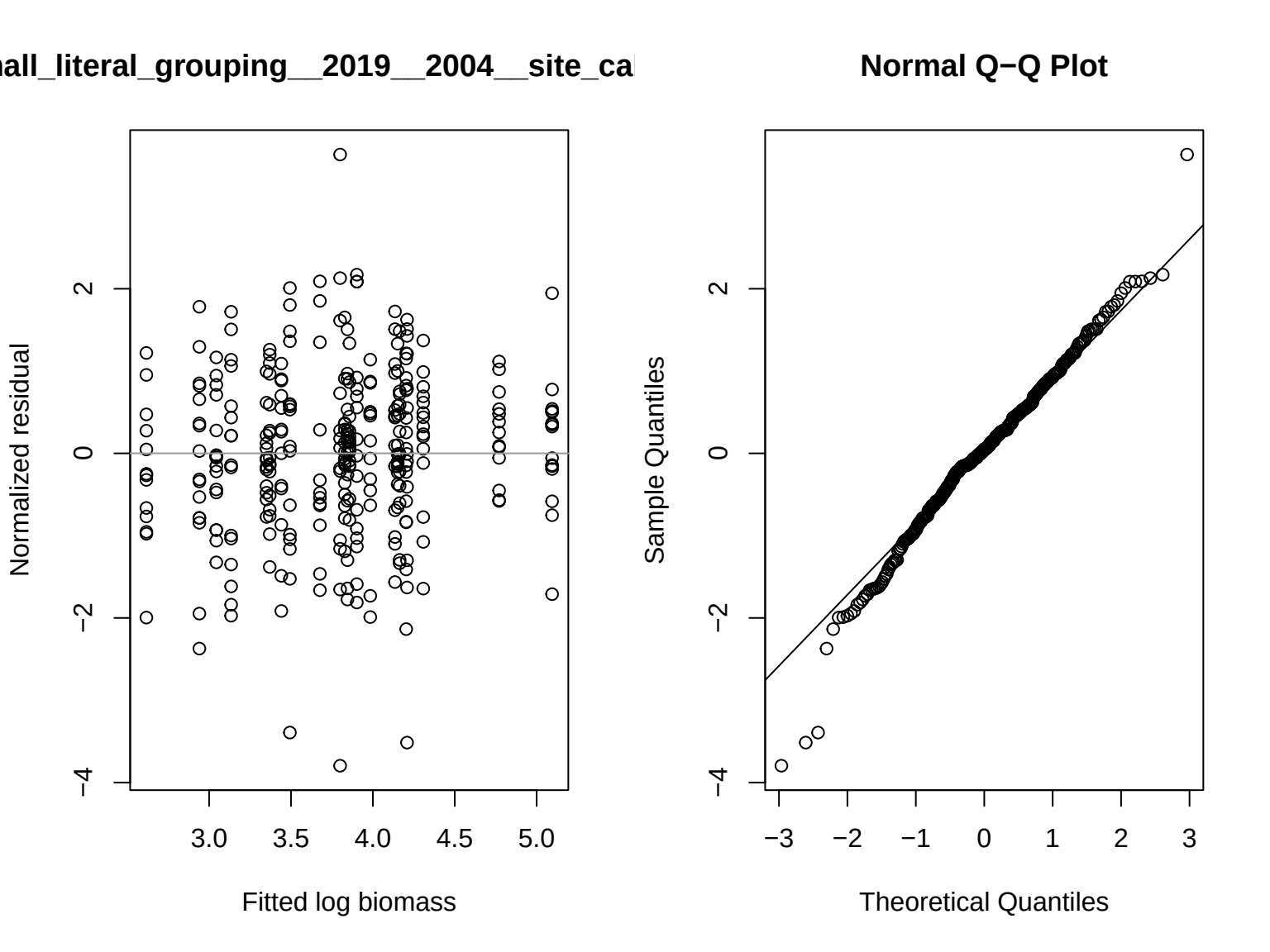

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

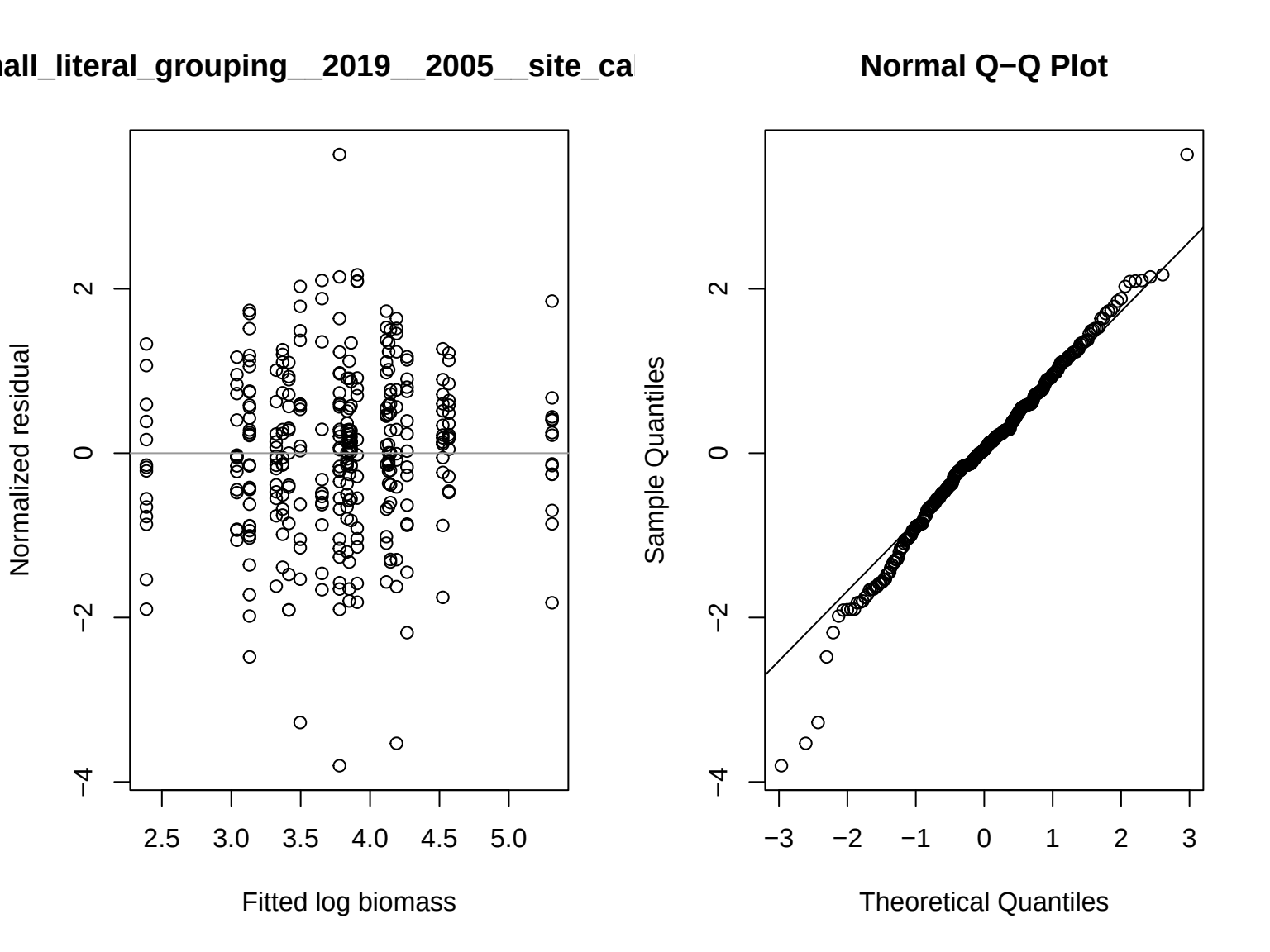

all_mass_corrected__2022__2005__site_ca
Normalized residual
Fitted log biomass
1.6
1.8
2.0
2.2
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

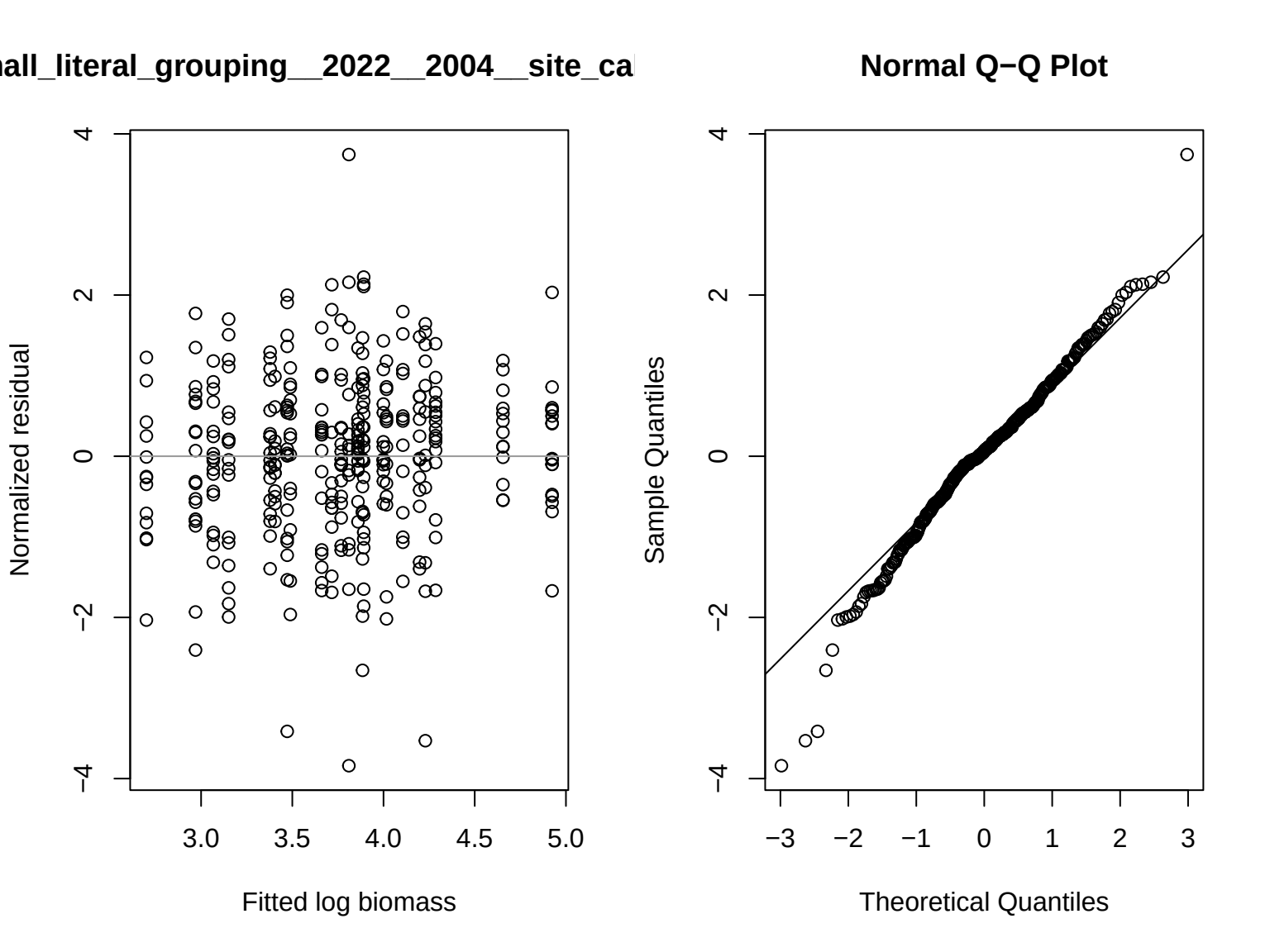

all_mass_corrected__2023__2004__site_ca
Normalized residual
4
2
0
−2
1.2
1.4
1.6
1.8
2.0
Fitted log biomass

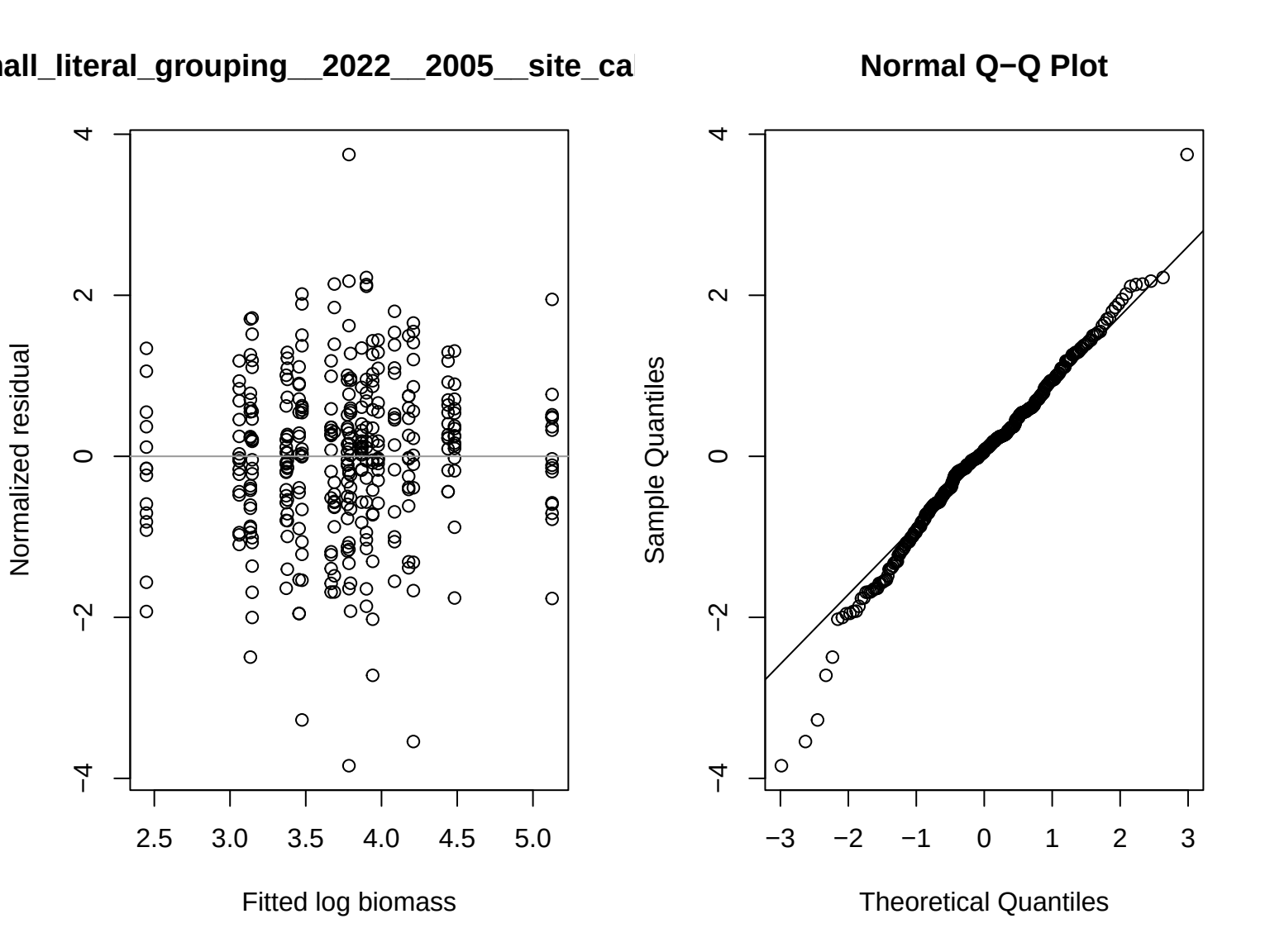

Normal Q−Q Plot
Sample Quantiles
4
2
0
−2
−3
−2
−1
0
1
2
3
Theoretical Quantiles

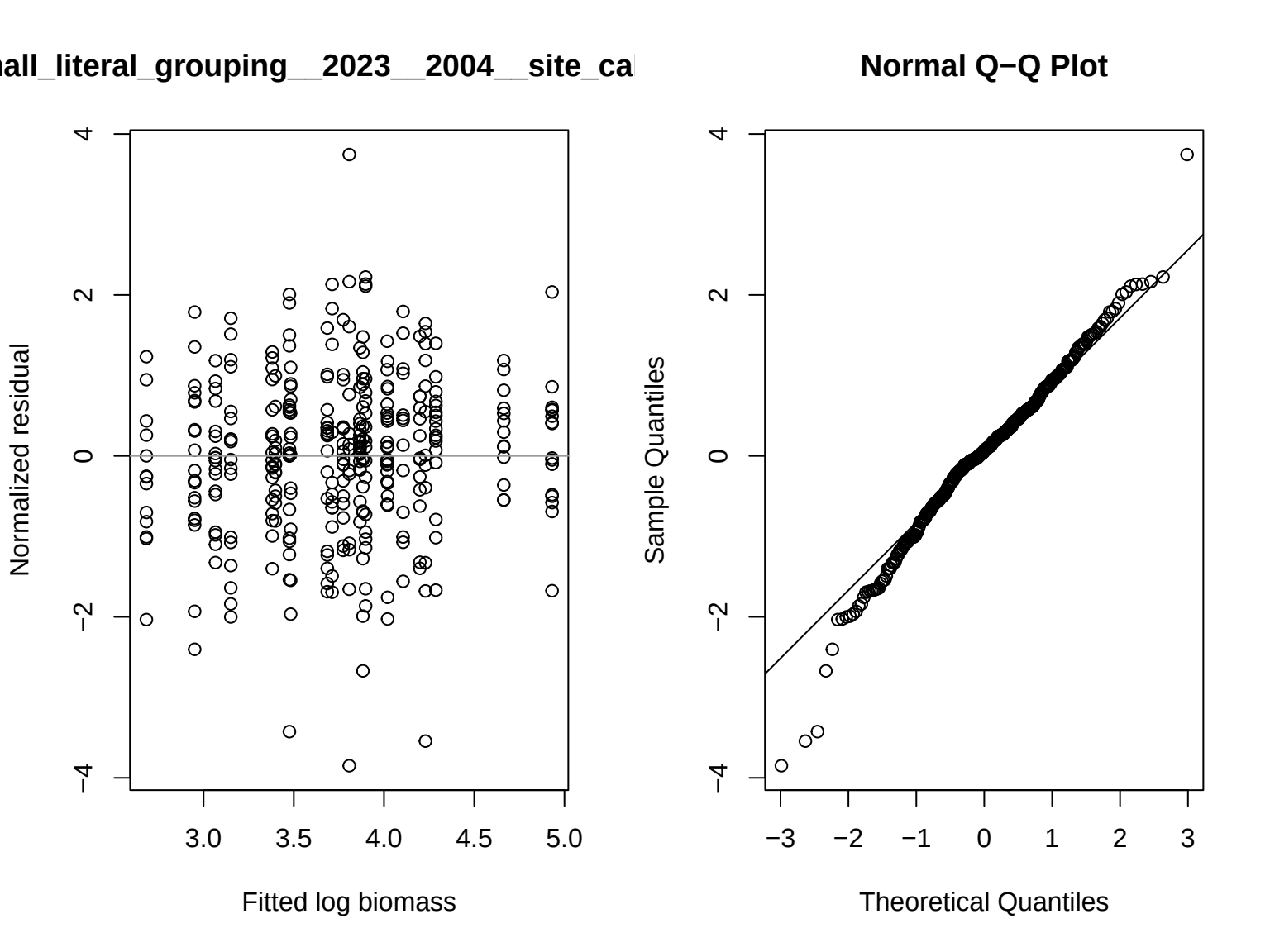

all_mass_corrected__2023__2005__site_ca
Normalized residual
Fitted log biomass
1.5
1.7
1.9
2.1
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

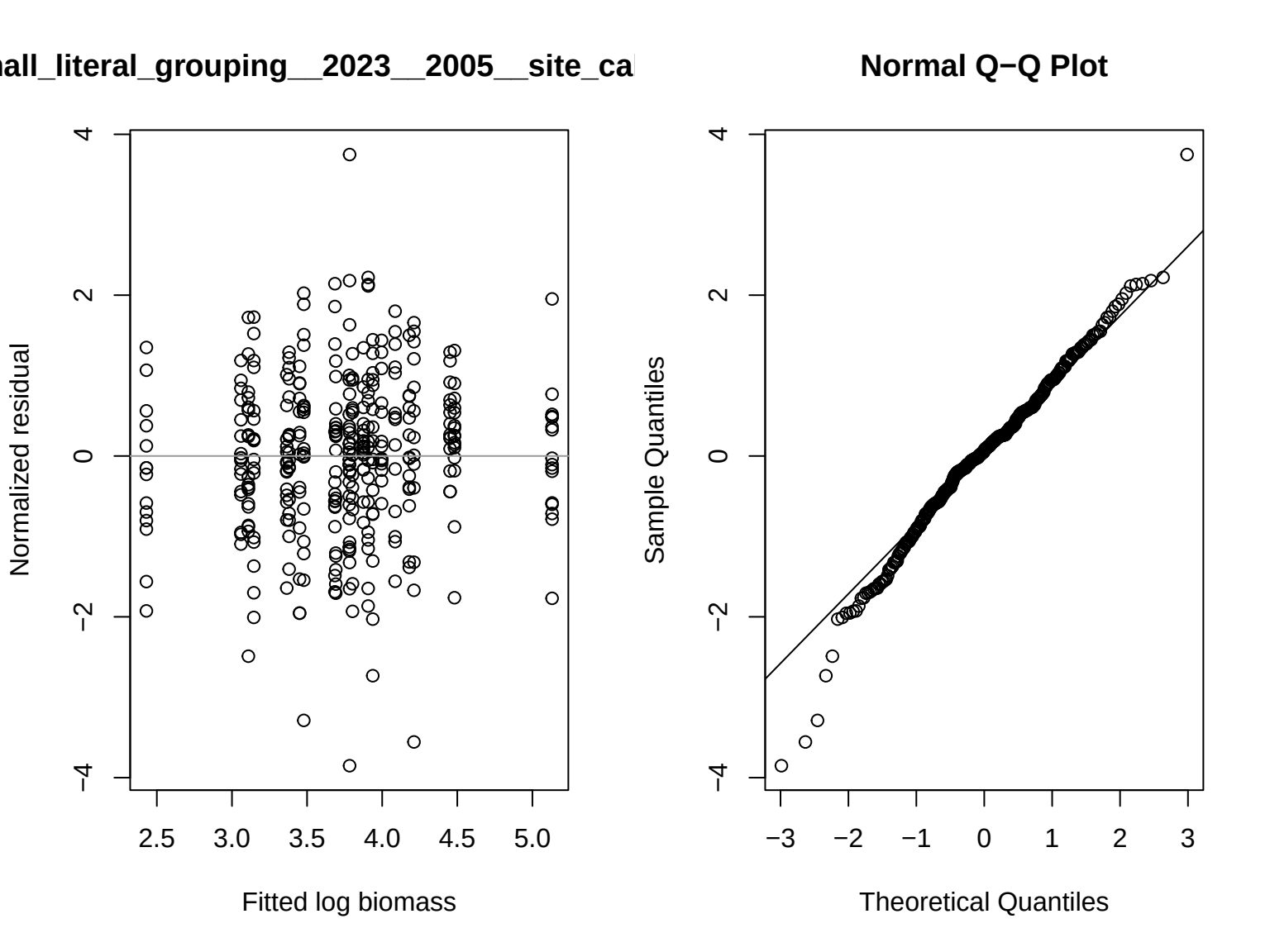

all_literal_grouping__2019__2004__site_ca
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

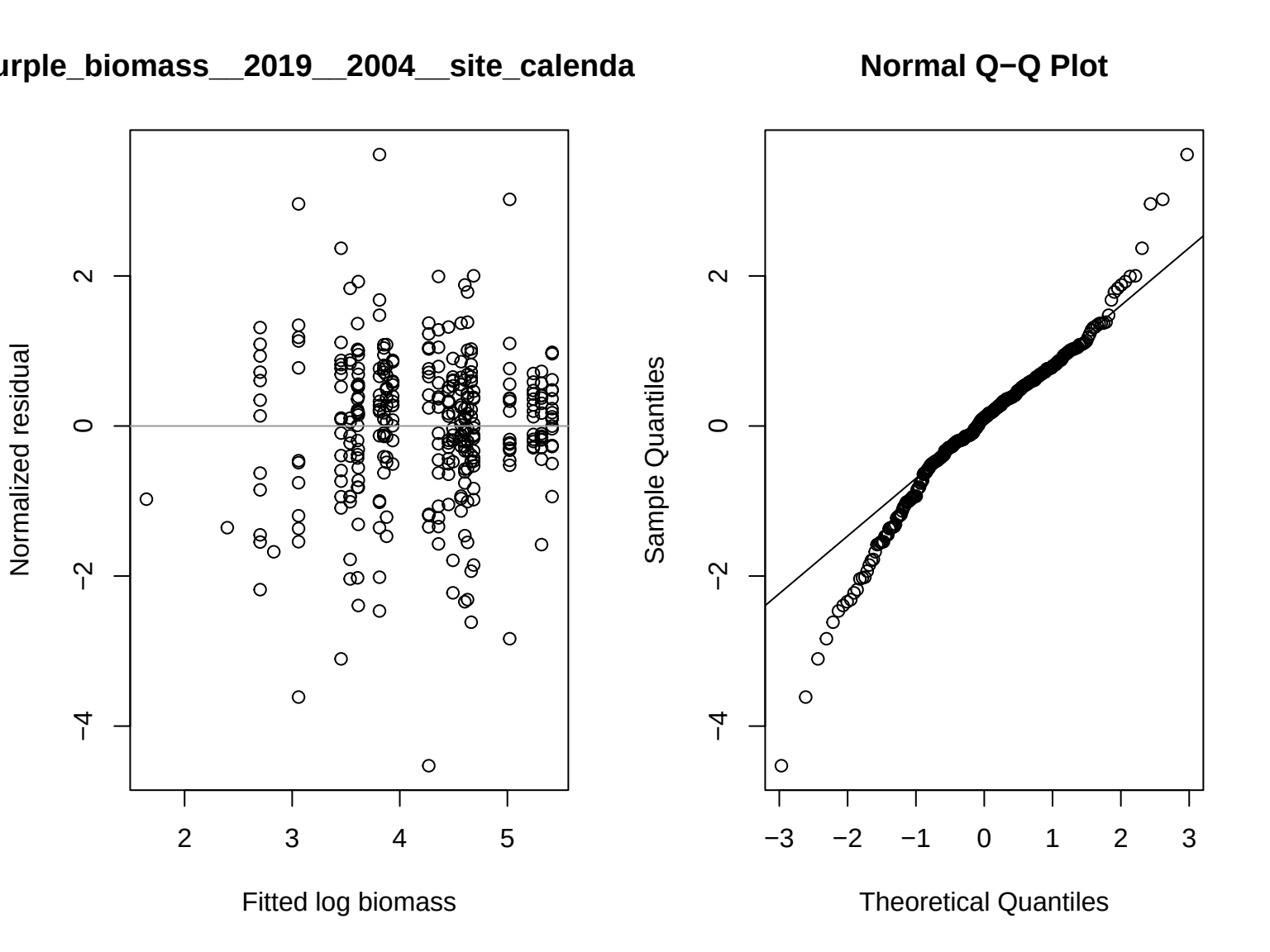

all_literal_grouping__2019__2005__site_ca
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

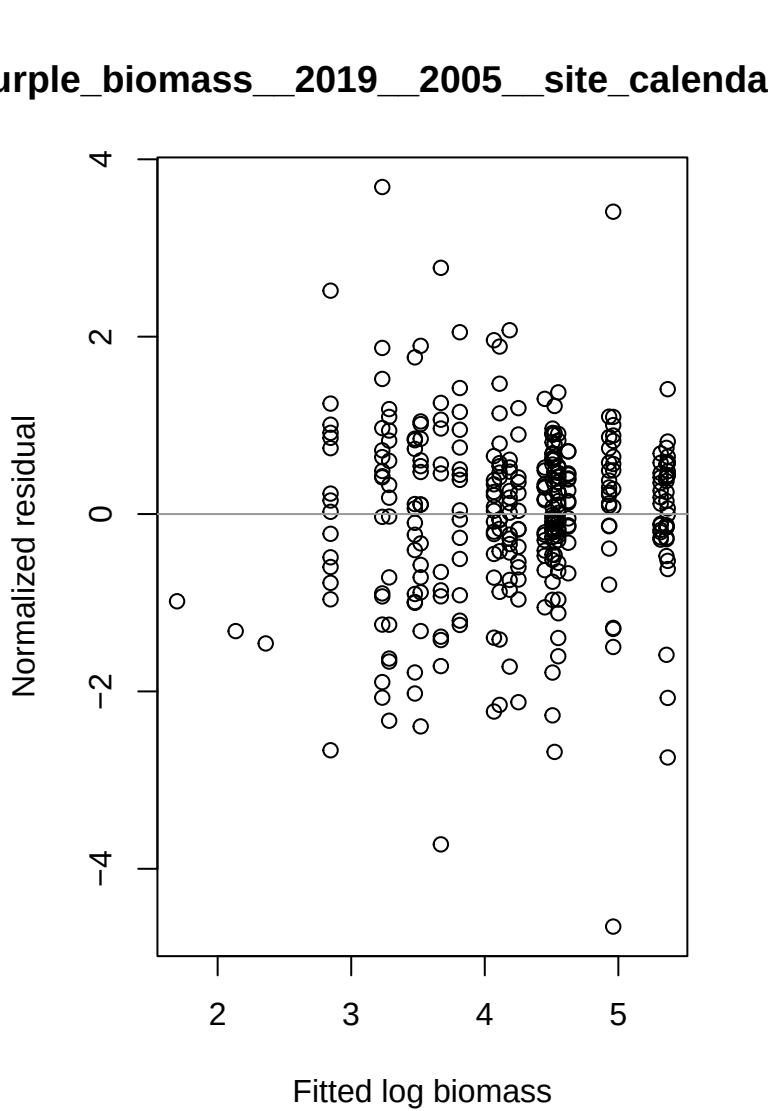

all_literal_grouping__2022__2004__site_ca
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

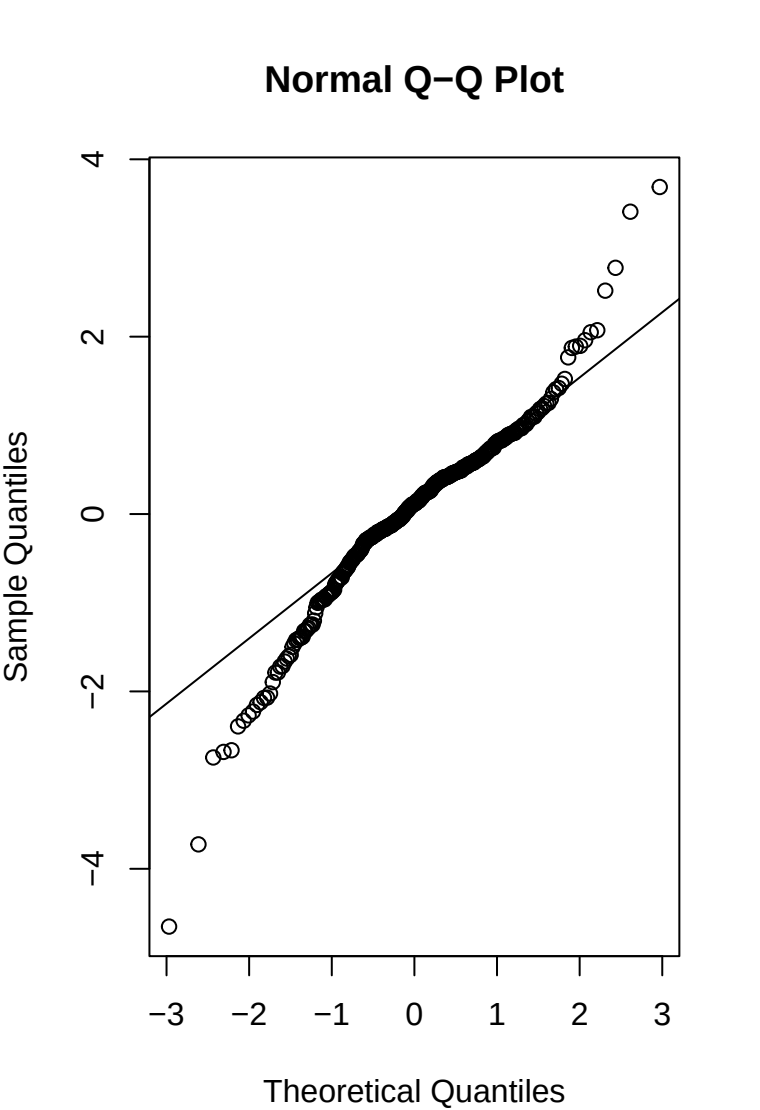

all_literal_grouping__2022__2005__site_ca
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

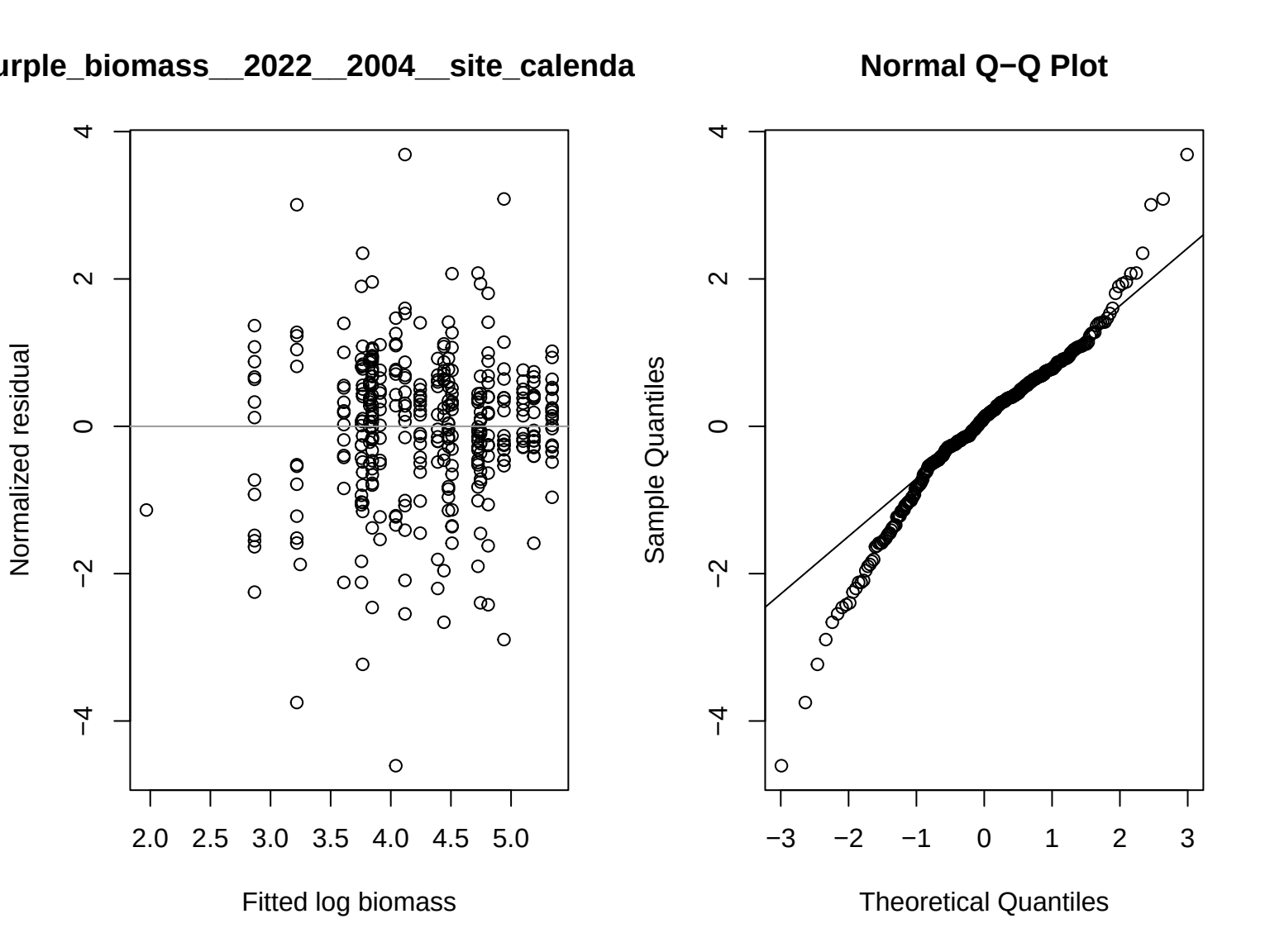

all_literal_grouping__2023__2004__site_ca
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

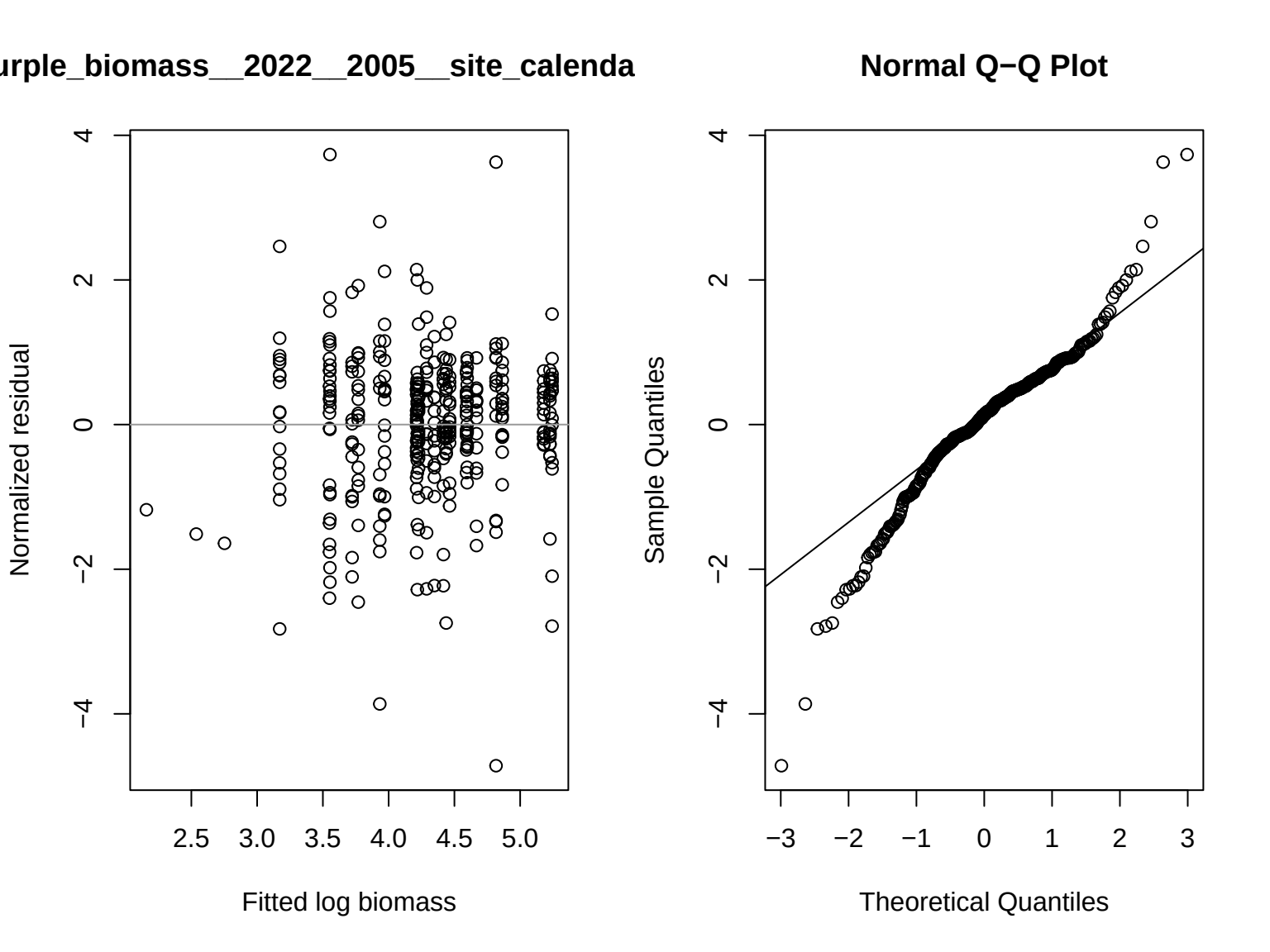

all_literal_grouping__2023__2005__site_ca
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

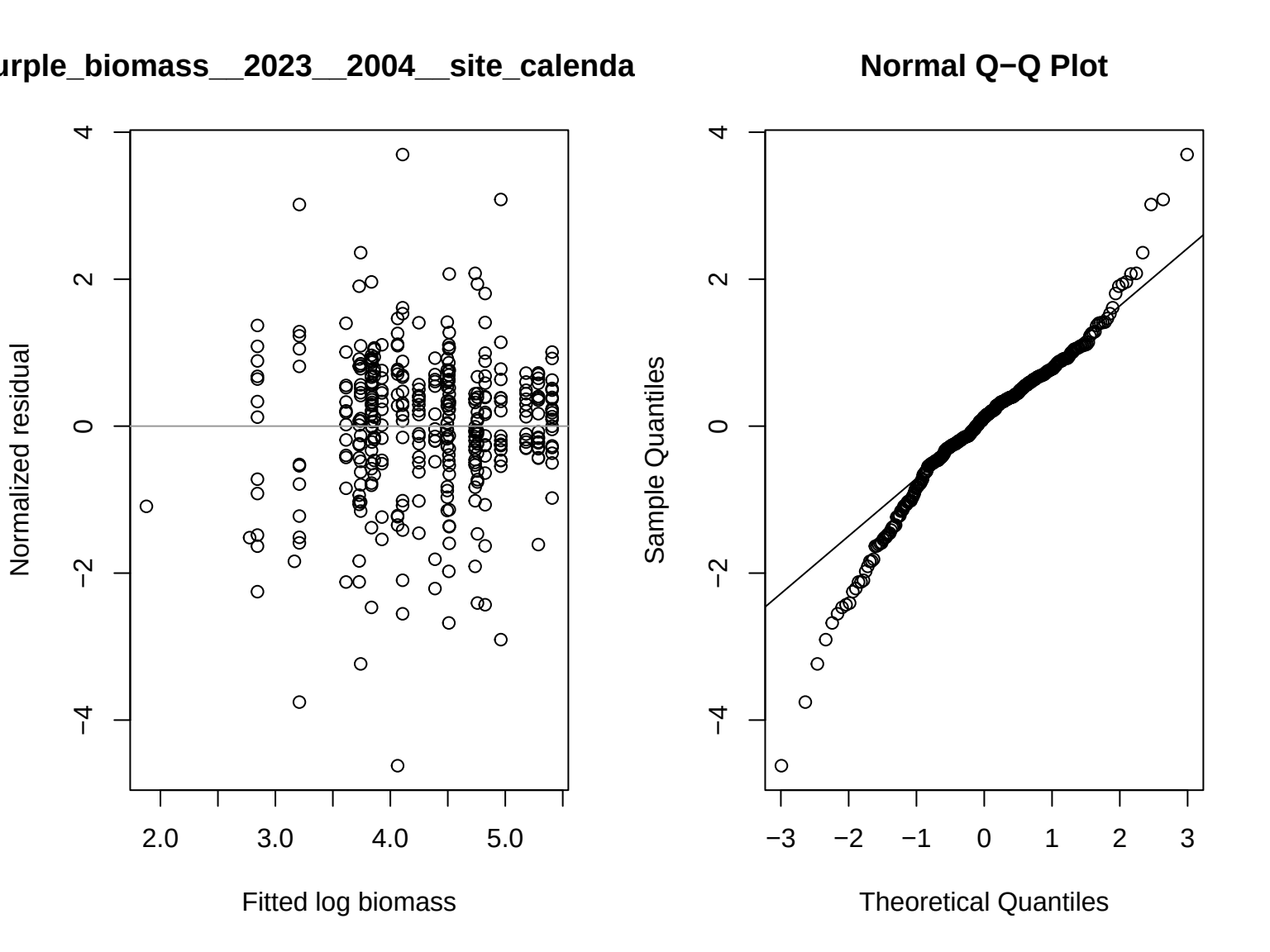

urple_biomass__2019__2004__site_calenda
Normalized residual
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

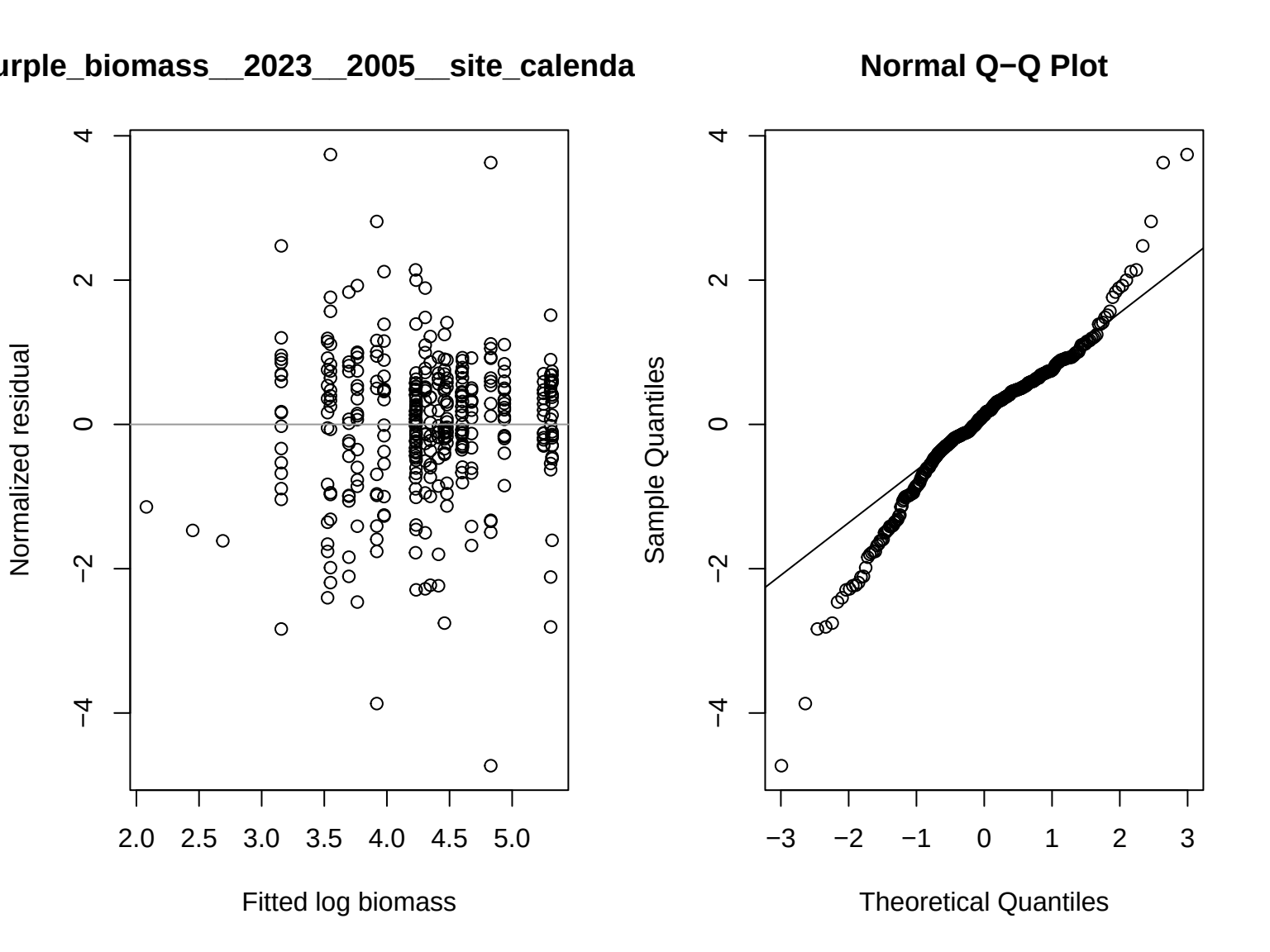

urple_biomass__2019__2005__site_calenda
Normalized residual
Fitted log biomass
4
2
0
−2
−4
2
3
4
5

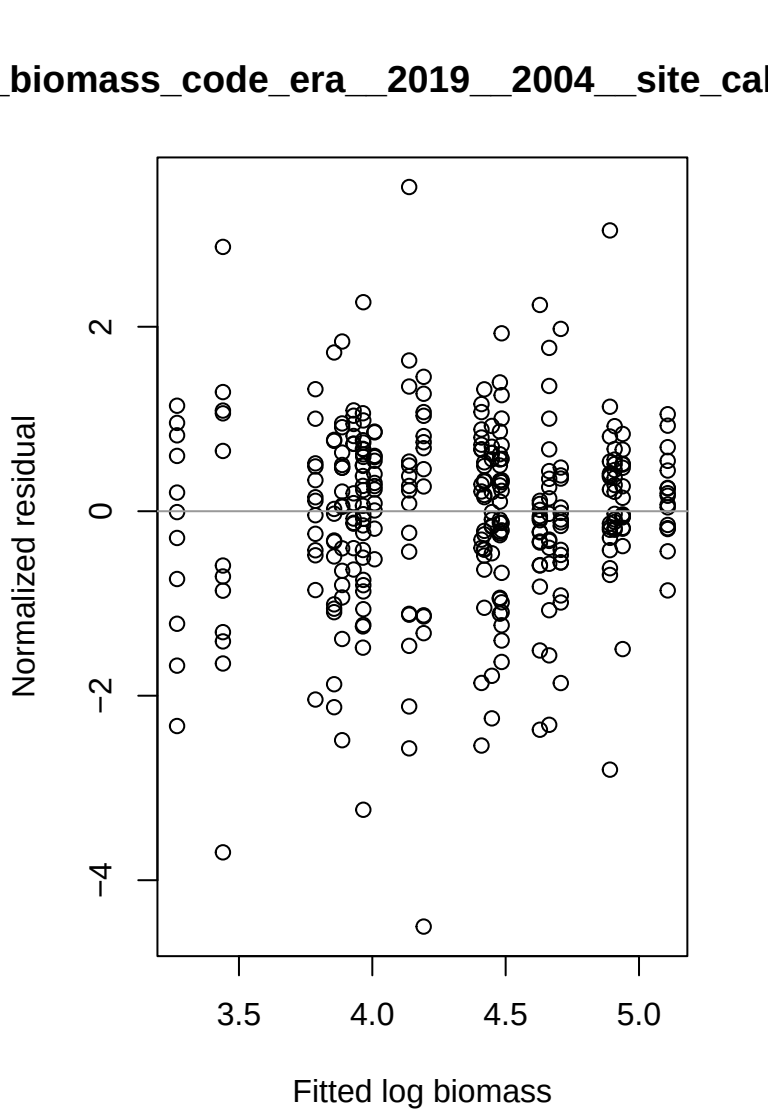

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
4
2
0
−2
−4
−3
−2
−1
0
1
2
3

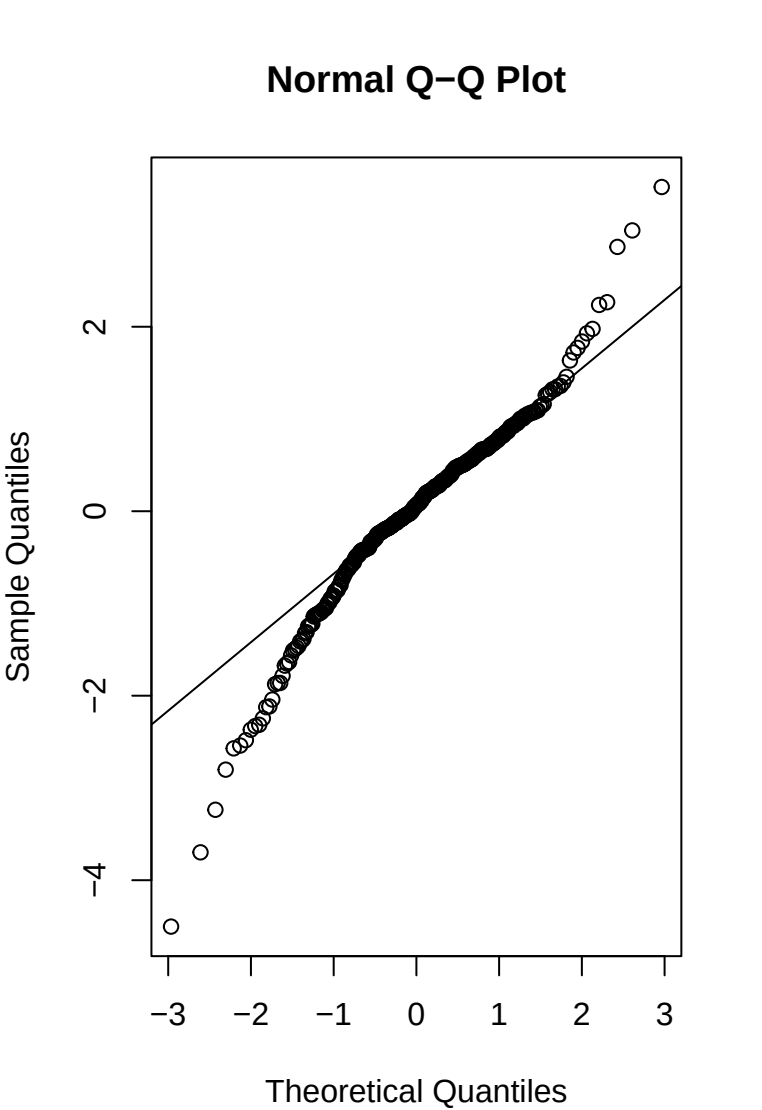

urple_biomass__2022__2004__site_calenda
Normalized residual
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

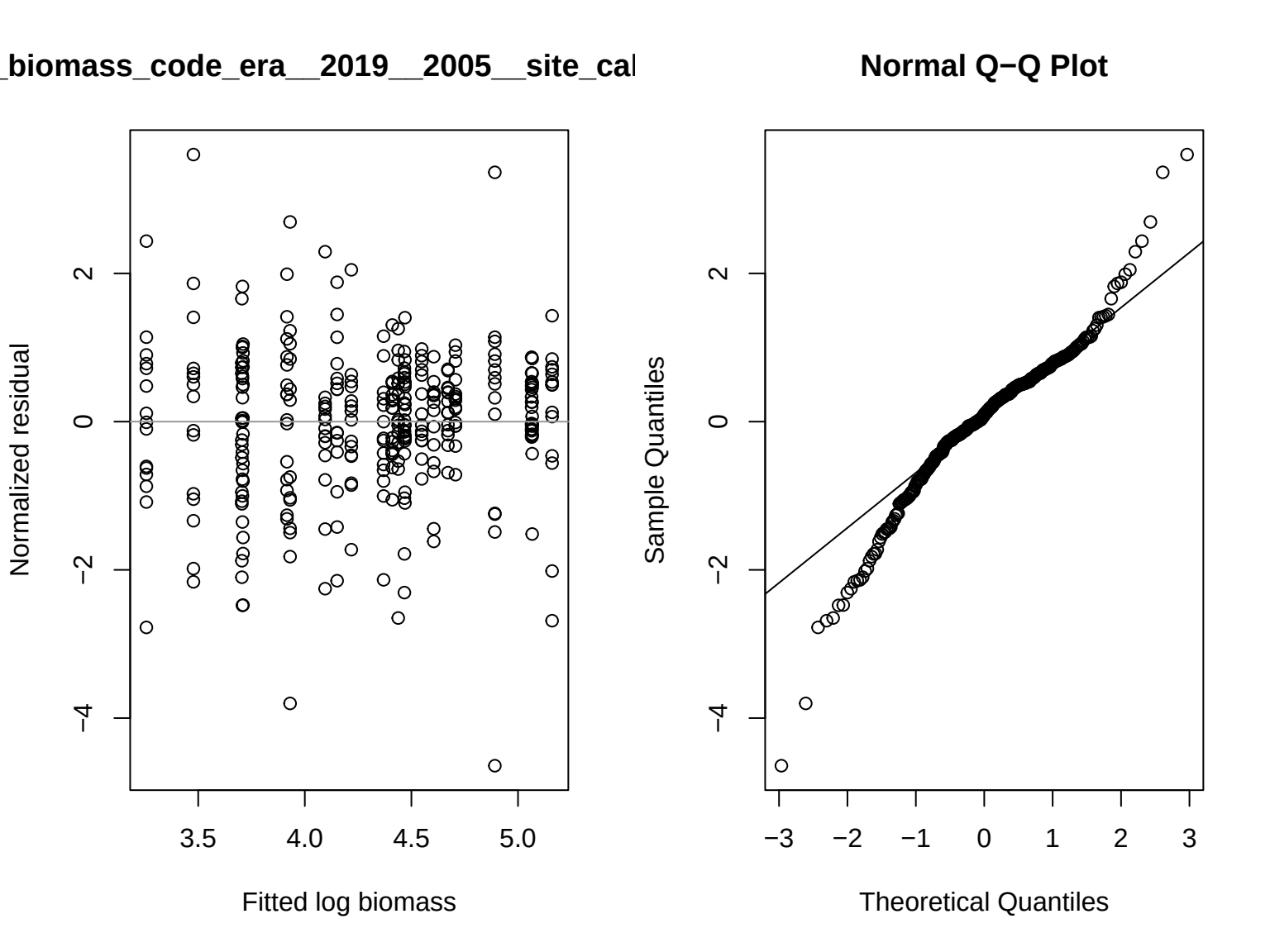

urple_biomass__2022__2005__site_calenda
Normalized residual
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

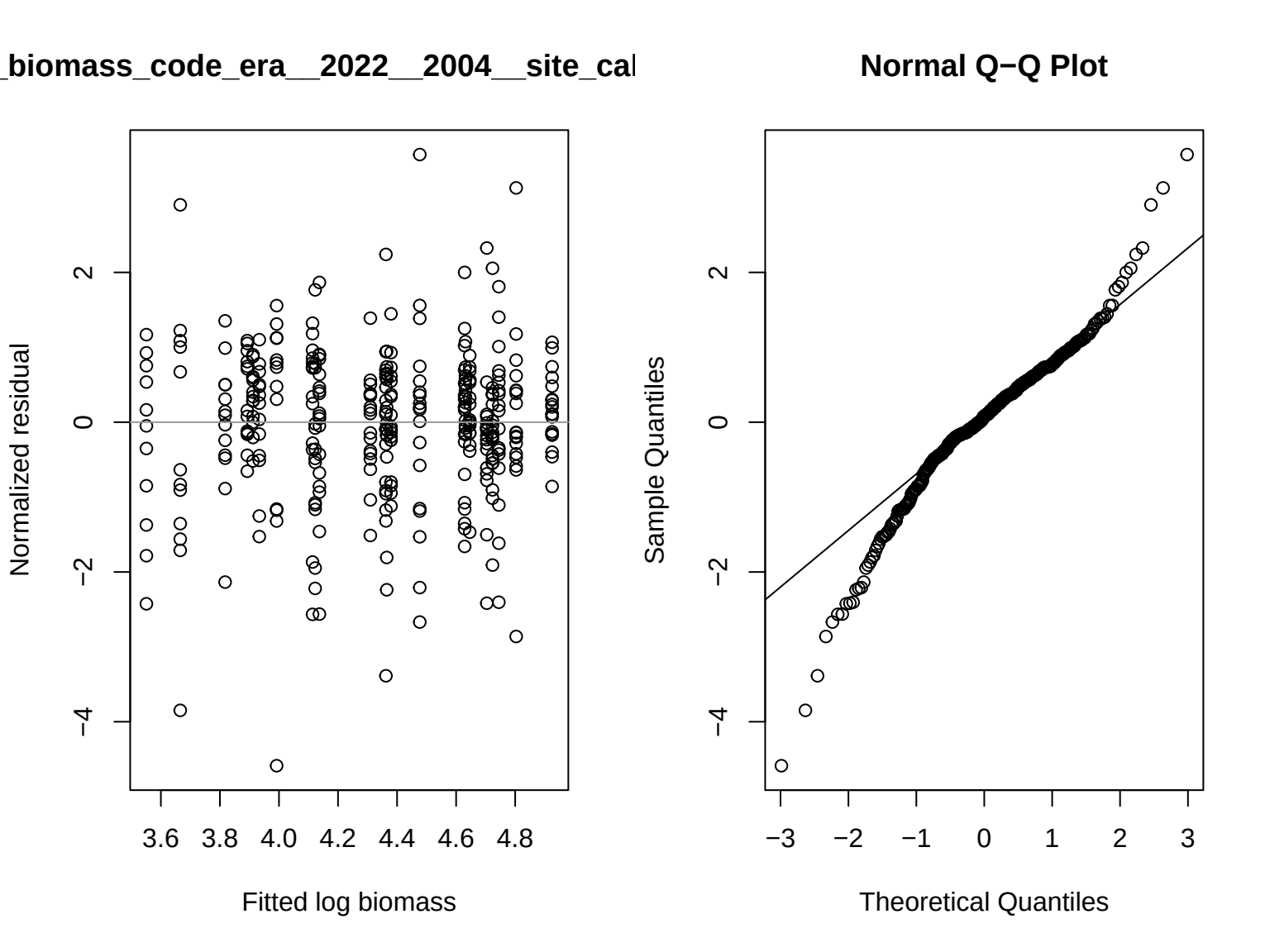

urple_biomass__2023__2004__site_calenda
Normalized residual
Fitted log biomass
2.0
3.0
4.0
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

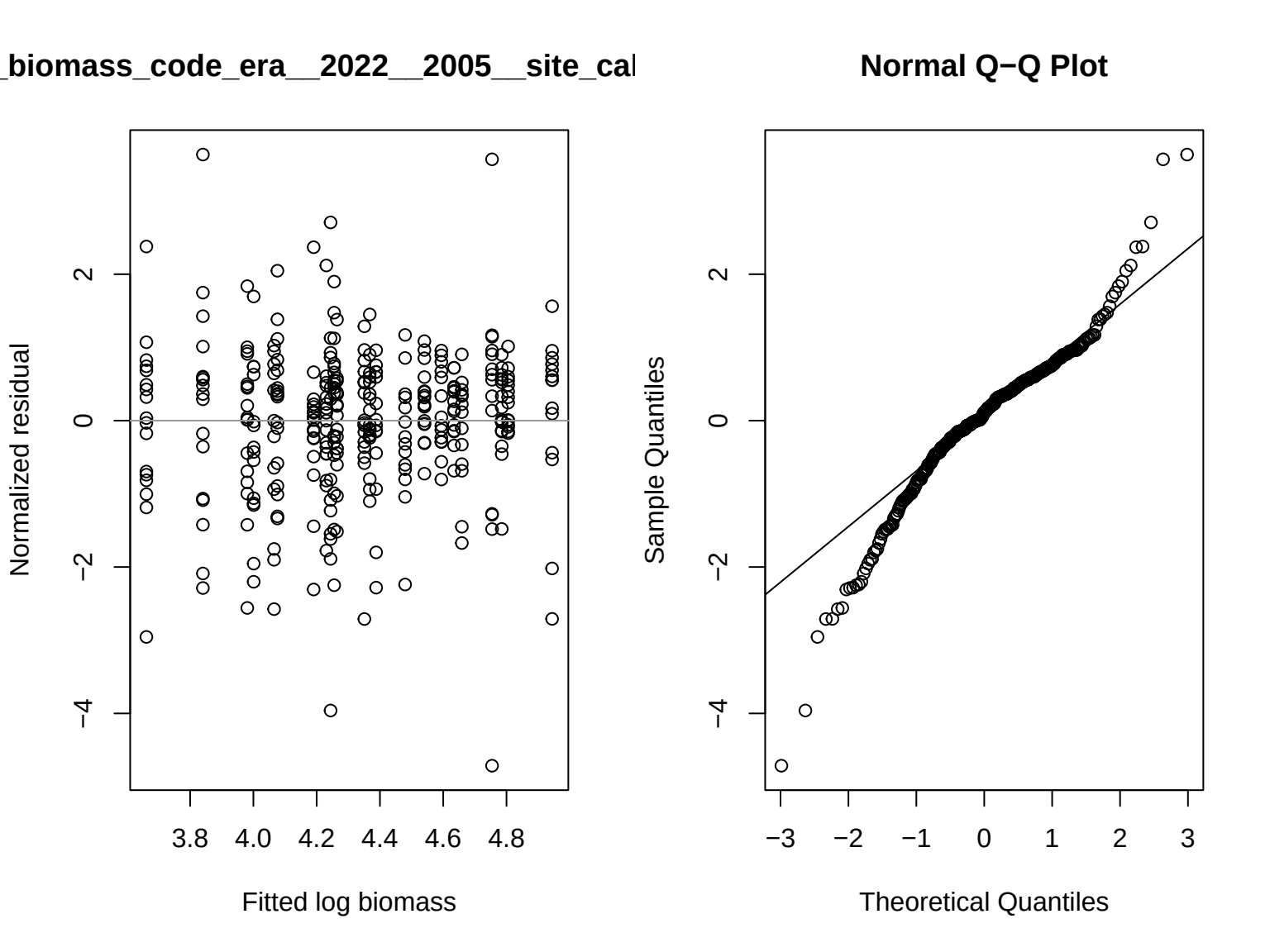

urple_biomass__2023__2005__site_calenda
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

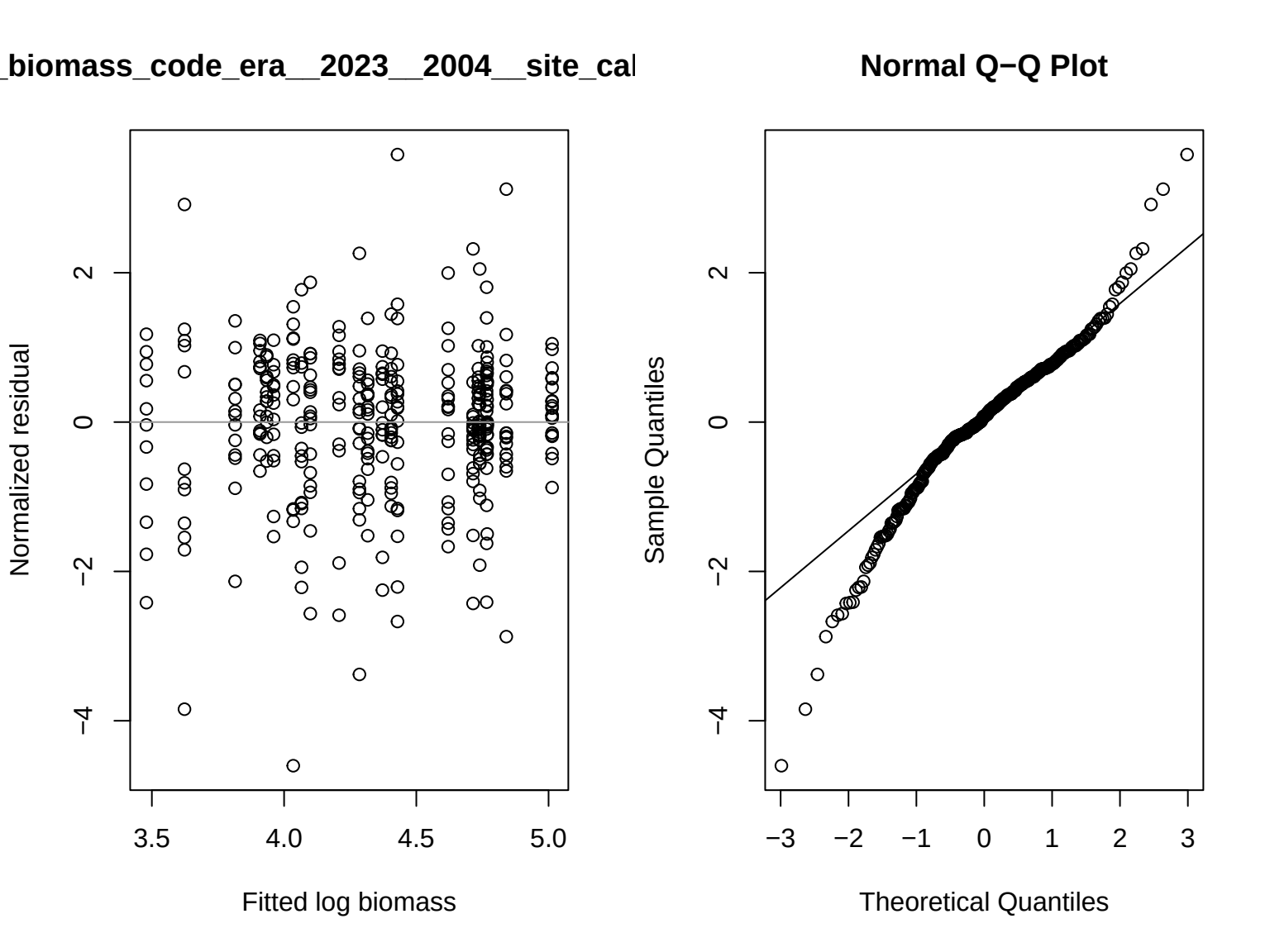

_biomass_code_era__2019__2004__site_cal
Normalized residual
Fitted log biomass
3.5
4.0
4.5
5.0
−4
−2
0
2

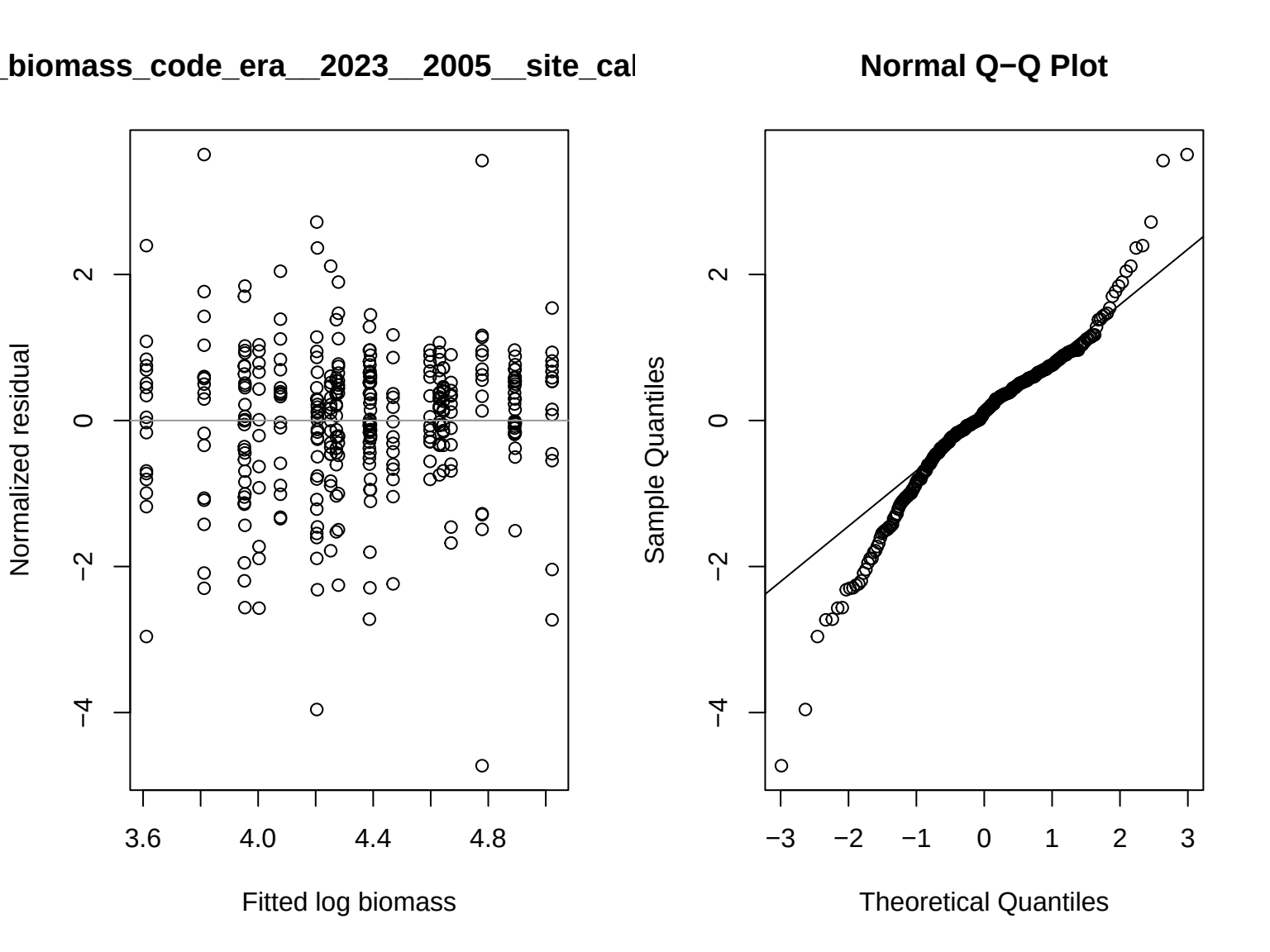

Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3
−4
−2
0
2

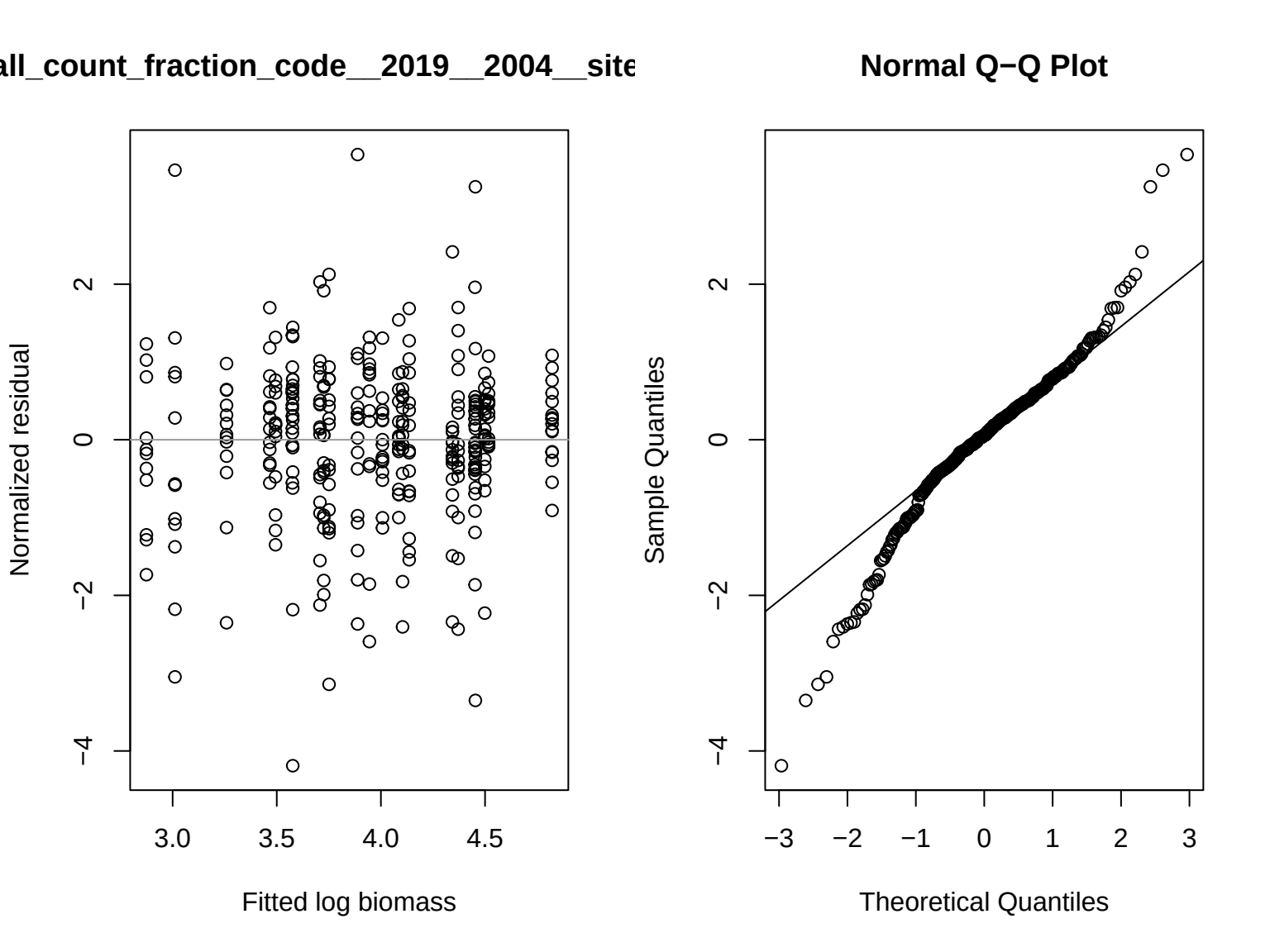

_biomass_code_era__2019__2005__site_cal
Normalized residual
Fitted log biomass
3.5
4.0
4.5
5.0
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

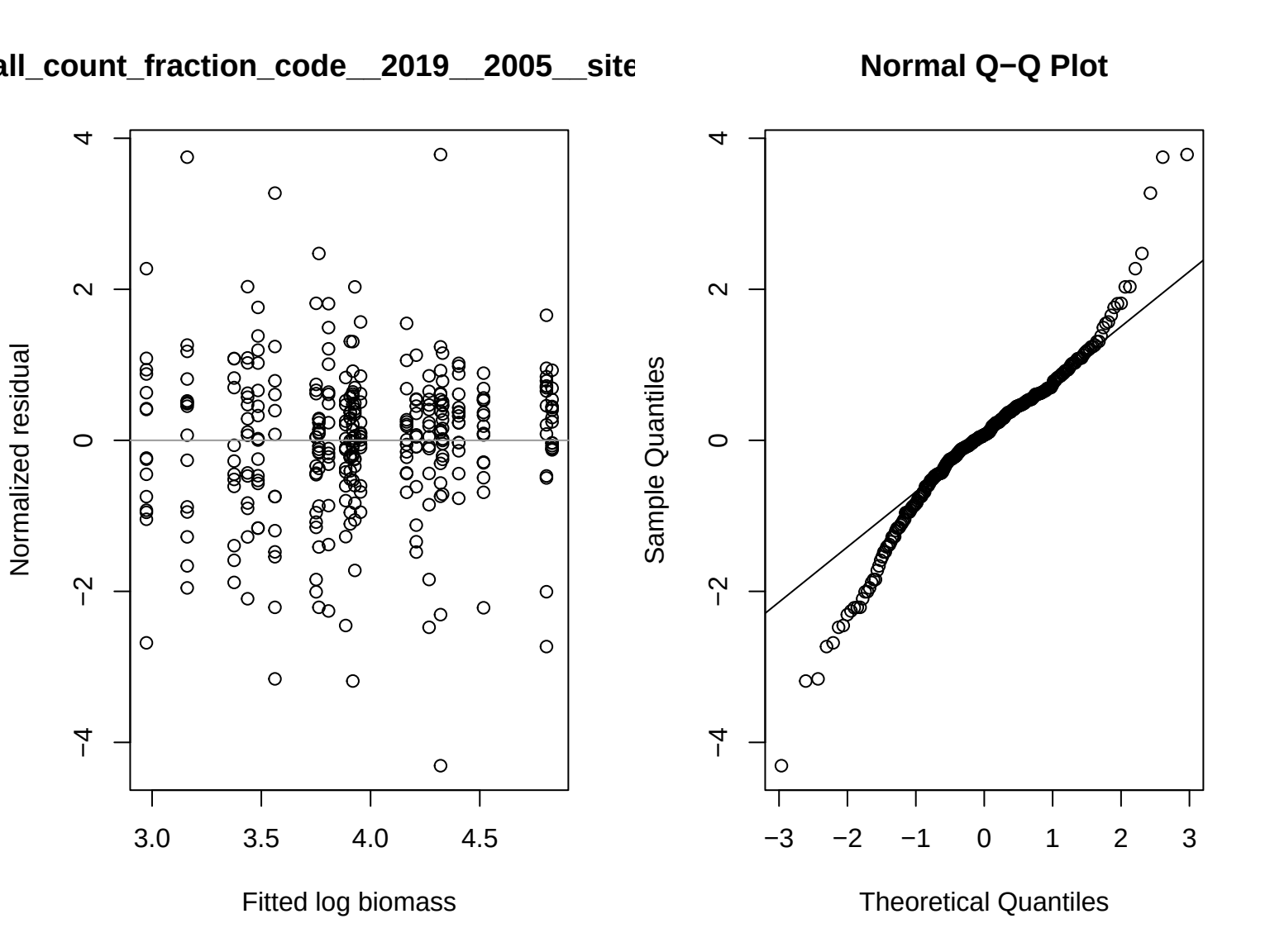

_biomass_code_era__2022__2004__site_cal
Normalized residual
Fitted log biomass
3.6
3.8
4.0
4.2
4.4
4.6
4.8
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

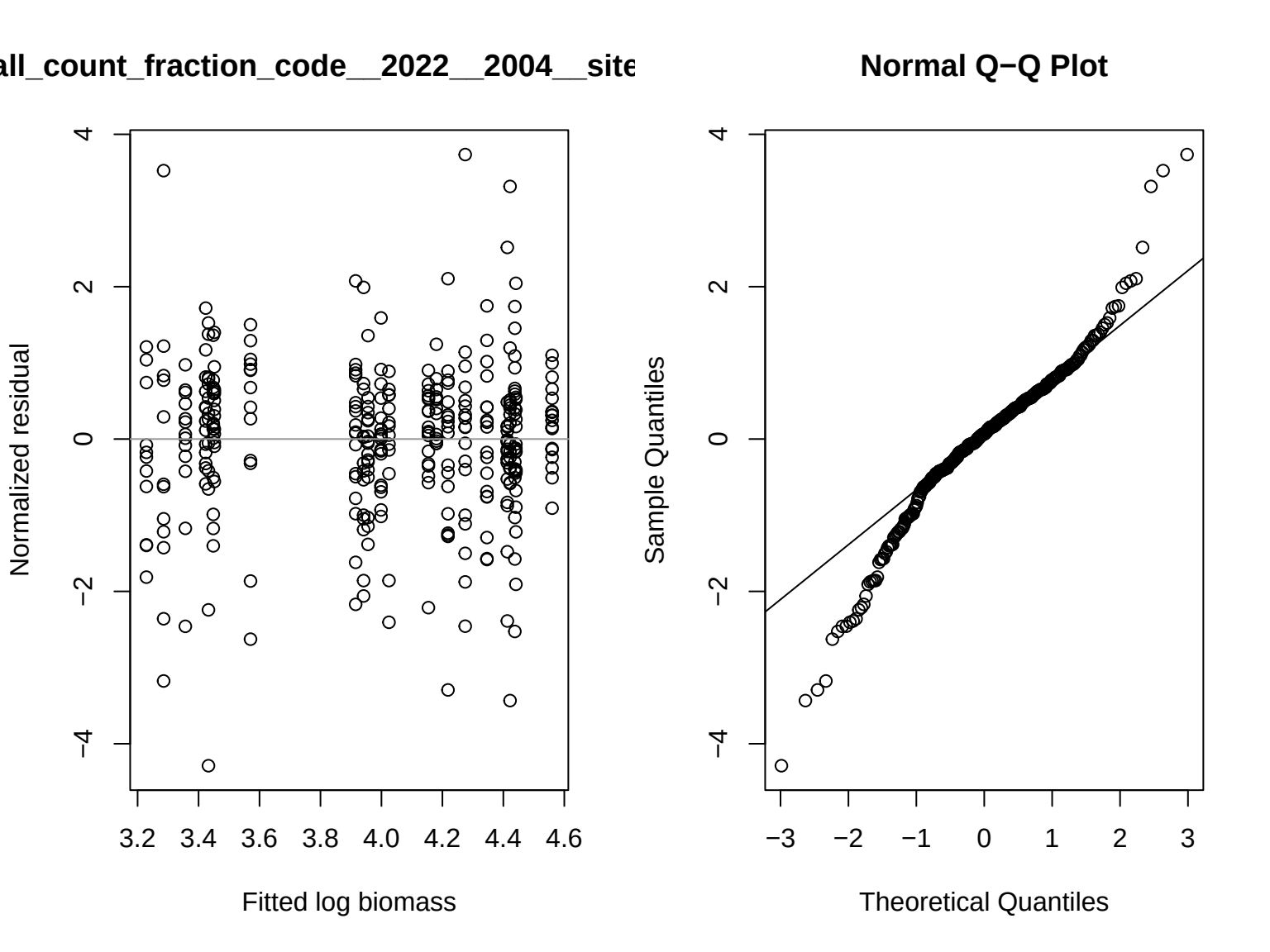

_biomass_code_era__2022__2005__site_cal
Normalized residual
Fitted log biomass
3.8
4.0
4.2
4.4
4.6
4.8
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

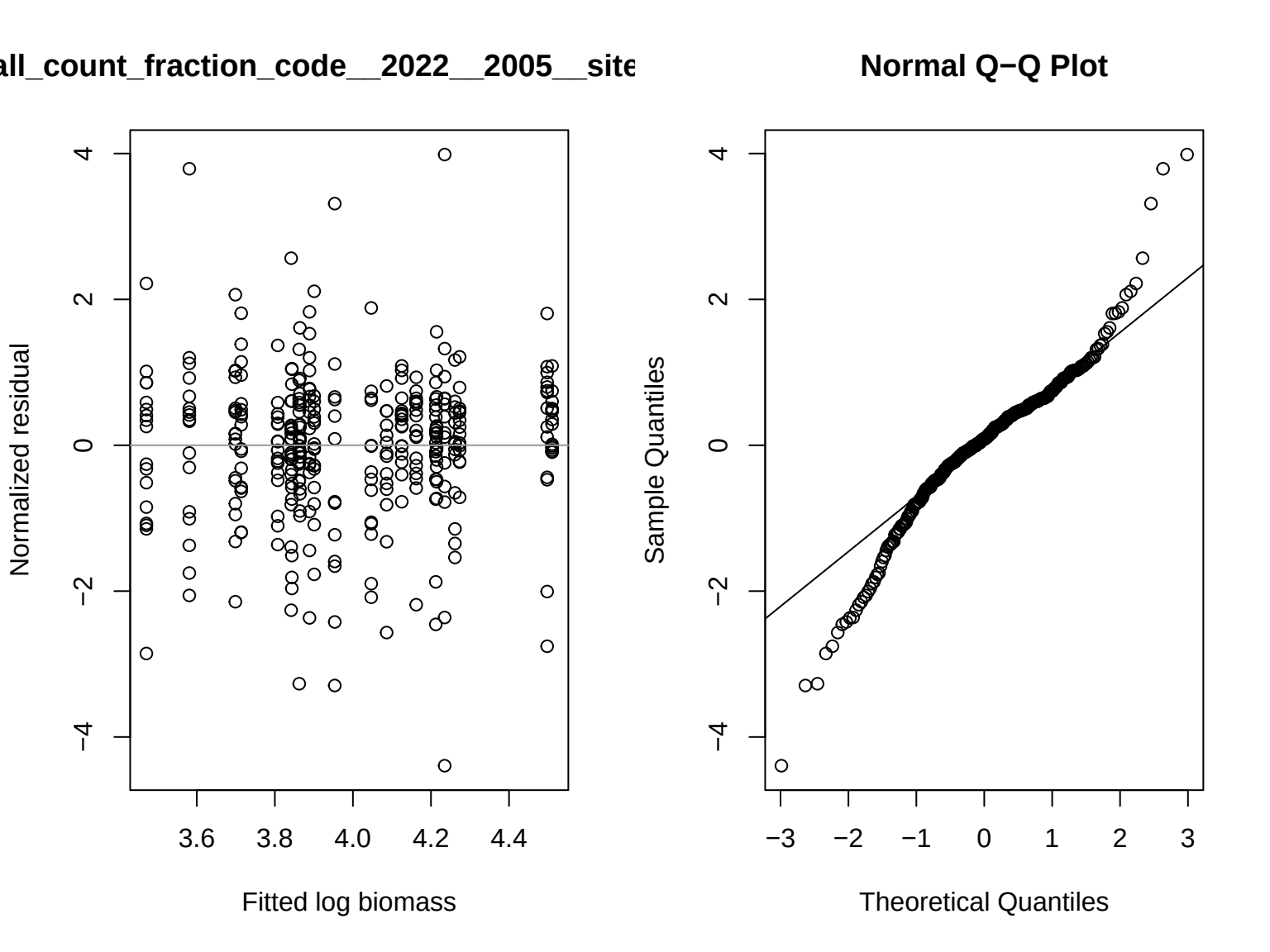

_biomass_code_era__2023__2004__site_cal
Normalized residual
Fitted log biomass
3.5
4.0
4.5
5.0
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

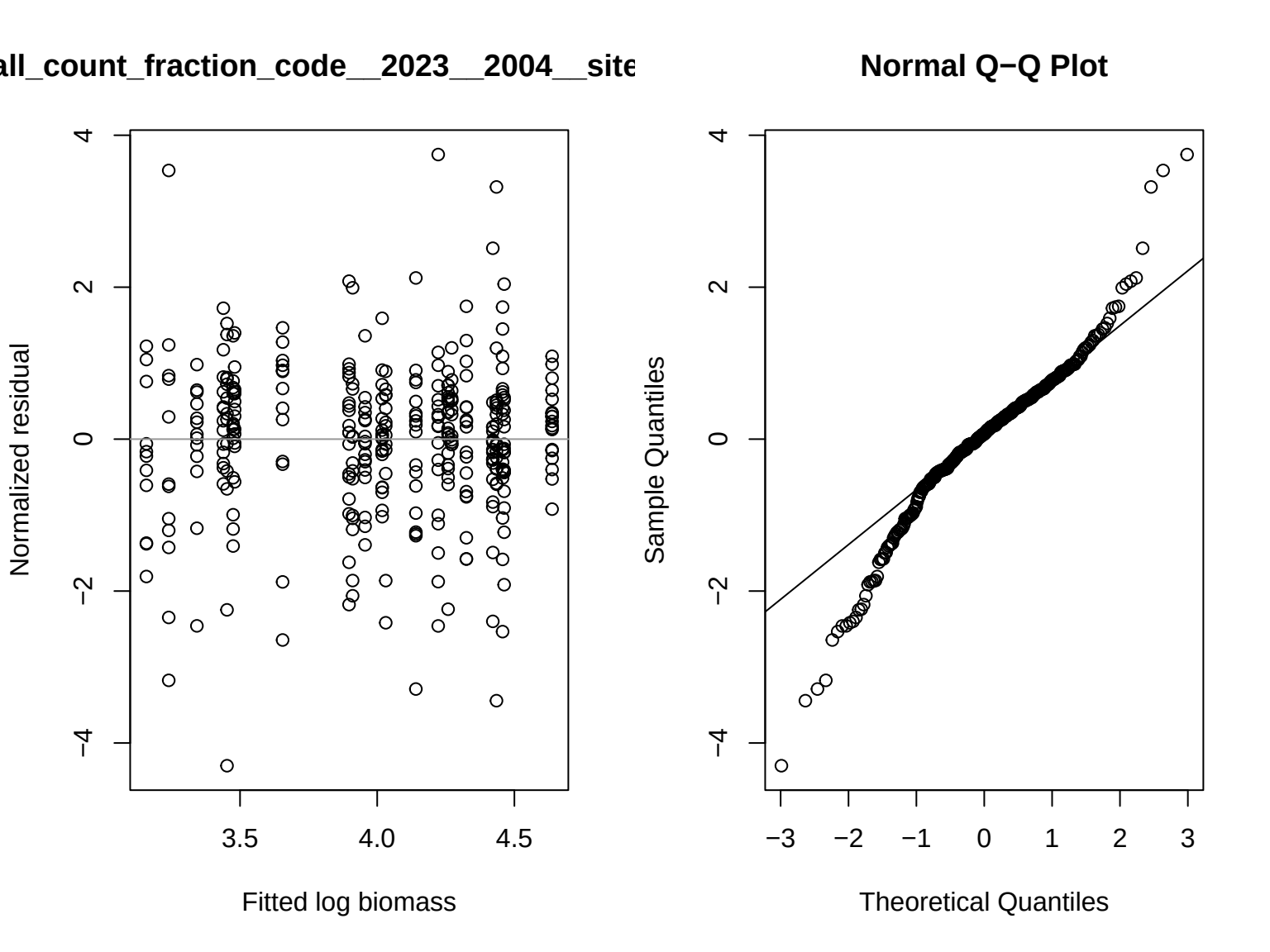

_biomass_code_era__2023__2005__site_cal
Normalized residual
Fitted log biomass
3.6
4.0
4.4
4.8
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

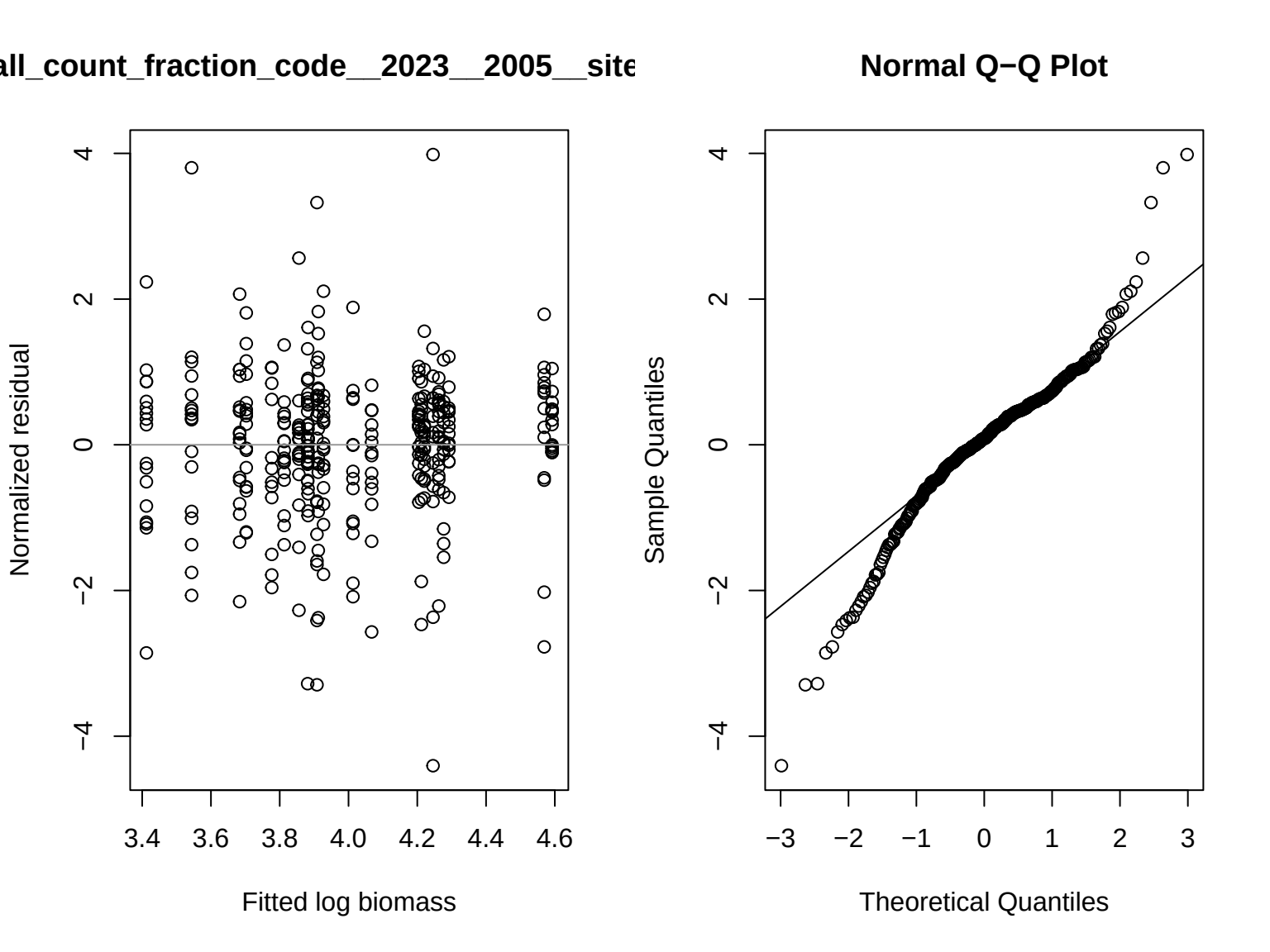

all_count_fraction_code__2019__2004__site
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

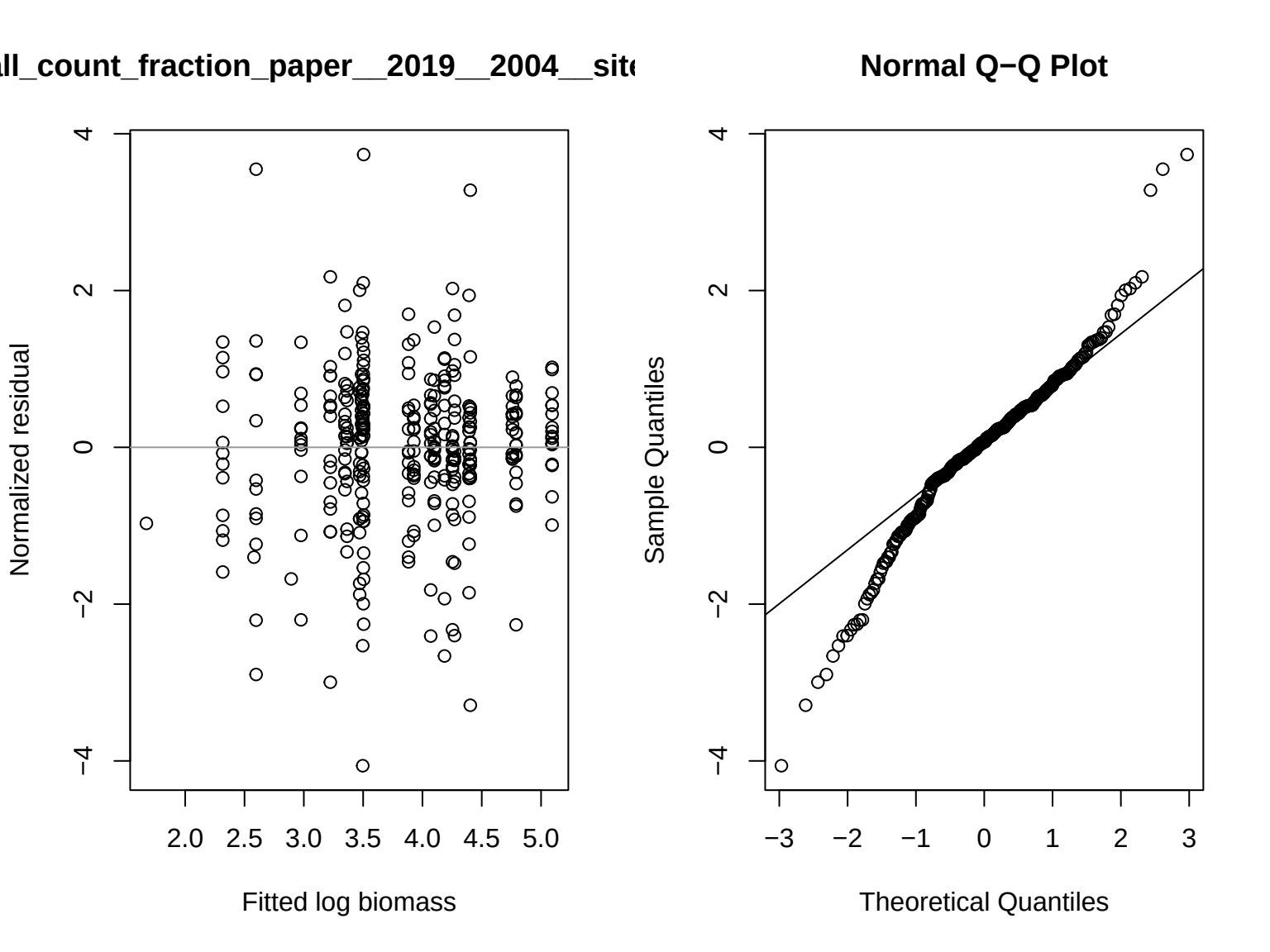

all_count_fraction_code__2019__2005__site
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

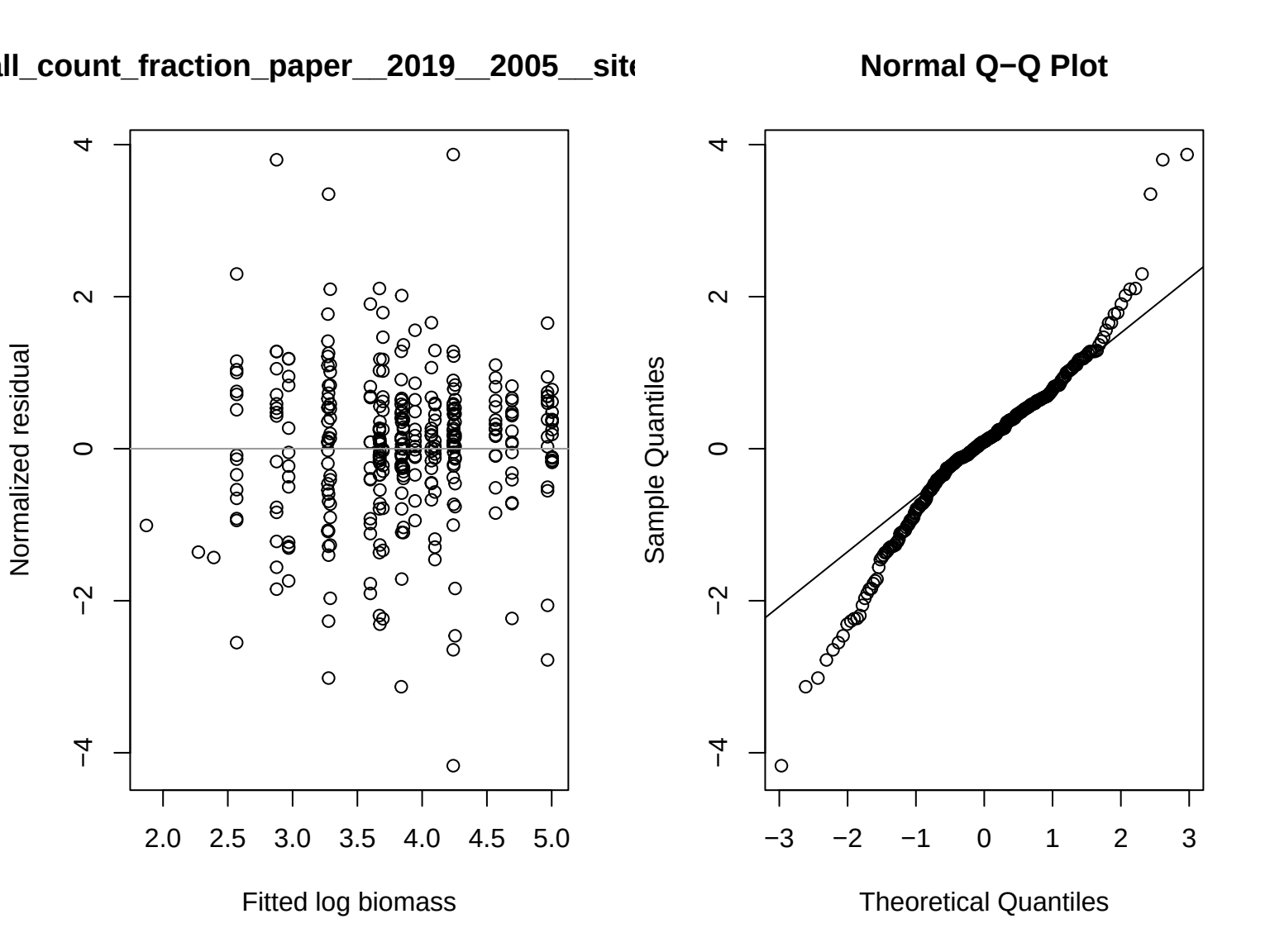

all_count_fraction_code__2022__2004__site
Normalized residual
Fitted log biomass
3.2
3.4
3.6
3.8
4.0
4.2
4.4
4.6
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

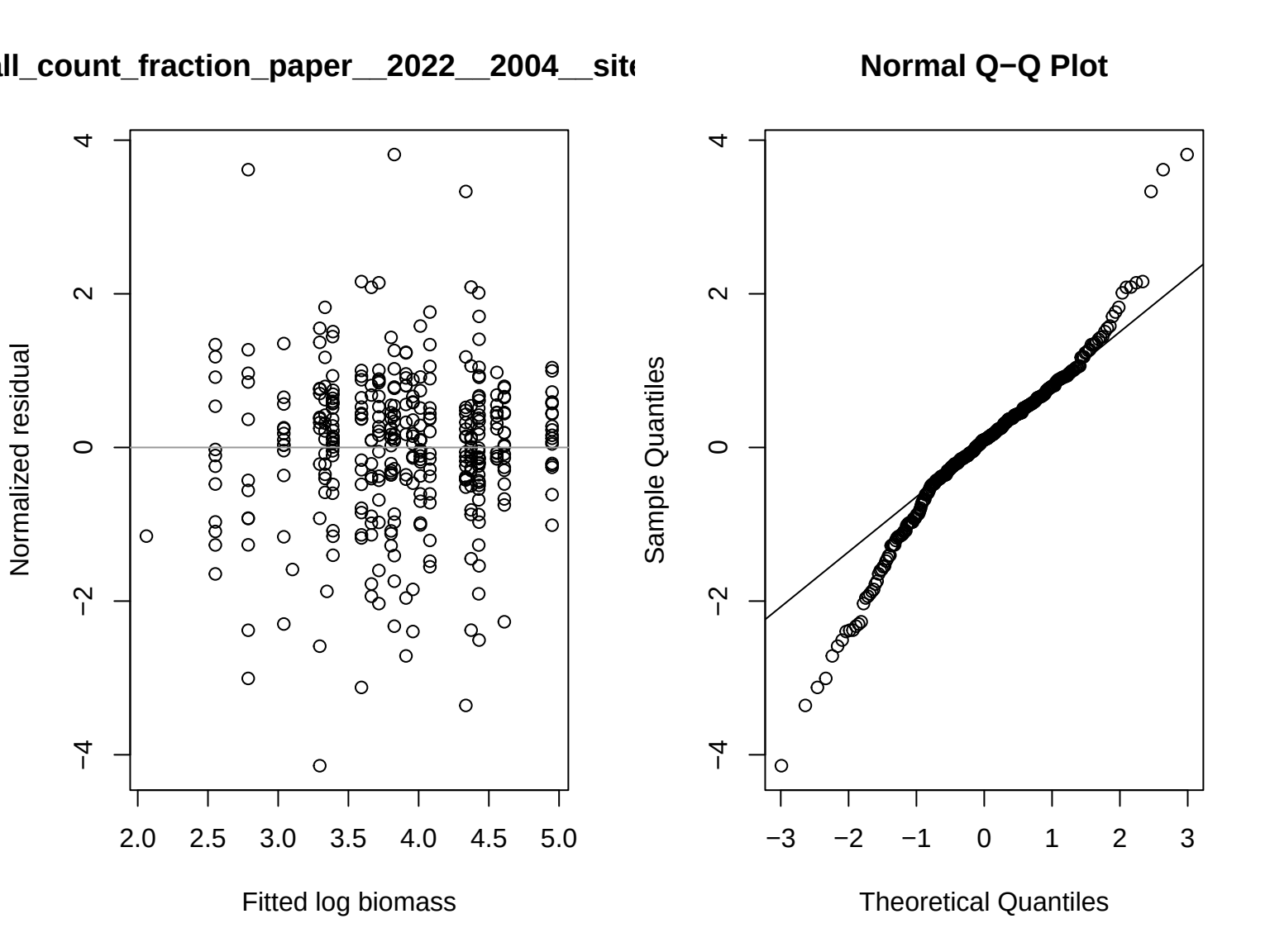

all_count_fraction_code__2022__2005__site
Normalized residual
Fitted log biomass
3.6
3.8
4.0
4.2
4.4
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

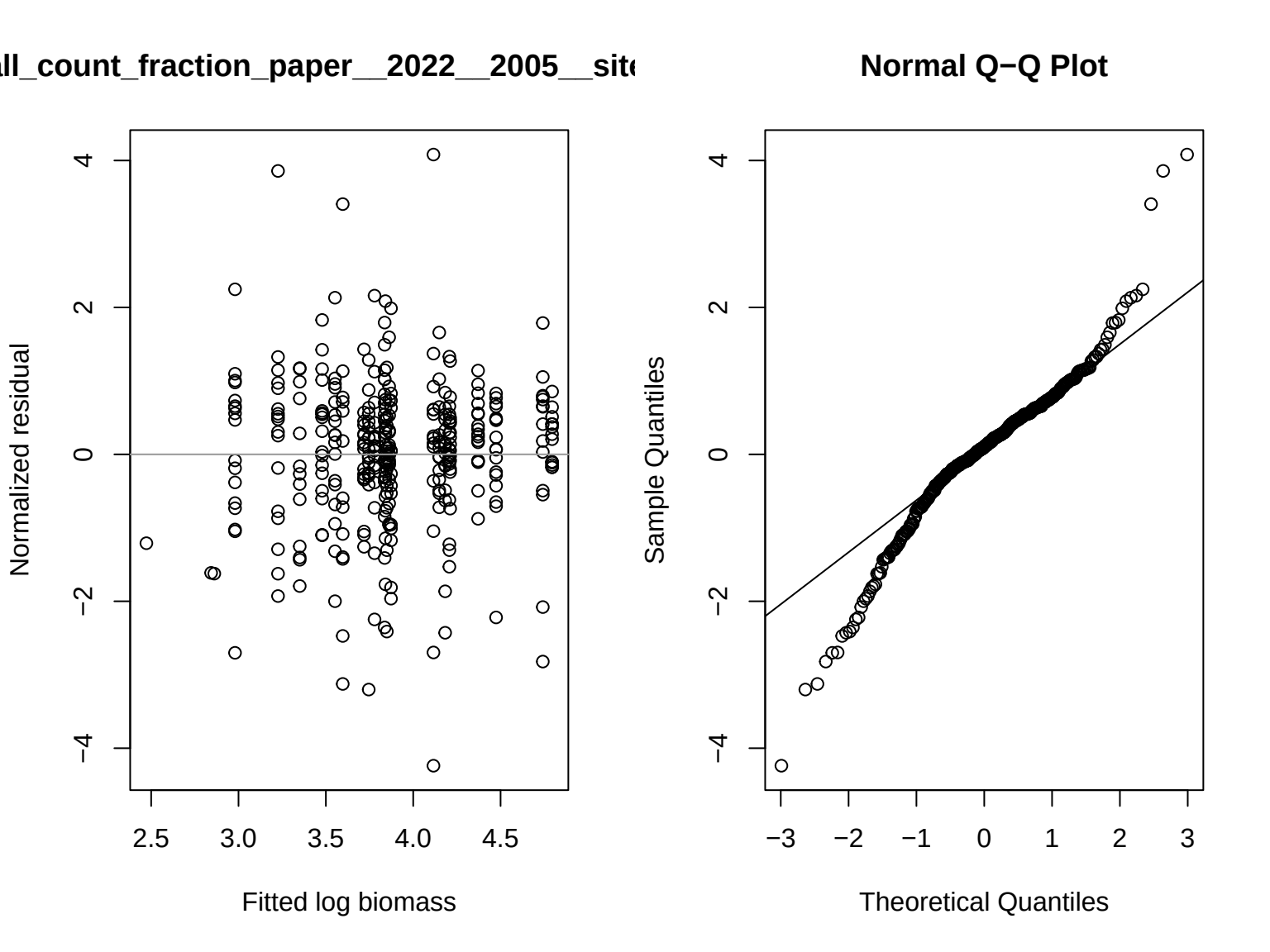

all_count_fraction_code__2023__2004__site
Normalized residual
Fitted log biomass
3.5
4.0
4.5
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

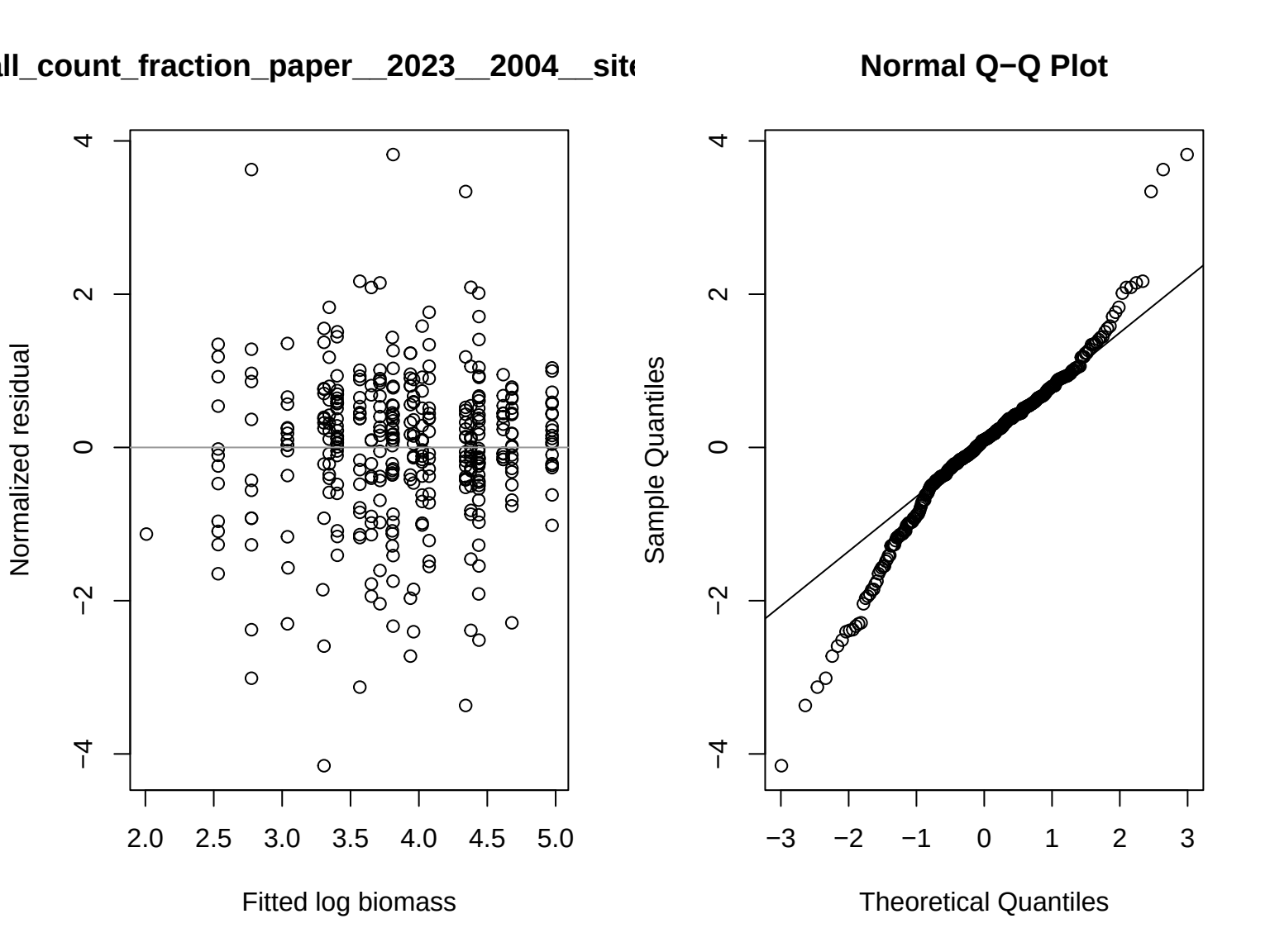

all_count_fraction_code__2023__2005__site
Normalized residual
Fitted log biomass
3.4
3.6
3.8
4.0
4.2
4.4
4.6
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

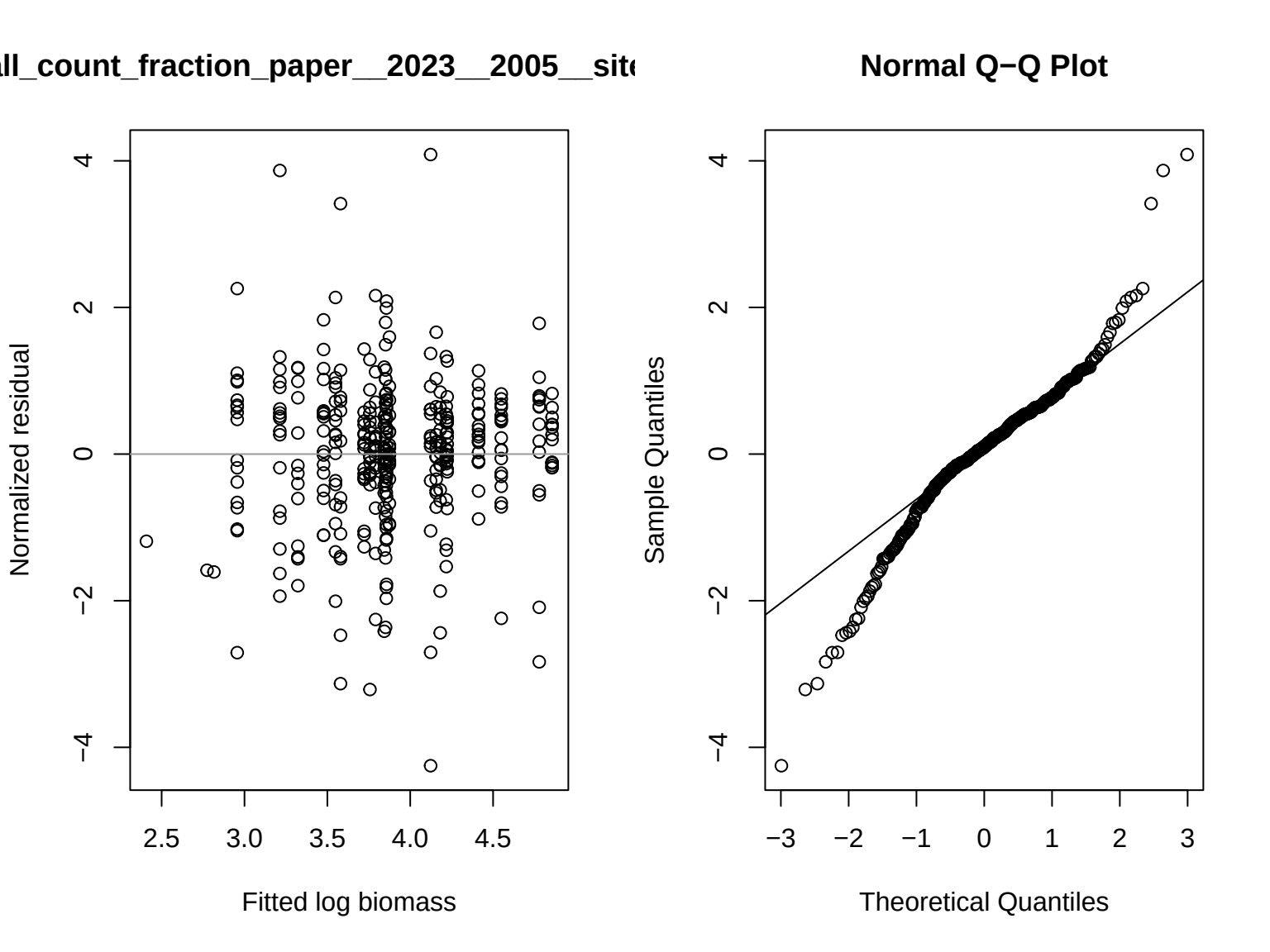

ll_count_fraction_paper__2019__2004__site
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

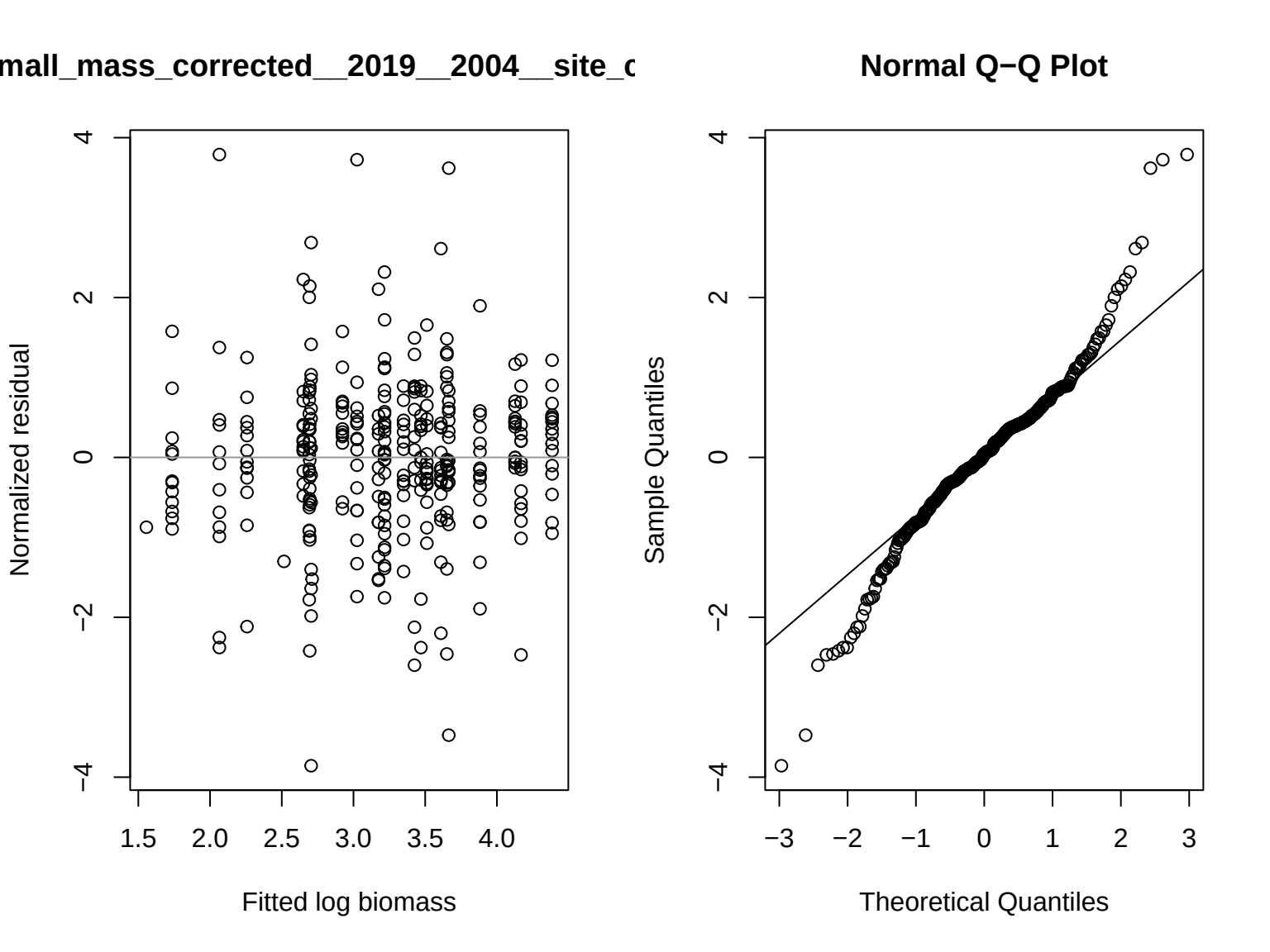

ll_count_fraction_paper__2019__2005__site
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

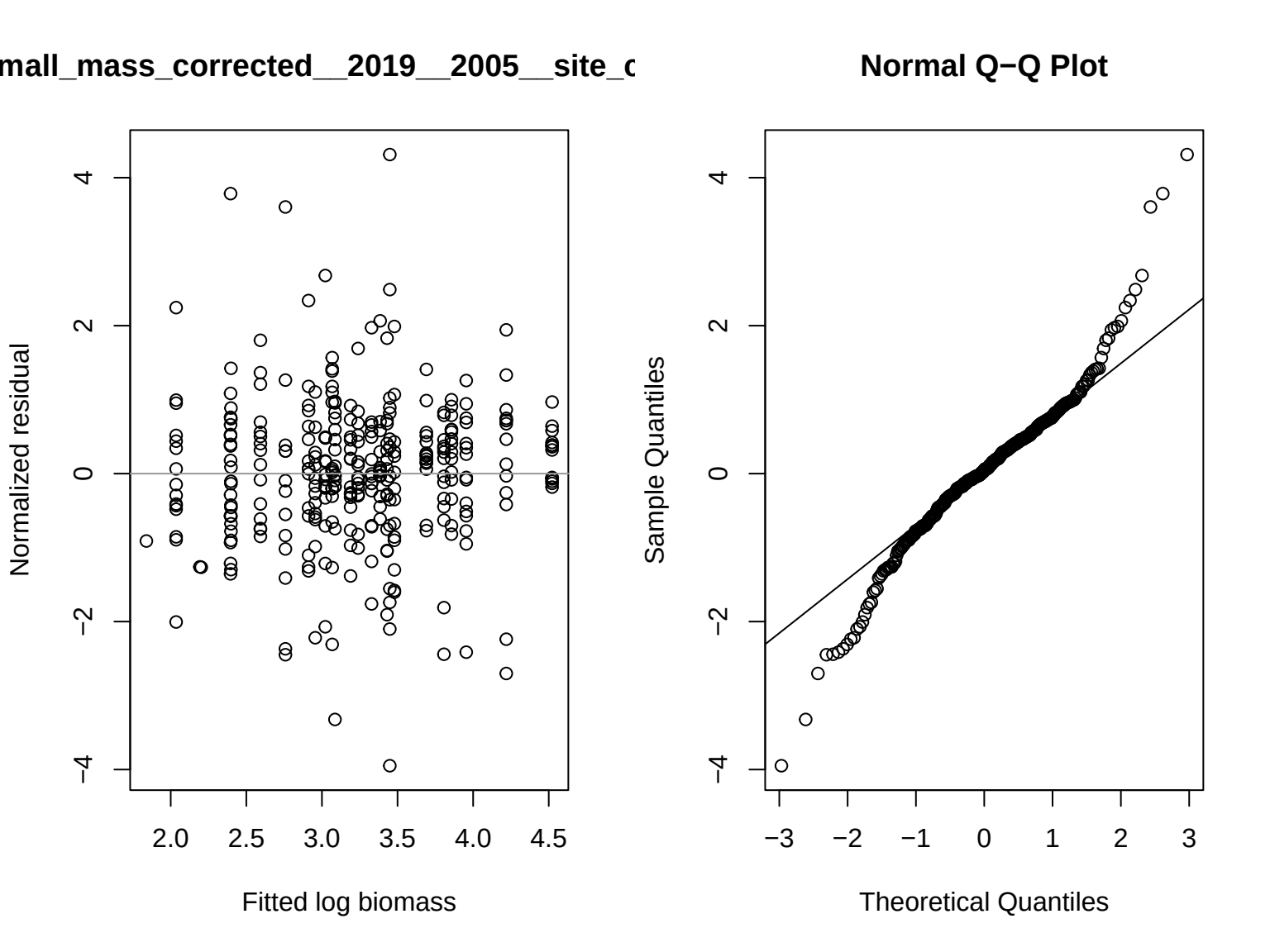

ll_count_fraction_paper__2022__2004__site
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4.5
5.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

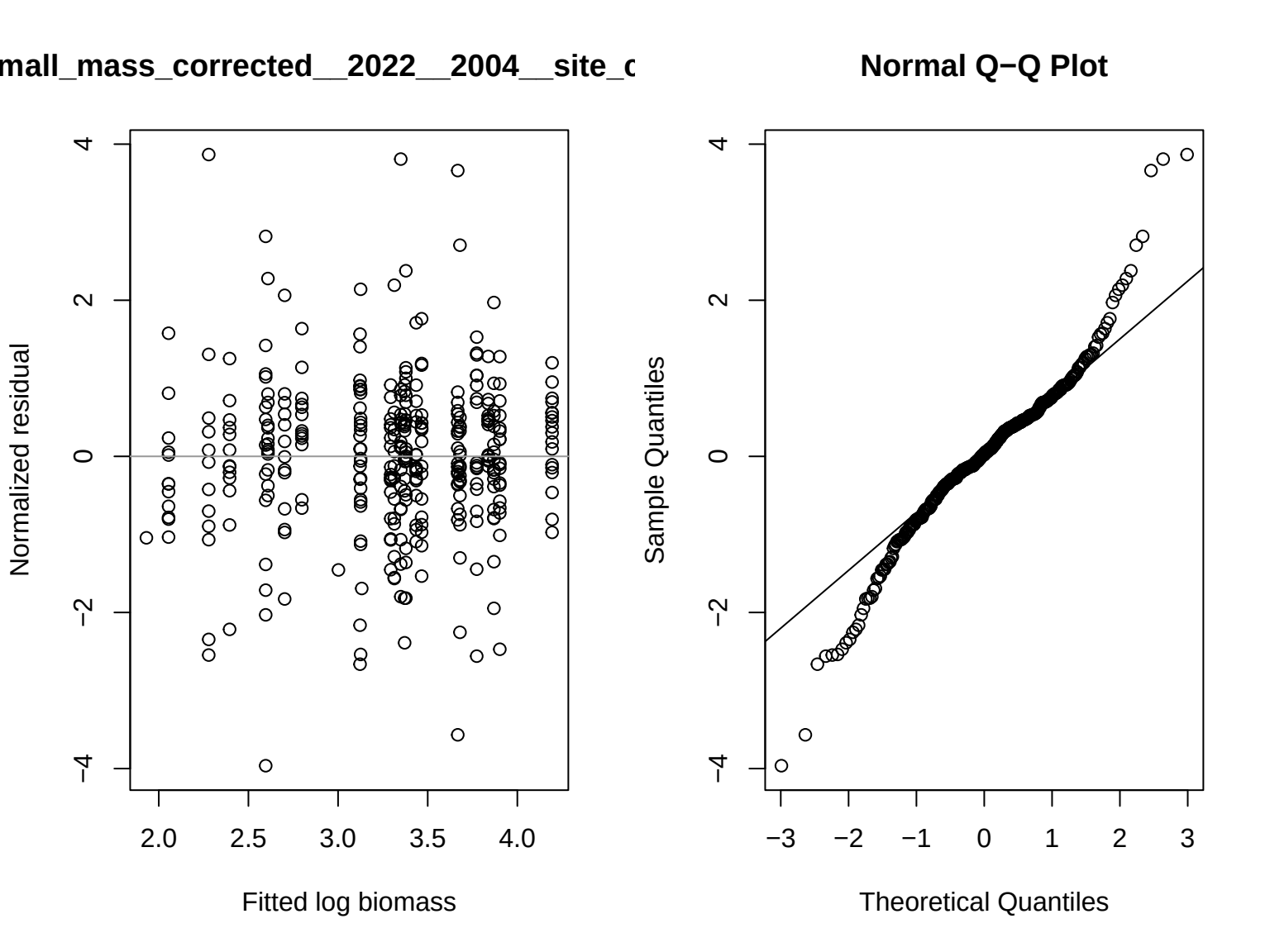

ll_count_fraction_paper__2022__2005__site
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
4.5
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

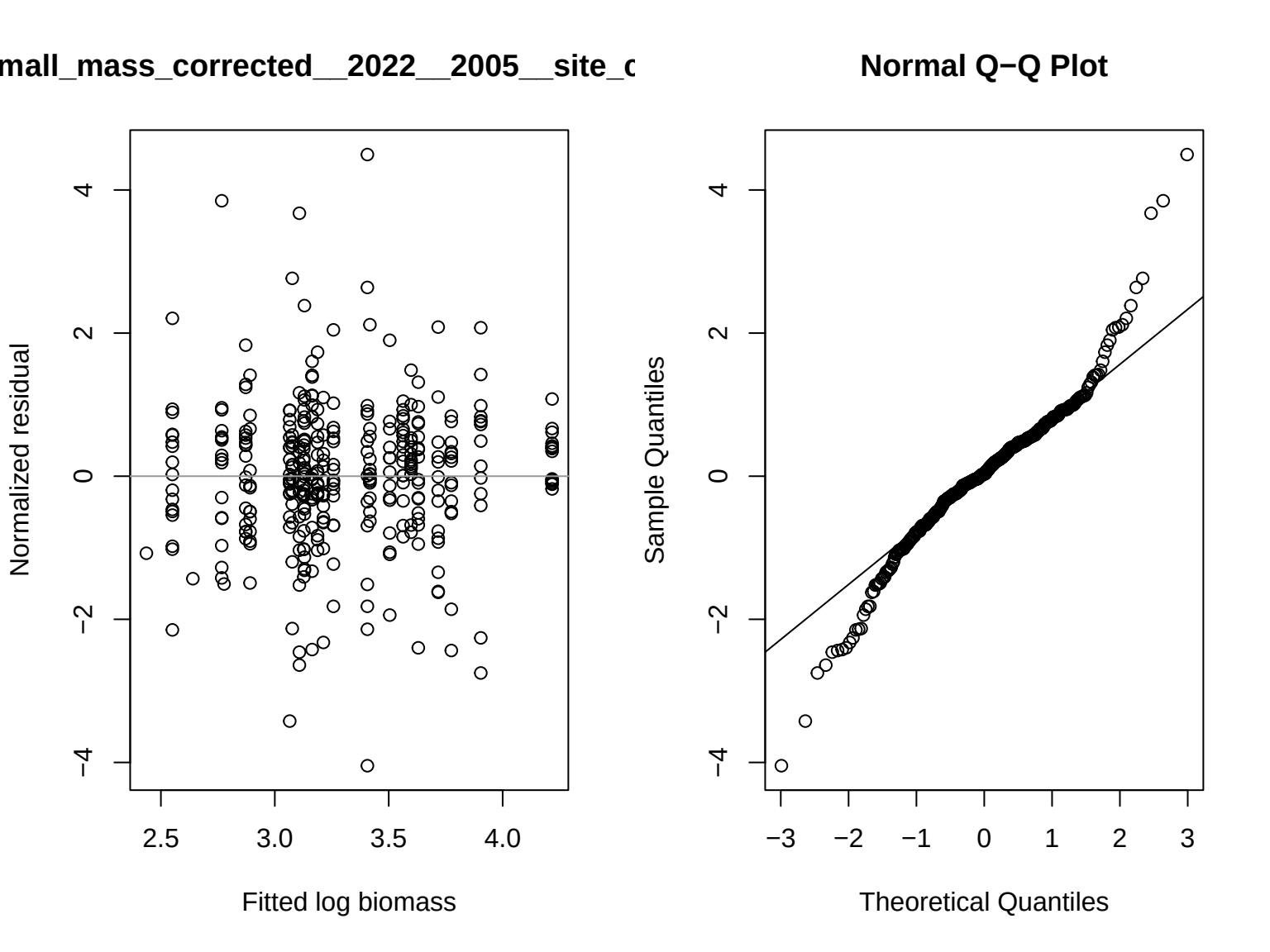

ll_count_fraction_paper__2023__2004__site
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4.5
5.0
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

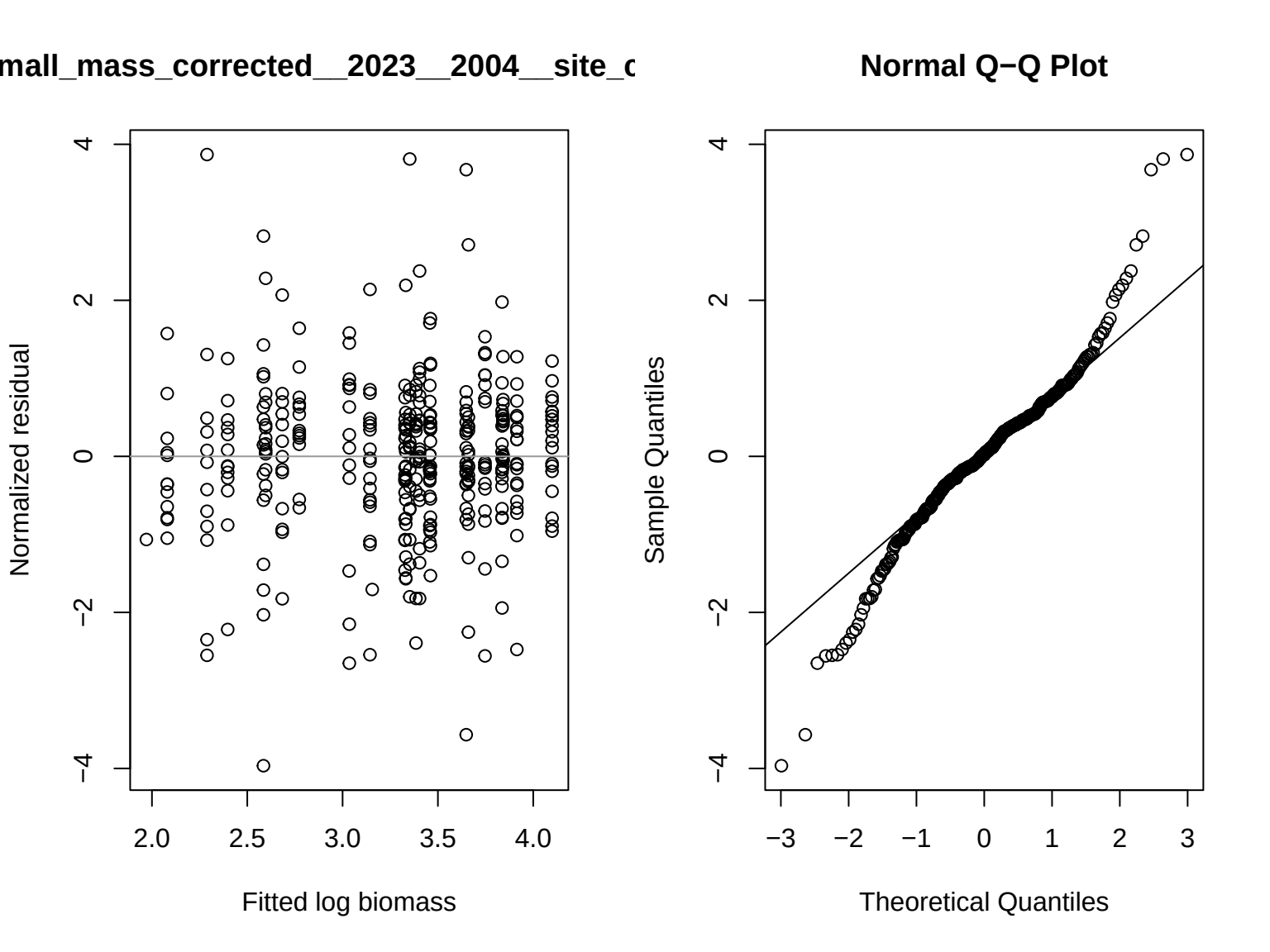

ll_count_fraction_paper__2023__2005__site
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
4.5
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

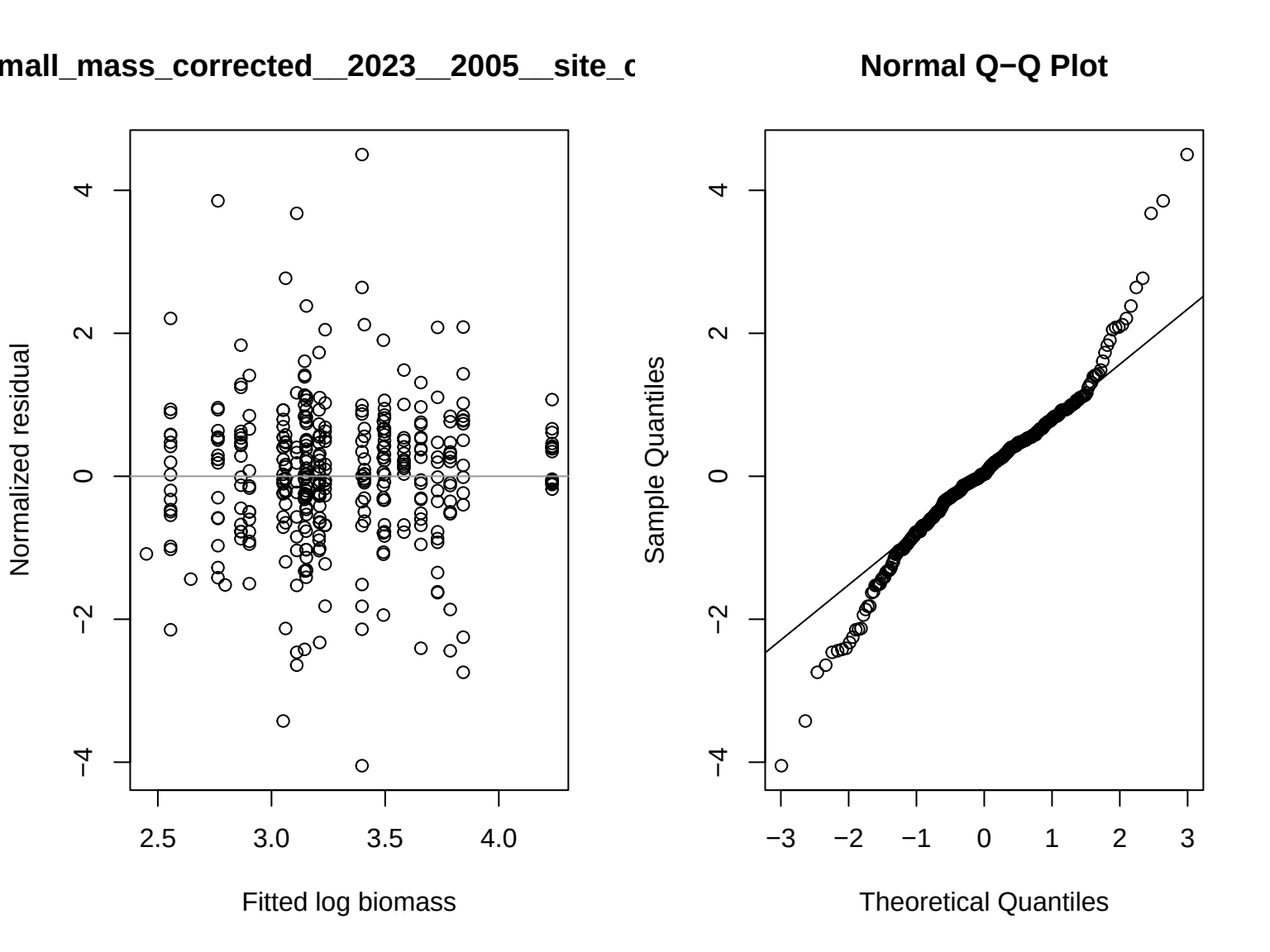

mall_mass_corrected__2019__2004__site_c
Normalized residual
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

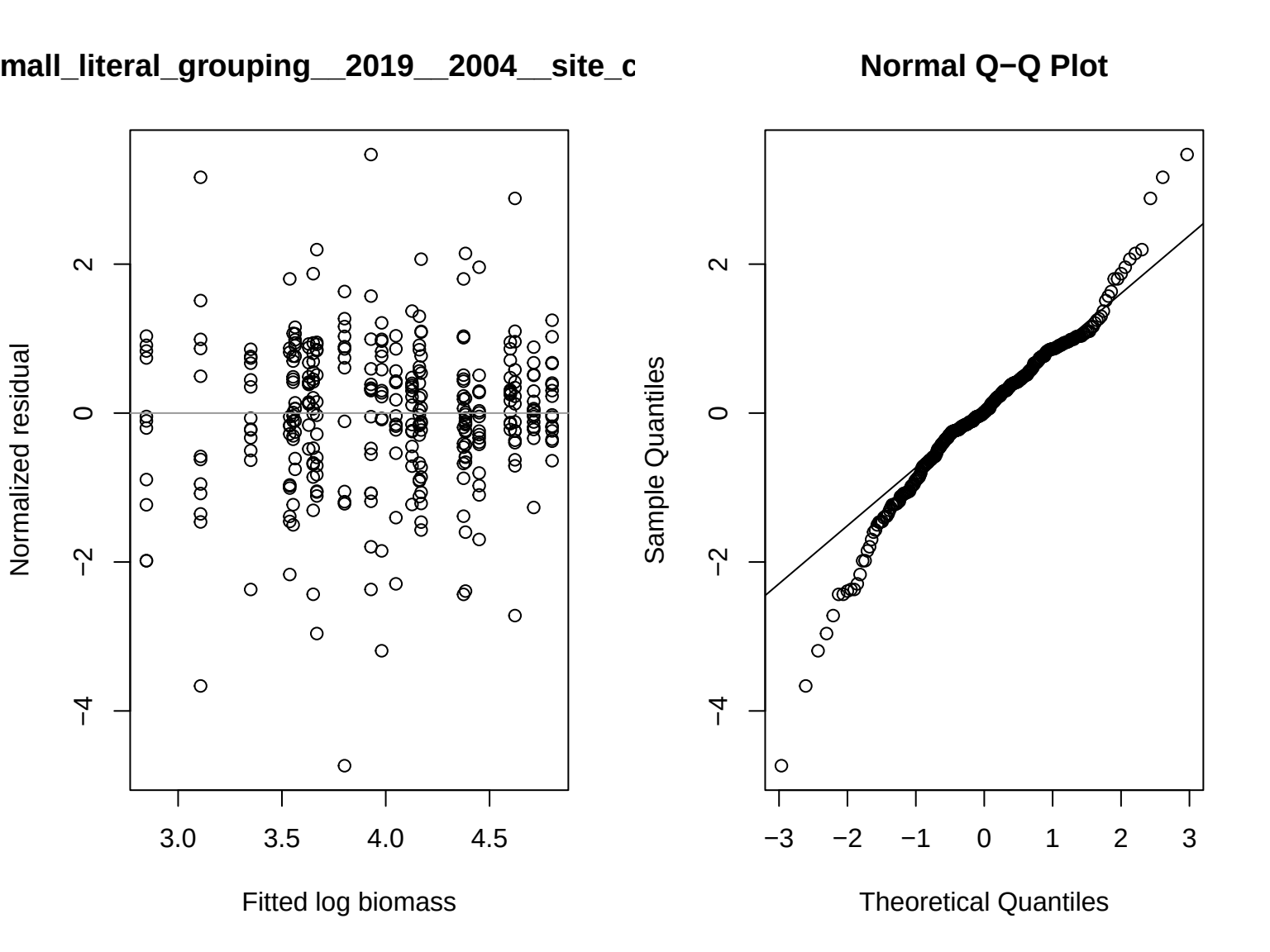

mall_mass_corrected__2019__2005__site_c
Normalized residual
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles

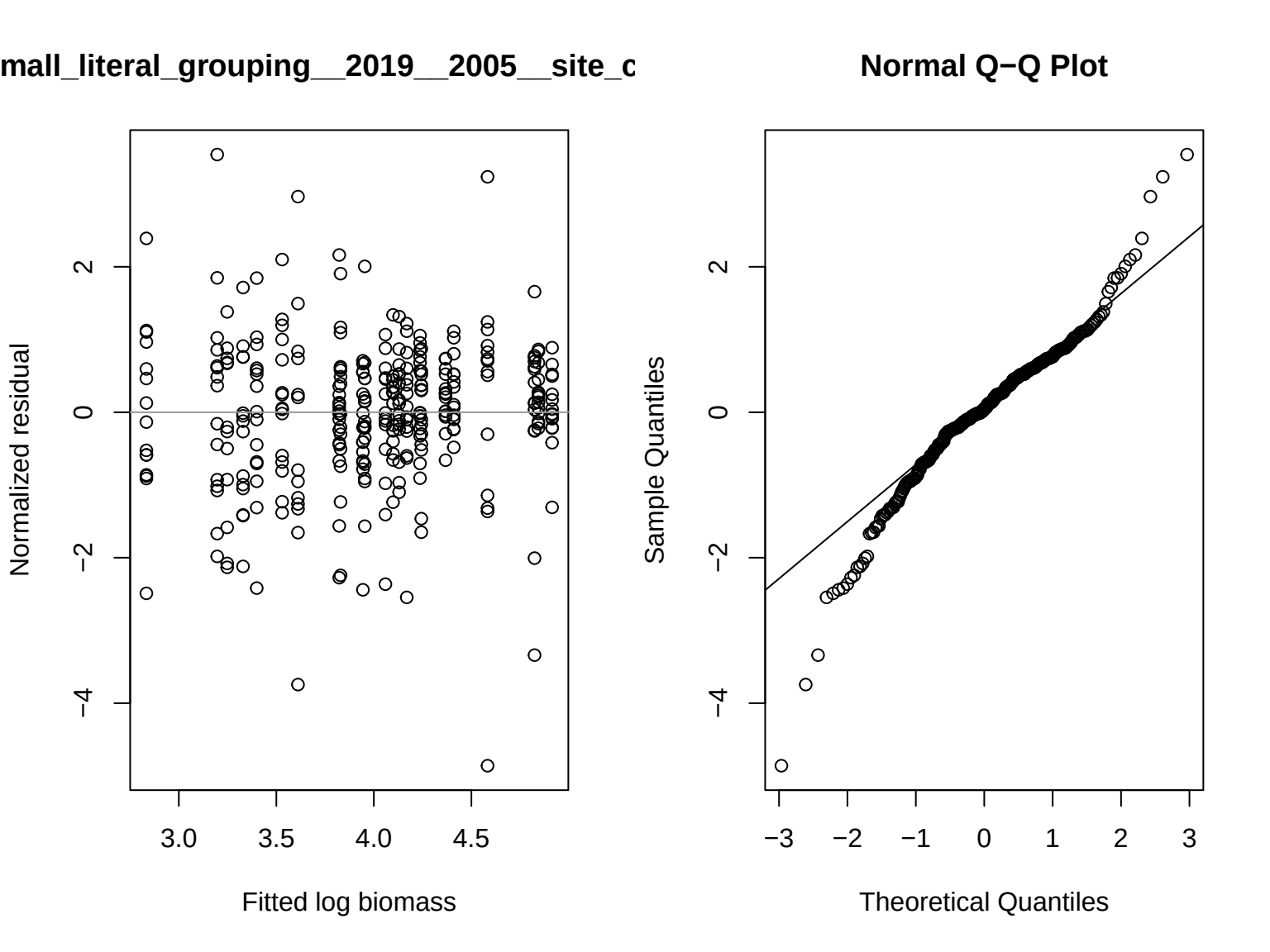

mall_mass_corrected__2022__2004__site_c
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

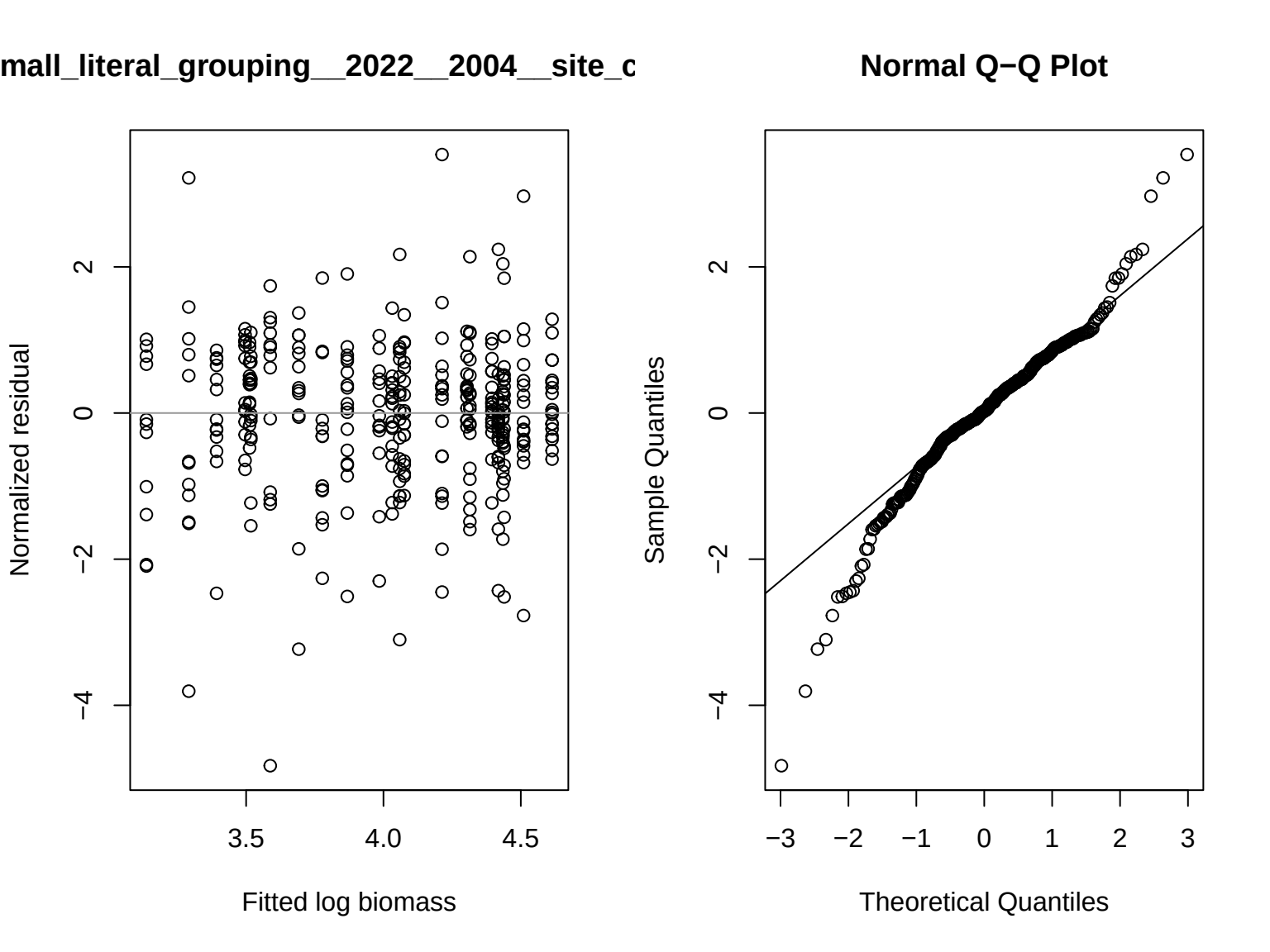

mall_mass_corrected__2022__2005__site_c
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

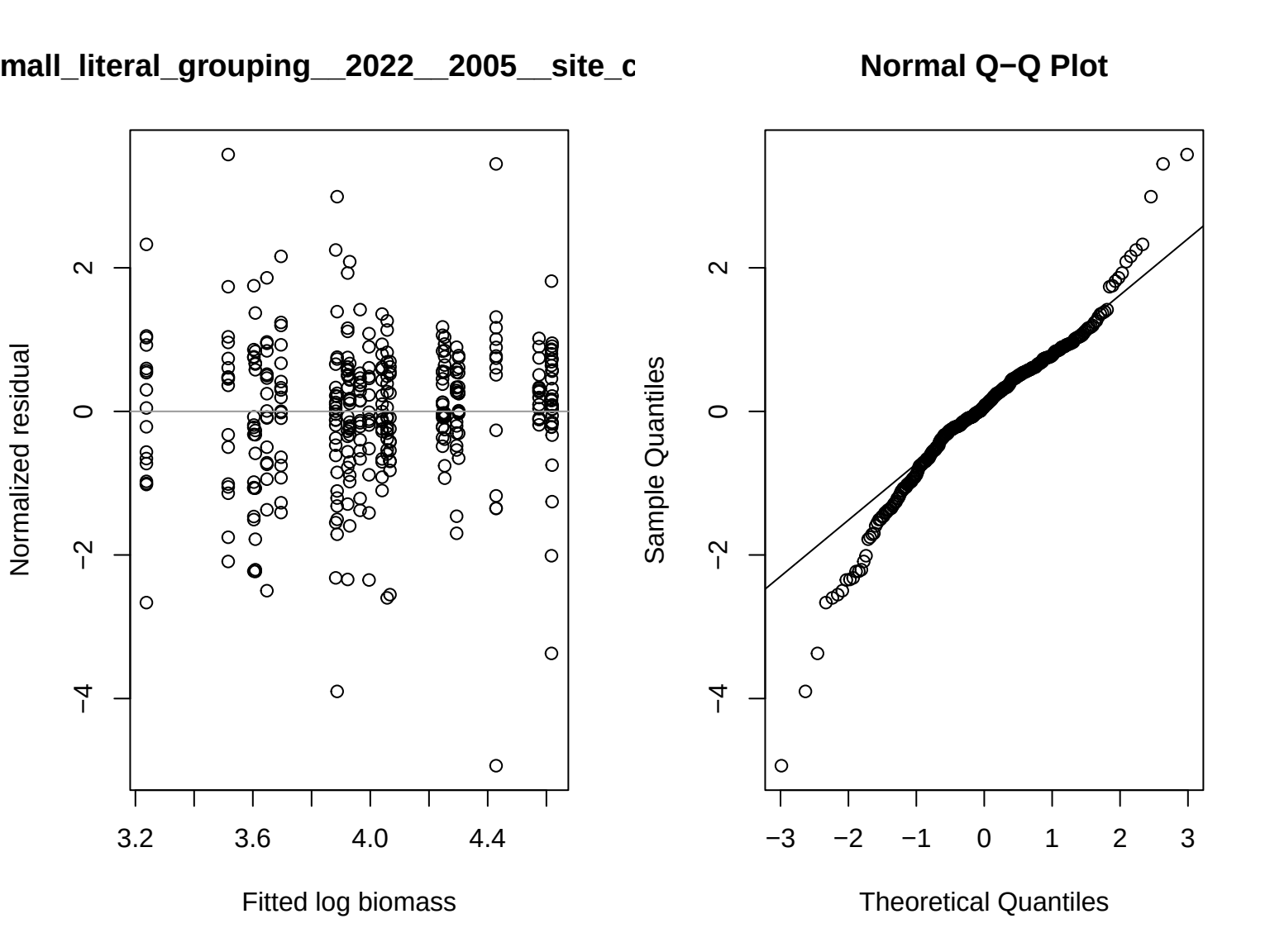

mall_mass_corrected__2023__2004__site_c
Normalized residual
Fitted log biomass
2.0
2.5
3.0
3.5
4.0
4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

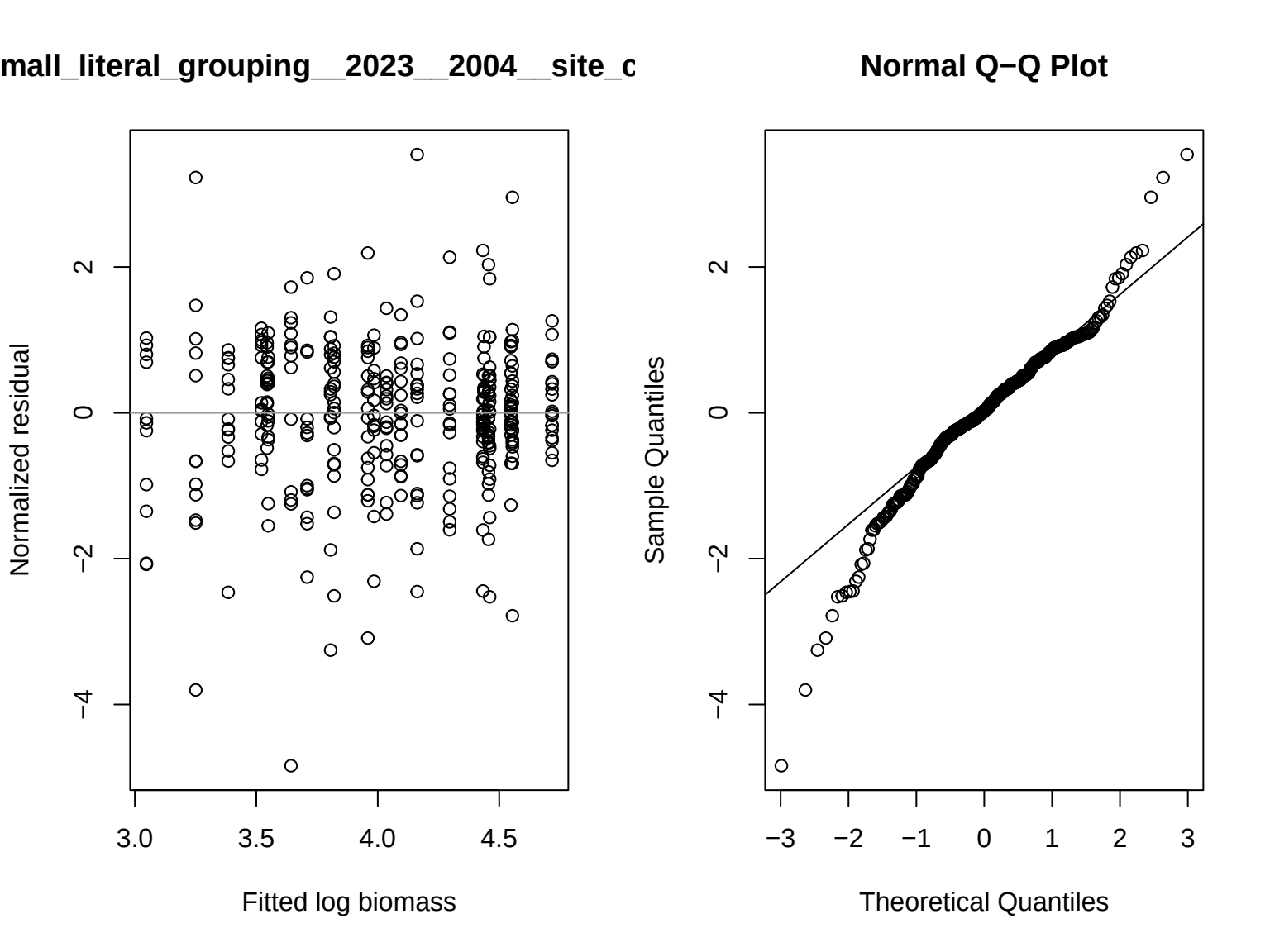

mall_mass_corrected__2023__2005__site_c
Normalized residual
Fitted log biomass
2.5
3.0
3.5
4.0
−4
−2
0
2
4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

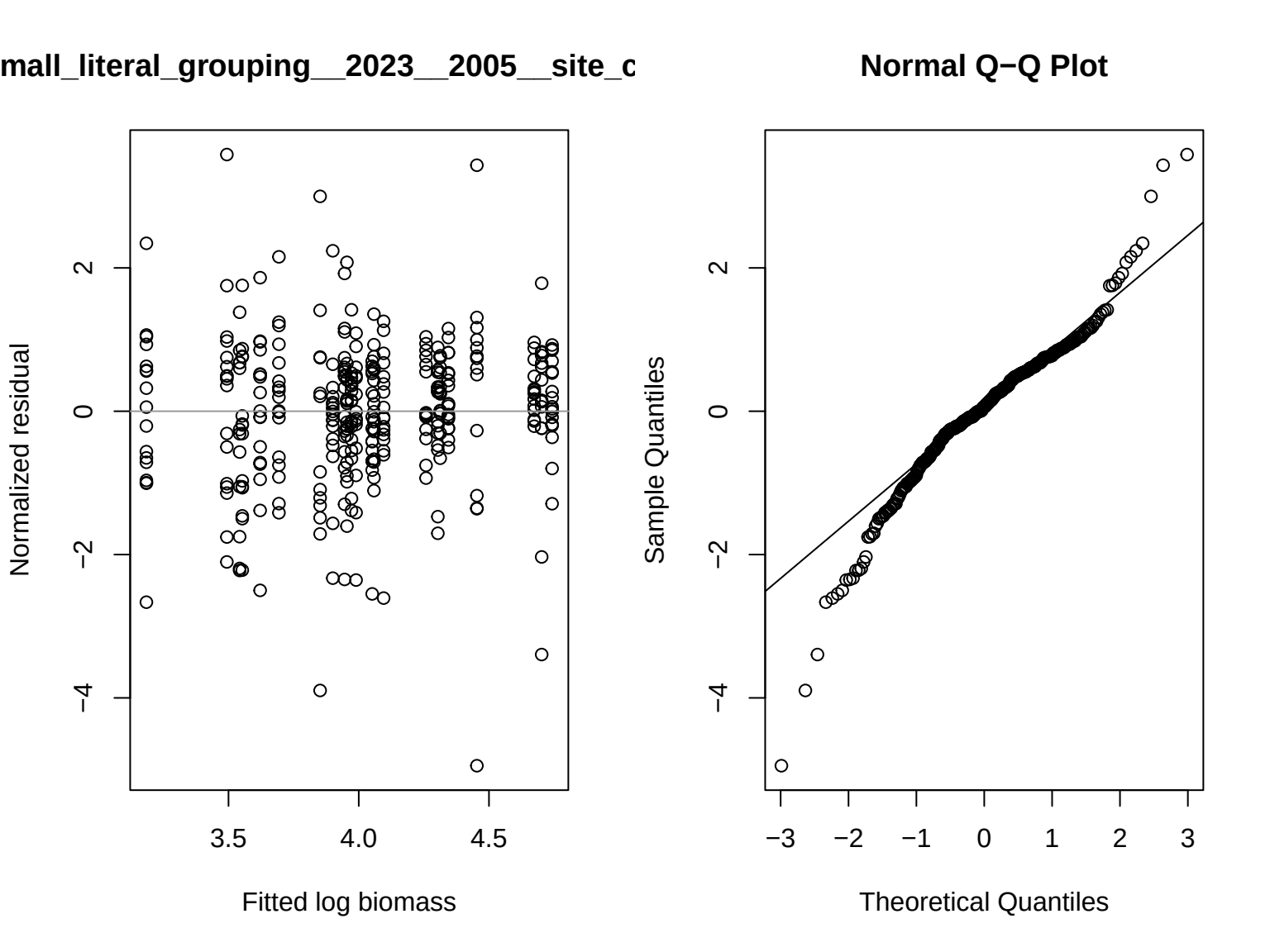

mall_literal_grouping__2019__2004__site_c
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

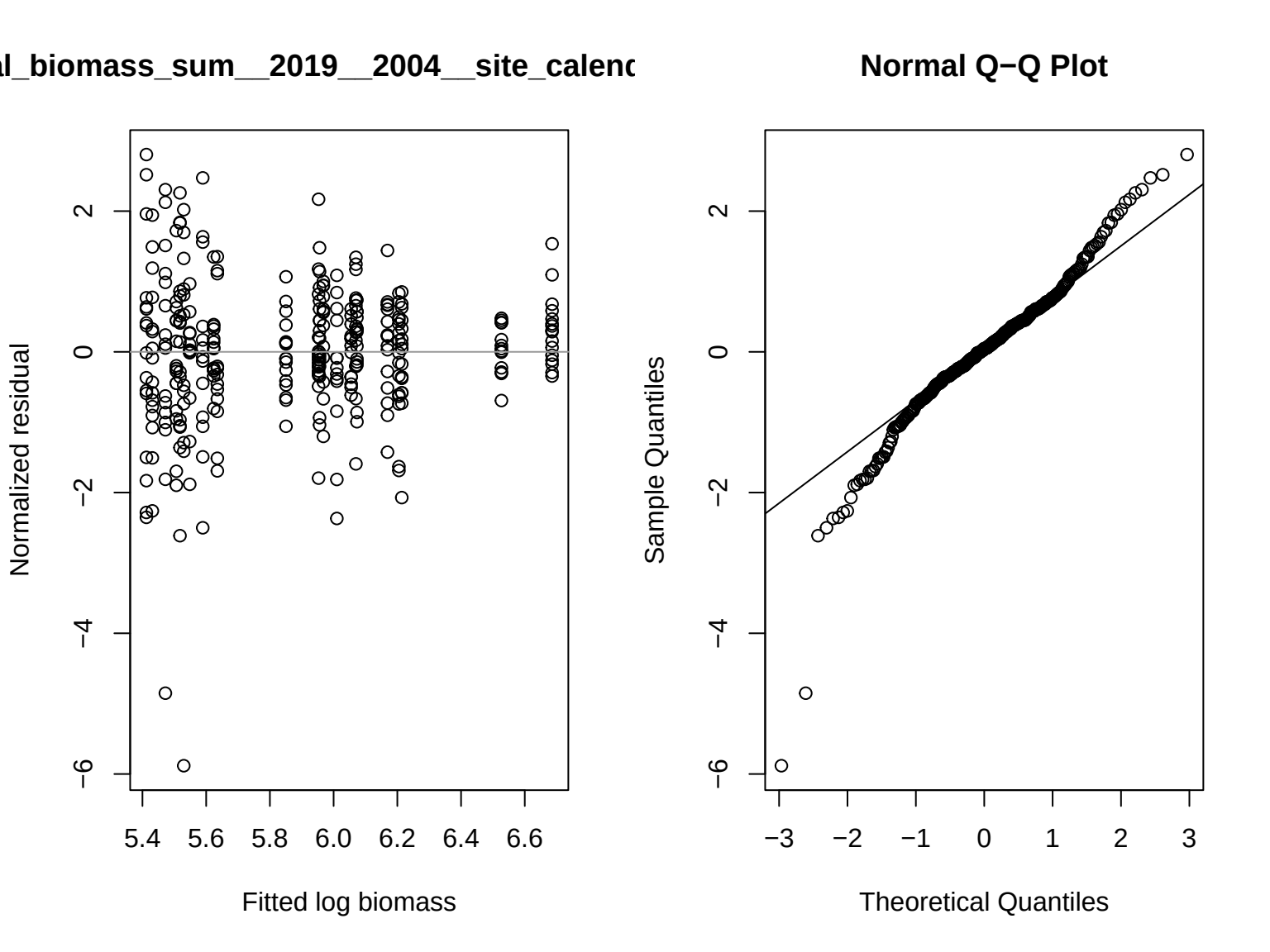

mall_literal_grouping__2019__2005__site_c
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

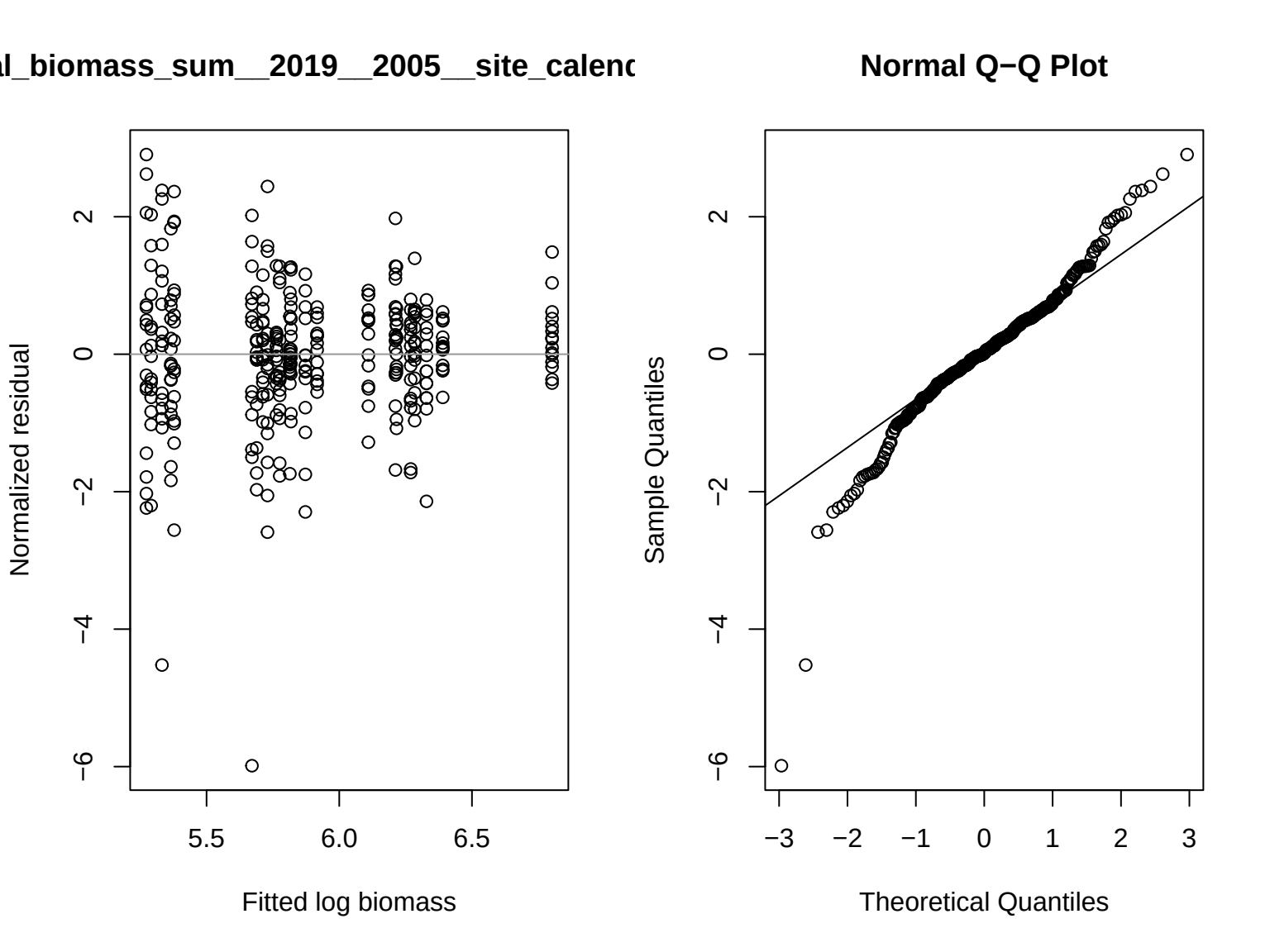

mall_literal_grouping__2022__2004__site_c
Normalized residual
Fitted log biomass
3.5
4.0
4.5
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

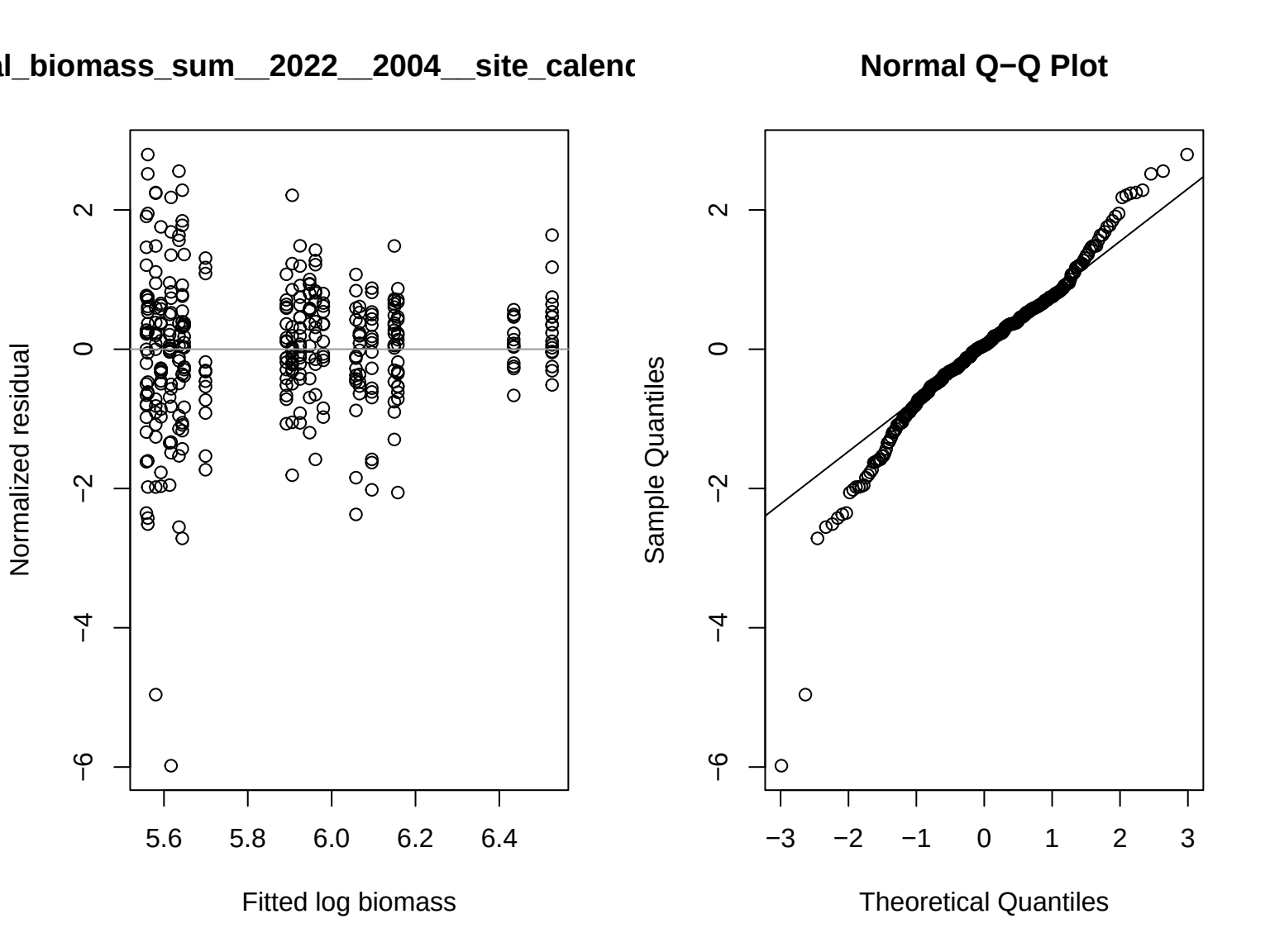

mall_literal_grouping__2022__2005__site_c
Normalized residual
Fitted log biomass
3.2
3.6
4.0
4.4
2
0
−2
−4
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

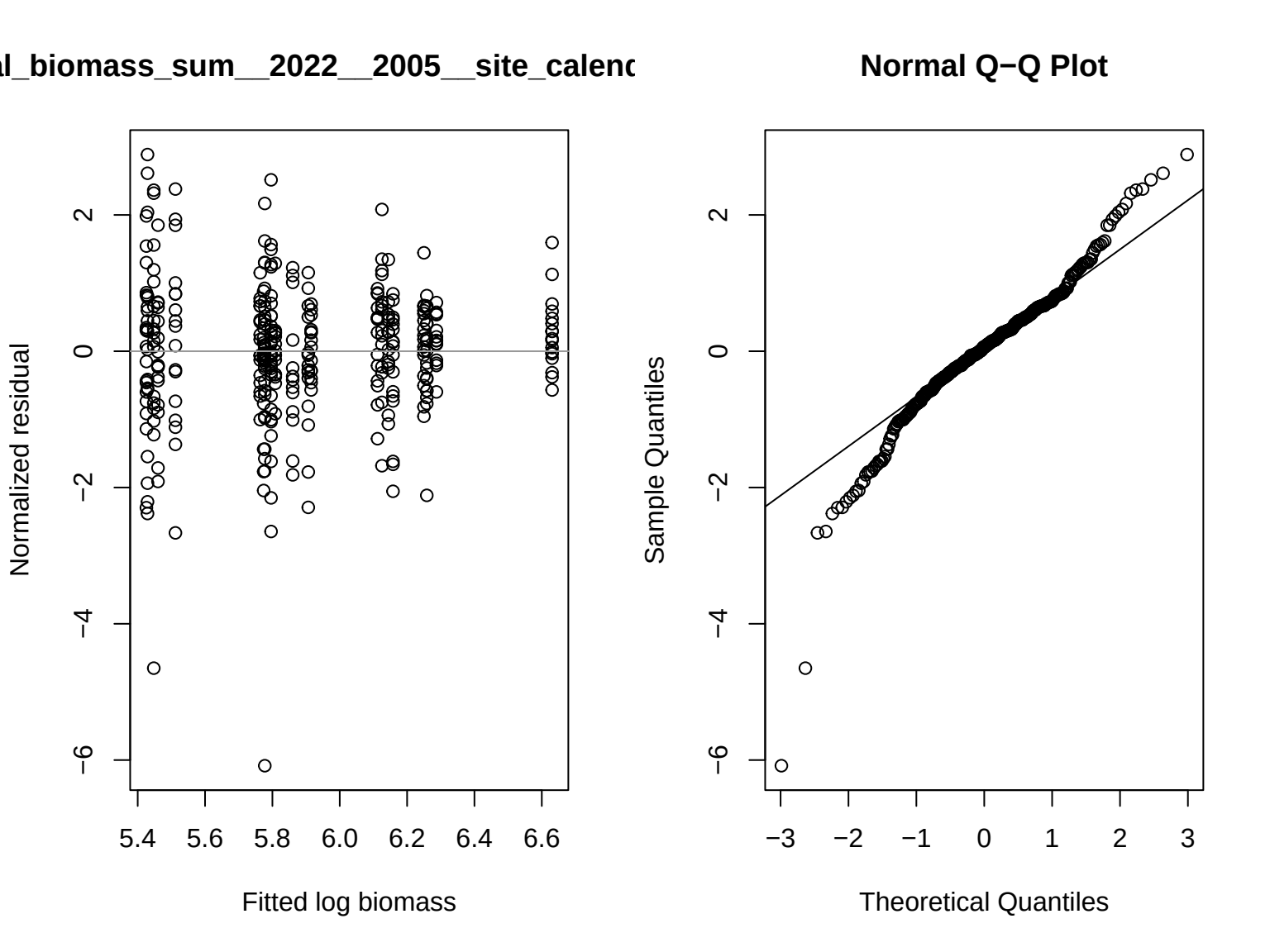

mall_literal_grouping__2023__2004__site_c
Normalized residual
Fitted log biomass
3.0
3.5
4.0
4.5
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

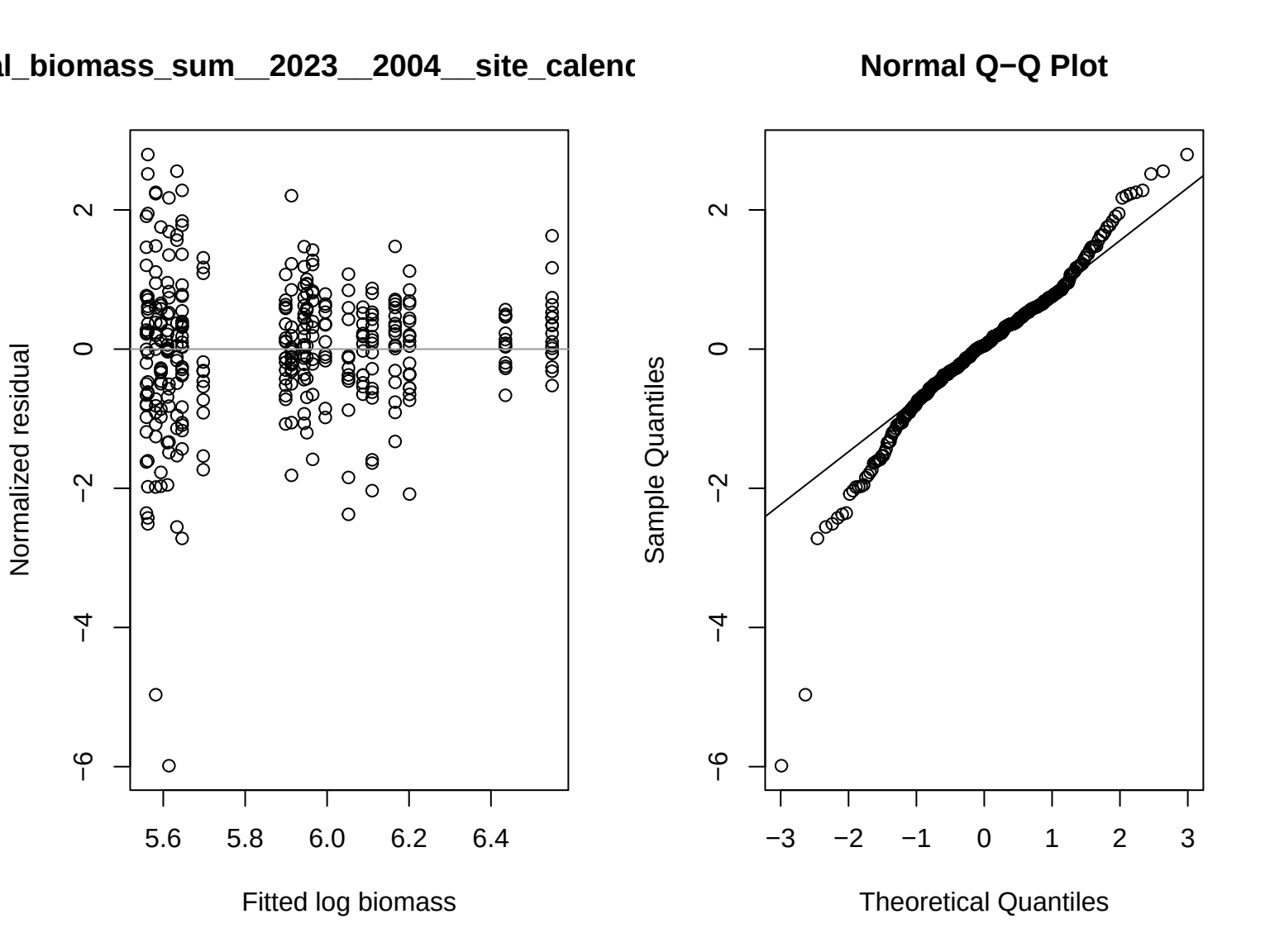

mall_literal_grouping__2023__2005__site_c
Normalized residual
Fitted log biomass
3.5
4.0
4.5
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

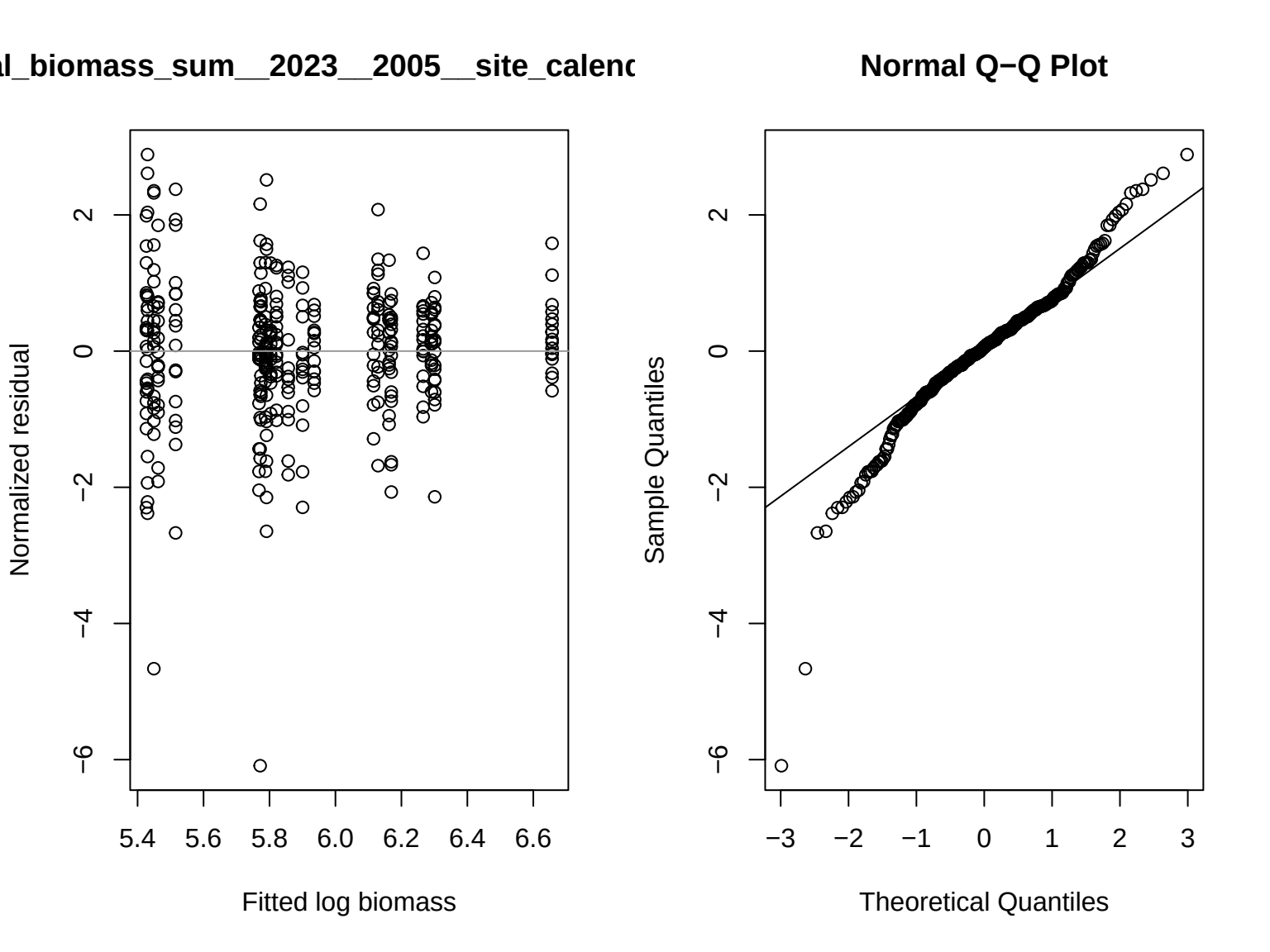

l_biomass_sum__2019__2004__site_calend
Normalized residual
Fitted log biomass
5.4
5.6
5.8
6.0
6.2
6.4
6.6
−6
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

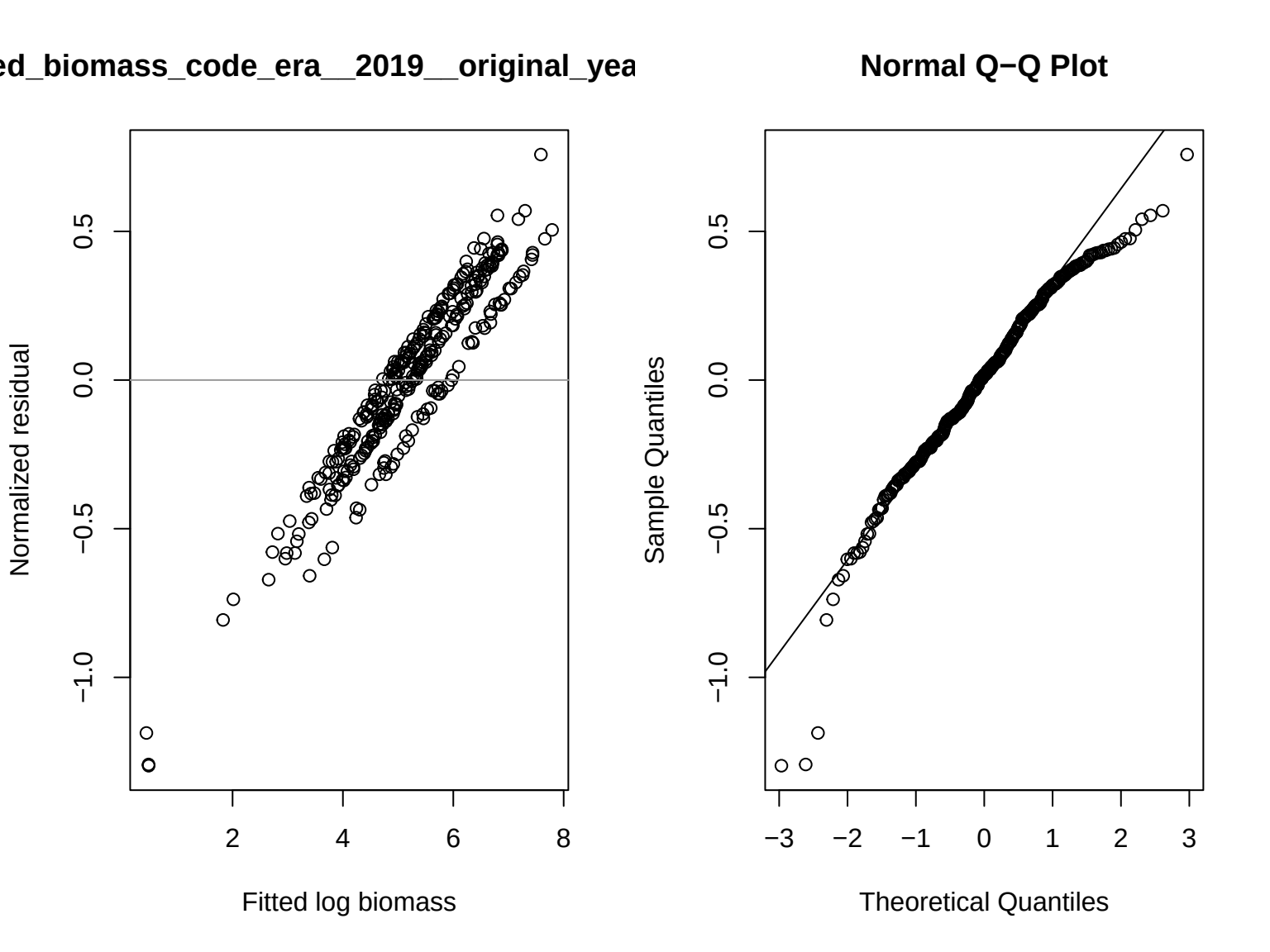

l_biomass_sum__2019__2005__site_calend
Normalized residual
Fitted log biomass
5.5
6.0
6.5
−6
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

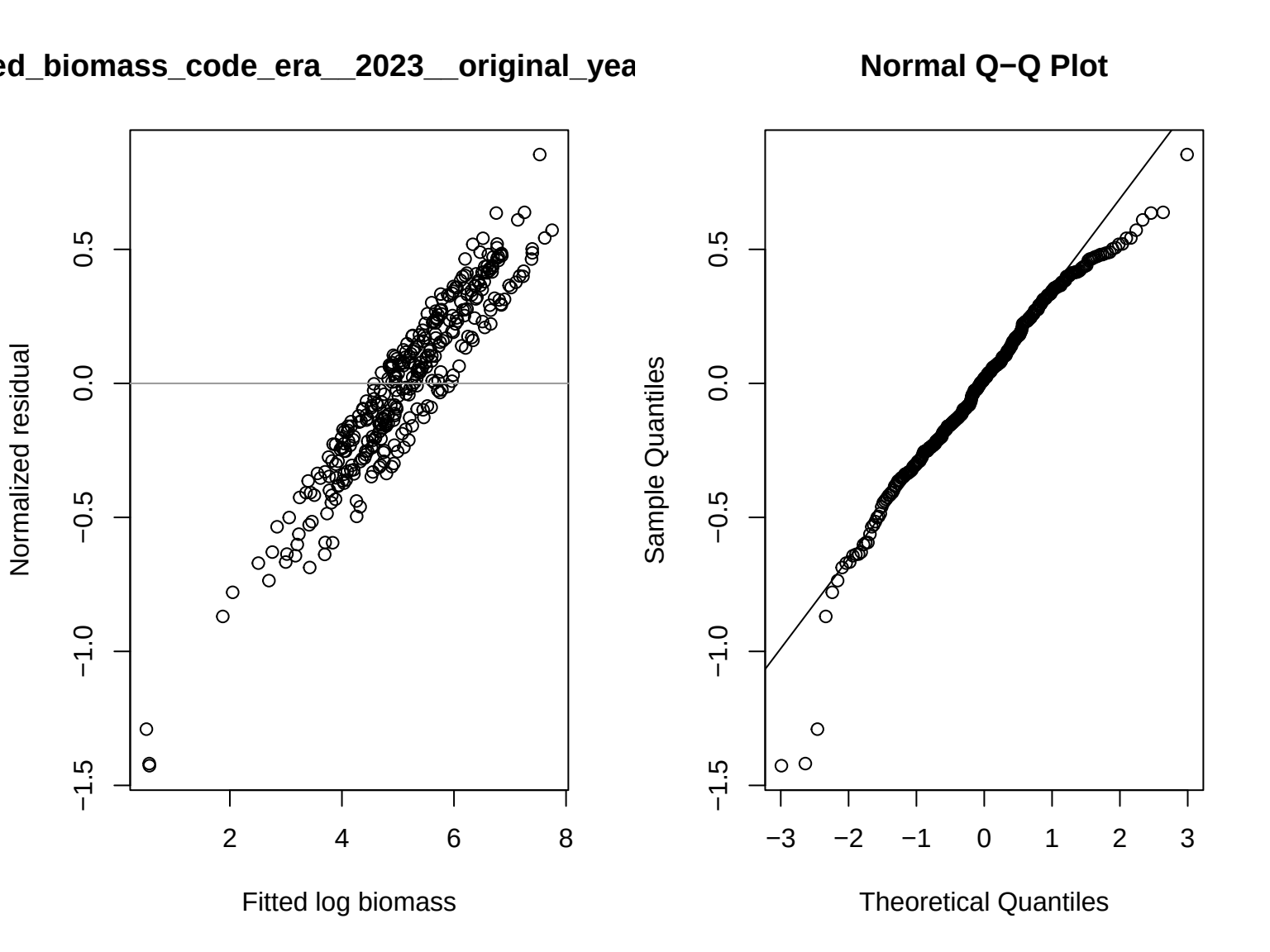

l_biomass_sum__2022__2004__site_calend
Normalized residual
Fitted log biomass
5.6
5.8
6.0
6.2
6.4
−6
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

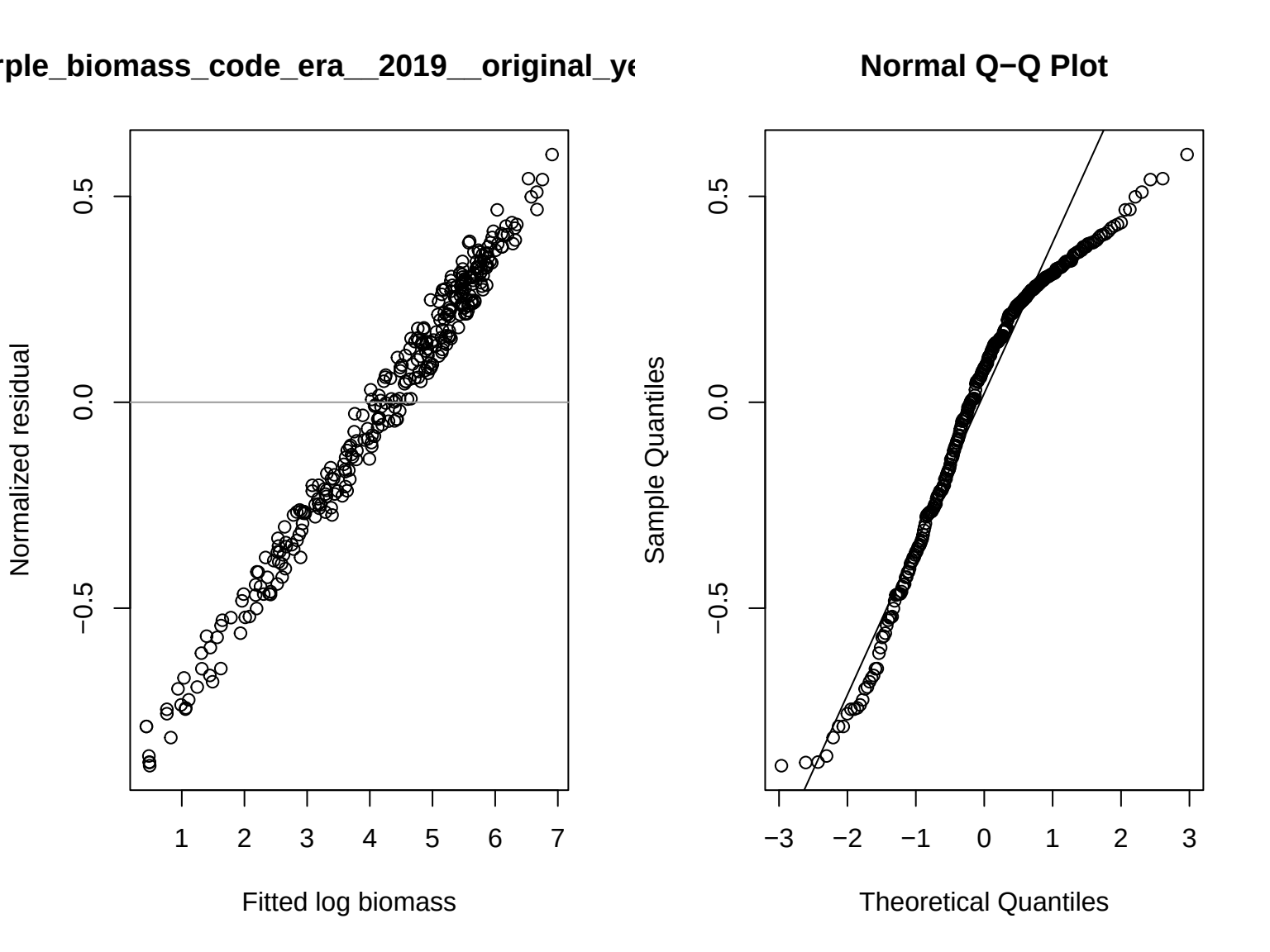

l_biomass_sum__2022__2005__site_calend
Normalized residual
Fitted log biomass
5.4
5.6
5.8
6.0
6.2
6.4
6.6
−6
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3

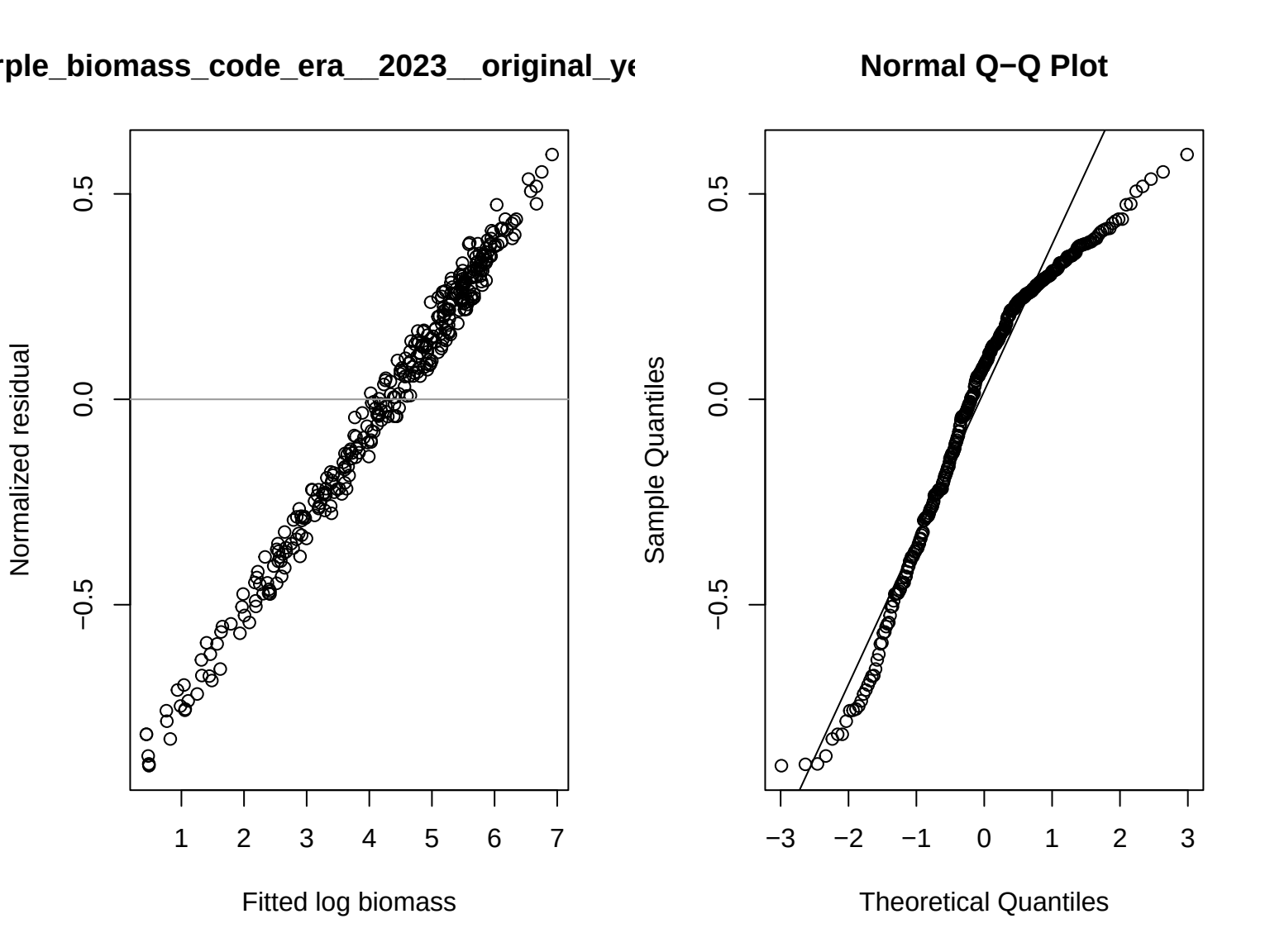

l_biomass_sum__2023__2004__site_calend
Normalized residual
Fitted log biomass
5.6
5.8
6.0
6.2
6.4
−6
−4
−2
0
2
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3


l_biomass_sum__2023__2005__site_calend
Normalized residual
Fitted log biomass
5.4
5.6
5.8
6.0
6.2
6.4
6.6
2
0
−2
−4
−6
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3


ed_biomass_code_era__2019__original_yea
Normalized residual
0.5
0.0
−0.5
−1.0
2
4
6
8
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
0.5
0.0
−0.5
−1.0
−3
−2
−1
0
1
2
3
Theoretical Quantiles


ed_biomass_code_era__2023__original_yea
Normalized residual
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
Theoretical Quantiles


rple_biomass_code_era__2019__original_ye
Normalized residual
0.5
0.0
−0.5
1
2
3
4
5
6
7
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
Theoretical Quantiles


rple_biomass_code_era__2023__original_ye
Normalized residual
0.5
0.0
−0.5
1
2
3
4
5
6
7
Fitted log biomass
Normal Q−Q Plot
Sample Quantiles
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
Theoretical Quantiles